1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

SCHEM,MLB_BAFFIN,X363G

REV	ECN	DESCRIPTION OF REVISION	CK APPD	DATE
10	0006897289	ENGINEERING RELEASED		2016-08-24

DRAWING
TITLE=MLB_BAFFIN
ABBREV=ABBREV
LAST_MODIFIED=Wed Aug 24 09:57:44 2016

LAST_MODIFICATION=Wed Aug 24 09:57:44 2016

Schematic / PCB #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
051-00647	1	SCHEM,MLB-BAFFIN,X363G	SCH	CRITICAL	
820-00281	1	PCBF,MLB-BAFFIN,X363G	PCB	CRITICAL	

This is the PVT design

DRAWING TITLE
SCHEM,MLB-BAFFIN,X363

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DRAWING NUMBER 051-00647 SIZE D

REVISION 10.0.0

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BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
685-00076	COMMON PARTS,MLB-BAFFIN,X363	X363_COMMON
985-00126	DEV,MLB-BAFFIN,X363	X363_DEVEL:PVT
985-00232	DEV,MLB-BAFFIN,PVT,X363	X363_DEVEL:PVT

639 BOMs have been moved to the end of the schematic

X363 BOM Groups

BOM GROUP	BOM OPTIONS
X363_COMMON	ALTERNATE,COMMON,X363_COMMON1,X363_COMMON2,X363_COMMON3,X363_COMMON4,X363_PROGPARTS
X363_COMMON1	SOC:HYNIX,SE:PROD,SKIP_5V3V3:AUDIBLE,DIPLEXER:MURATA,T208_PROG:REV5,BOARD_ID:17,VCCHDA:S0
X363_COMMON2	XDP:YES,SAMCONN,SOC_BOOT:SPI,DPMUX_XTAL:NO,GPUCLK:OSC,BAFFIN,AP_TEMP,VCCPLLOC:S3,WIFI_SAK:NO
X363_COMMON3	CPUTHRM:ALRT,TBTTHRM:ALRT,LOADRC:NO,OTHERRC:YES,DDRRC:YES,TBTRC:YES,TPADRC:YES,LID_FEATURE_ON
X363_COMMON4	EDP:YES,CPUPEG:X8X4X4,TBTTHRM_SNS,GPUTHRM_SNS,S3_STATE:YES,GPU_ROM:YES,SVID_PU:CORE
X363_PROGPARTS	BOOTROM_PROG:DVT,BT_PROG:DVT,WIFI_PROG:DVT,UPCROM_PROG:DVT,SMC_PROG:PVT,DPMUXMCU:PROG,PCC:NO
X363_DEVEL:ENG	ALTERNATE,ENGISNS,DBGLED,XDP_CONN,USBC_DBG,DBG_BTN,DBG_FAN,DBG_XTAL,DPMUX_DEBUG,WIFI_DBG,SSD_DEBUG,GPUROM:BLANK,PCC:YES
X363_DEVEL:DVT	ALTERNATE,ENGISNS,DBGLED,XDP_CONN,USBC_DBG,DBG_BTN,DBG_FAN,DBG_XTAL,DPMUX_DEBUG,WIFI_DBG,SSD_DEBUG
X363_DEVEL:PVT	ALTERNATE,XDP_CONN,USBC_DBG
ENGISNS	TBTISNS,LOADISNS,TPADISNS,DDRISNS,OTHERISNS

Module Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
337S00227	1	CPU,SKY,SR2FT,R1,PRQ,4/2,2.9,BGA1440	U0500	CRITICAL	CPU_SKL:2.9
337S00228	1	CPU,SKY,SR2FU,R1,PRQ,4/2,2.7,BGA1440	U0500	CRITICAL	CPU_SKL:2.7
337S00229	1	CPU,SKY,SR2FQ,R1,PRQ,4/2,2.6,BGA1440	U0500	CRITICAL	CPU_SKL:2.6
998-04701	1	INTERPOSER,INTEL,BGA1440,MM940989	U0500	CRITICAL	CPU_SKL:SOCKET
337S00258	1	IC,SKL PCH-H,SFF,SR2NH,PRQ,D1,BGA939	U1100	CRITICAL	
353S00961	4	IC,CD3215,ACE,C0,USB PWR SW,BLNK,BGA96	U3100,U3200,UB300,UB400	CRITICAL	
338S00254	2	IC,TBT,ALPINE RIDGE DP,QT5S,QS,C1,BGA337	U2800,UB000	CRITICAL	
353S01016	1	IC,ISL9239HIZ,PMU,TUBA,WCSP40,2.1X3.3MM	U7000	CRITICAL	
338S00221	1	IC,PMU,P650839,7X7MM.BGA168	U7800	CRITICAL	
338S00142	1	IC,CODEC,CLIFDEN,CS42L83A,B0,WLCSP49	U6300	CRITICAL	
337S00330	1	IC,GPU,BAFFIN,ULA,A1,PS,BGA769	UA000	CRITICAL	BAFFIN_ULA
337S00331	1	IC,GPU,BAFFIN,PROA,A1,PS,BGA769	UA000	CRITICAL	BAFFIN_PROA
337S00332	1	IC,GPU,BAFFIN,LEA,A1,PS,BGA769	UA000	CRITICAL	BAFFIN_LEA
998-04866	1	INTERPOSER,AMD,C989,BGA769,VDDCI/MVDD	UA000	CRITICAL	STARDUST:VDDCI_MVDD
998-04867	1	INTERPOSER,AMD,C988,BGA769,VDDC	UA000	CRITICAL	STARDUST:VDDC
677-04532	2	SUBASSY (T&R) PCBA, AMR, INTERPOSER, X363	J5250,J5260	CRITICAL	

Development/Base BOMs

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
685-00076	1	COMMON PARTS,MLB-BAF,X363	BASE	CRITICAL	BASE_BOM
985-00126	1	DEV,MLB-BAF,X363	DEVEL	CRITICAL	DEVEL_BOM

WIFI/BT Diplexers

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
155S0979	3	FLTR,DIPLEXER,2.45/5.54GHZ,0805	U3810,U3820,U3830	CRITICAL	DIPLEXER:MURATA

Strategic Silicon

PART#	STRATEGIC VALUE	COMMENT
337S00227	08	CPU
337S00228	08	CPU
337S00229	08	CPU
333S00050	07	MAIN MEMORY
333S00070	07	MAIN MEMORY
335S00149	02	SSD NAND
335S00204	02	SSD NAND
335S00205	02	SSD NAND
335S00219	02	SSD NAND
339S00154	02	SSD CONTROLLER
339S00155	02	SSD CONTROLLER
338S00166	02	SSD PMIC
337S00225	08	GPU
337S00285	08	GPU
337S00286	08	GPU
333S00044	07	VIDEO MEMORY
333S00043	07	VIDEO MEMORY
333S00078	07	VIDEO MEMORY
333S00074	07	VIDEO MEMORY
333S00075	07	VIDEO MEMORY
343S00135	10	T208
343S00136	10	T208
343S00137	10	T208
338S00138	10	T208
338S00193	09	BERKELIUM
353S3978	02	MOJAVE
338S00097	02	SECURE ELEMENT
338S00254	08	ALPINE RIDGE
353S00961	09	ACE
338S00142	09	CLIFDEN
353S00604	07	AUDIO AMP
353S4316	08	BAYSIDE
338S00221	08	BANJO
353S00853	09	TUBA
339S00056	05	ICEBOCK
359S00006	08	GREEN CLOCK
353S00795	09	DEBUG MUX

GPU Options

BOM GROUP	BOM OPTIONS
2GB_MC_BAFFIN	FB_2GB_MICRON,VRAM:GRP1
2GB_HY_BAFFIN	FB_2GB_HYNIX,VRAM:GRP1
2GB_SM_BAFFIN	FB_2GB_SAMSUNG,VRAM:GRP2
4GB_SM_BAFFIN	FB_4GB_SAMSUNG,VRAM:GRP1
4GB_MC_BAFFIN	FB_4GB_MICRON,VRAM:GRP1

Main DRAM Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
333S00050	4	IC,SDRAM,LPDDR3-2133,32GBIT,20NM,BGA178	U2300,U2400,U2500,U2600	CRITICAL	16G_SAMSUNG_2133
333S00070	4	IC,SDRAM,LPDDR3-2133,32GBIT,20NM,BGA178	U2300,U2400,U2500,U2600	CRITICAL	16G_MICRON_2133

Main DRAM SPD Straps

BOM GROUP	BOM OPTIONS
RAM_16G_SAMSUNG_2133	16G_SAMSUNG_2133,RAMCFG4:L,RAMCFG3:L,RAMCFG0:L
RAM_16G_MICRON_2133	16G_MICRON_2133,RAMCFG4:L,RAMCFG3:L,RAMCFG1:L

FB VDRAM Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
333S00044	4	IC,GDDR5,4Gb,7Gbps,1.5V,25NM,A,170 BGA	UA400,UA450,UA500,UA550	CRITICAL	FB_2GB_MICRON
333S00043	4	IC,GDDR5,4Gb,7Gbps,1.5V,25NM,A,170 BGA	UA400,UA450,UA500,UA550	CRITICAL	FB_2GB_HYNIX
333S00078	4	IC,GDDR5,8Gb,7Gbps,1.5V,25NM,B,170 BGA	UA400,UA450,UA500,UA550	CRITICAL	FB_2GB_SAMSUNG
333S00074	4	IC,GDDR5,8Gb,7Gbps,1.5V,25NM,B,170 BGA	UA400,UA450,UA500,UA550	CRITICAL	FB_4GB_SAMSUNG
333S00075	4	IC,GDDR5,8Gb,7Gbps,1.5V,25NM,A,170 BGA	UA400,UA450,UA500,UA550	CRITICAL	FB_4GB_MICRON

Sub-BOM DIPLEXER

BOM NUMBER	BOM NAME	BOM OPTIONS
685-00085	DIPLEXERS,MURATA,X363G	DIPLEXER:MURATA

SYNC_MASTER=J80_MLB SYNC_DATE=07/07/2015

PAGE TITLE
BOM Configuration

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Programmable Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	
338S1231	1	IC,SMC12,40MHZ/50DMIPS MCU,7X7,168BGA	U5000	CRITICAL	SMC_PROG:BLANK	Blank
341S00701	1	IC,SMC-B1,EXT (v2.37F7) PVT,X363G	U5000	CRITICAL	SMC_PROG:PVT	TI
335S00013	1	IC,SPI SERIAL FLASH,64M BITS,3V,8P SOIC,QE=1	U6100	CRITICAL	BOOTROM_PROG:BLANK	Blank
341S00699	1	IC,EFI ROM (V0193), DVT, X363G	U6100	CRITICAL	BOOTROM_PROG:DVT	Macronix/Winbond
353S00926	2	IC,CD3215,ACE,B03,BLNK,BGA96	U2890,UB090	CRITICAL	UPCROM_PROG:BLANK	Blank
341S00707	1	T29,AR1 (V10.5) PVT, X363G	U2890	CRITICAL	UPCROM_PROG:DVT	Winbond
341S00708	1	T29,AR2 (V10.5) PVT, X363G	UB090	CRITICAL	UPCROM_PROG:DVT	Winbond
335S00024	1	IC,SERIAL-FLASH,2MBIT,4V,8-USON,2x3x,6MM	U3750	CRITICAL	BT_PROG:BLANK	Blank
341S00695	1	IC,BT ROM (V28), DVT, X362/X363	U3750	CRITICAL	BT_PROG:DVT	Macronix/Winbond
341S00709	1	WIFI ROM (P107) DVT,NEW,WW1,X362/X363	U3710	CRITICAL	WIFI_PROG:DVT	Rohm/On Semi
341S3565	1	IC, EDP MUX-95C, (RENESAS) V3.2.8,DVB,D2	U9800	CRITICAL	DPMUXMCU:PROG	
335S0724	1	IC,1Mbit SERIAL FLASH 2X3X0.6MM UFDFPN8 PKG	UA701	CRITICAL	GPUROM:BLANK	Blank

SYNC_MASTER=J80_MLB_BAFFIN_CLEAN SYNC_DATE=12/02/2015

PAGE TITLE
BOM Configuration

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Shield Can Omit Table

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
806-08023	1	SHIELD,FENCE,DRAM,X378	SH0400	CRITICAL	
806-08019	1	SHIELD,FENCE,ALPINE RIDGE,RIGHT,X378	SH0410	CRITICAL	
806-08021	1	SHIELD,FENCE,ALPINE RIDGE,LEFT,X378	SH0420	CRITICAL	
806-07918	1	SHIELD,NAND,TOP,ALT,X363	SH0440	CRITICAL	
806-07917	1	SHIELD,NAND,BOTTOM,ALT,X363	SH0450	CRITICAL	
806-08024	1	SHIELD,DIPLEX,EG,X378	SH0460	CRITICAL	
806-08026	1	FENCE,VRAM,EG,X378	SH0470	CRITICAL	

BOM_COST_GROUP=MECHANICALS

SYNC_MASTER=J80_MLB SYNC_DATE=11/16/2015

PAGE TITLE: PD Parts

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DRAWING NUMBER: 051-00647 SIZE: D

REVISION: 10.0.0

BRANCH: dvt-fab10

CPU Daisy-Chain Strategy:

Each corner of CPU has two testpoints.
Other corner test signals connected in
daisy-chain fashion. Continuity should
exist between both TP's on each corner.

BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE
CPU DMI/PEG/FDI/RSVD

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CFG [7] :PEG DEFER TRAINING 1 = (DEFAULT) IMMEDIATELY AFTER xxRESETB 0 = WAIT FOR BIOS
CFG [6:5] :PCIE BIFURCATION 11 = 1 X16 (DEFAULT) 10 = 2 X8 01 = RSVD 00 = X8, X4, X4
CFG [4] :eDP ENABLE/DISABLE 1 = DISABLED 0 = ENABLED
CFG [3] :PCIE x4 LANE REVERSAL 1 = NORMAL OPERATION 0 = LANES REVERSED
CFG [2] :PCIE x16 LANE REVERSAL 1 = NORMAL OPERATION 0 = LANES REVERSED

BOM GROUP	BOM OPTIONS
CPUPEG:X8X8	CPUCFG5_PD
CPUPEG:X8X4X4	CPUCFG6_PD,CPUCFG5_PD

To use PEG X16 configuration, simply remove CPUPEG:X8X8 and CPUPEG:X8X4X4 from BOMs.

BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE
CPU Clock/Misc/JTAG/CFG

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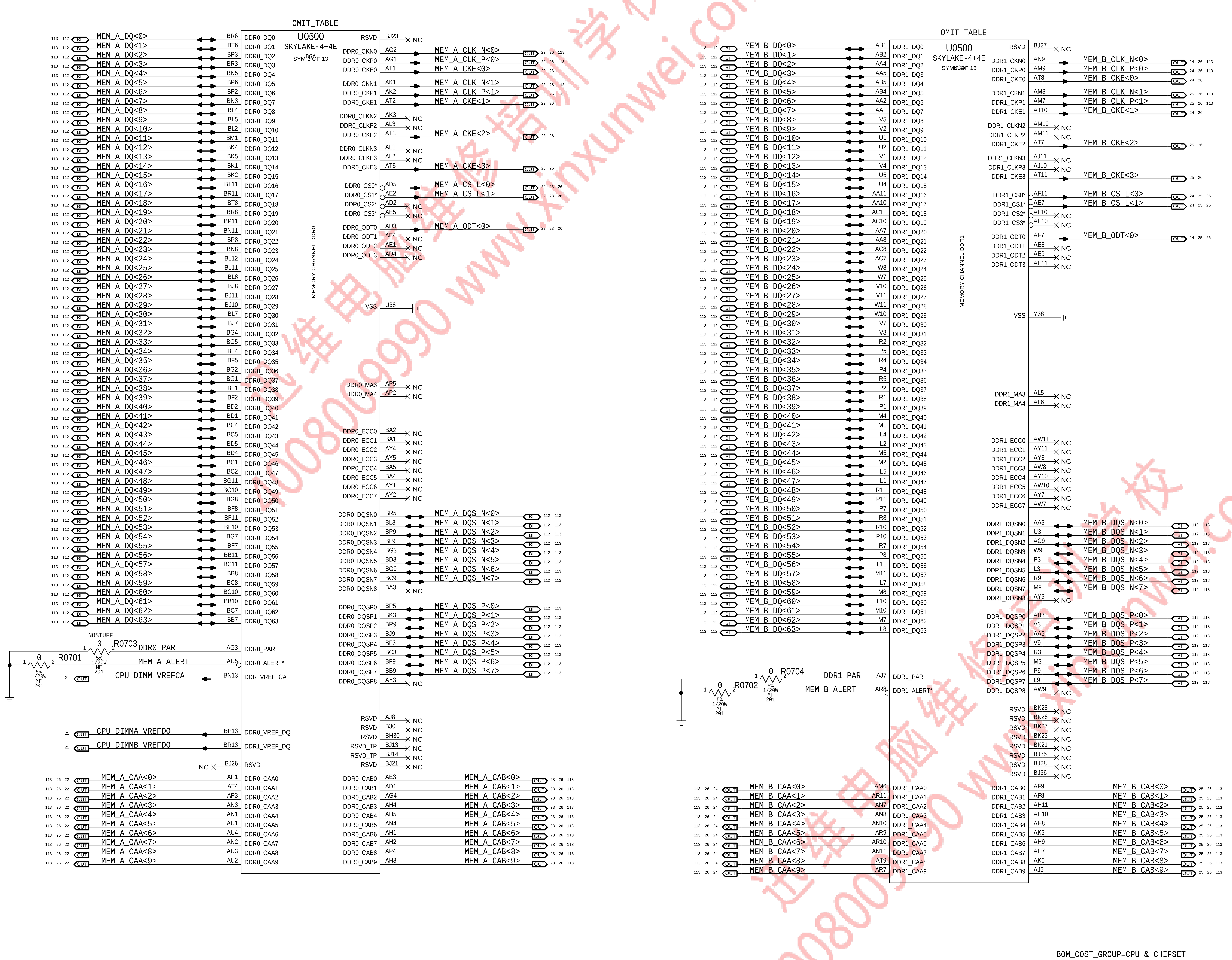

SYNC_MASTER=J80_MLB
SYNC_DATE=11/06/2015

PAGE TITLE
CPU DDR3 Interfaces

Apple Inc.

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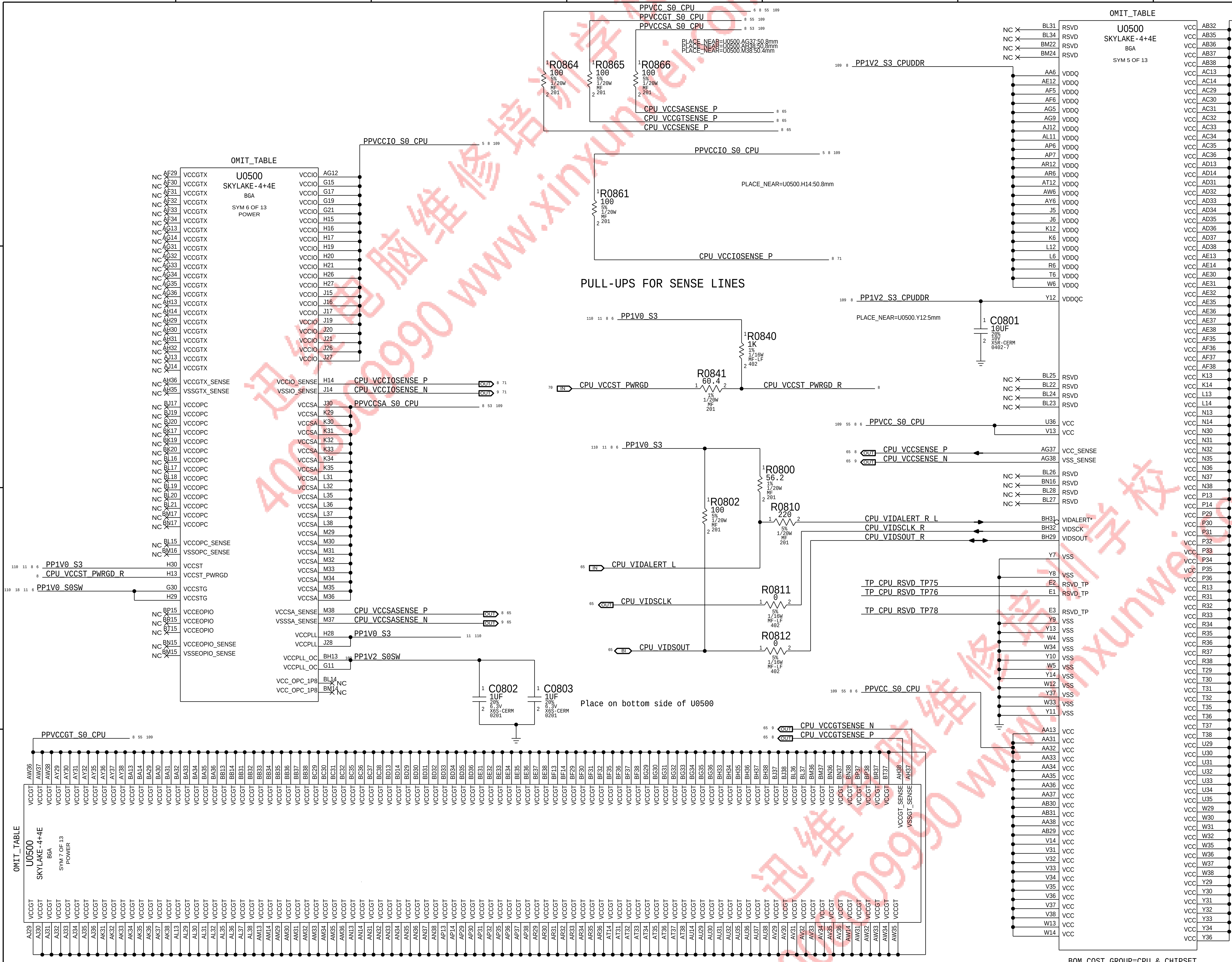

CPU Power

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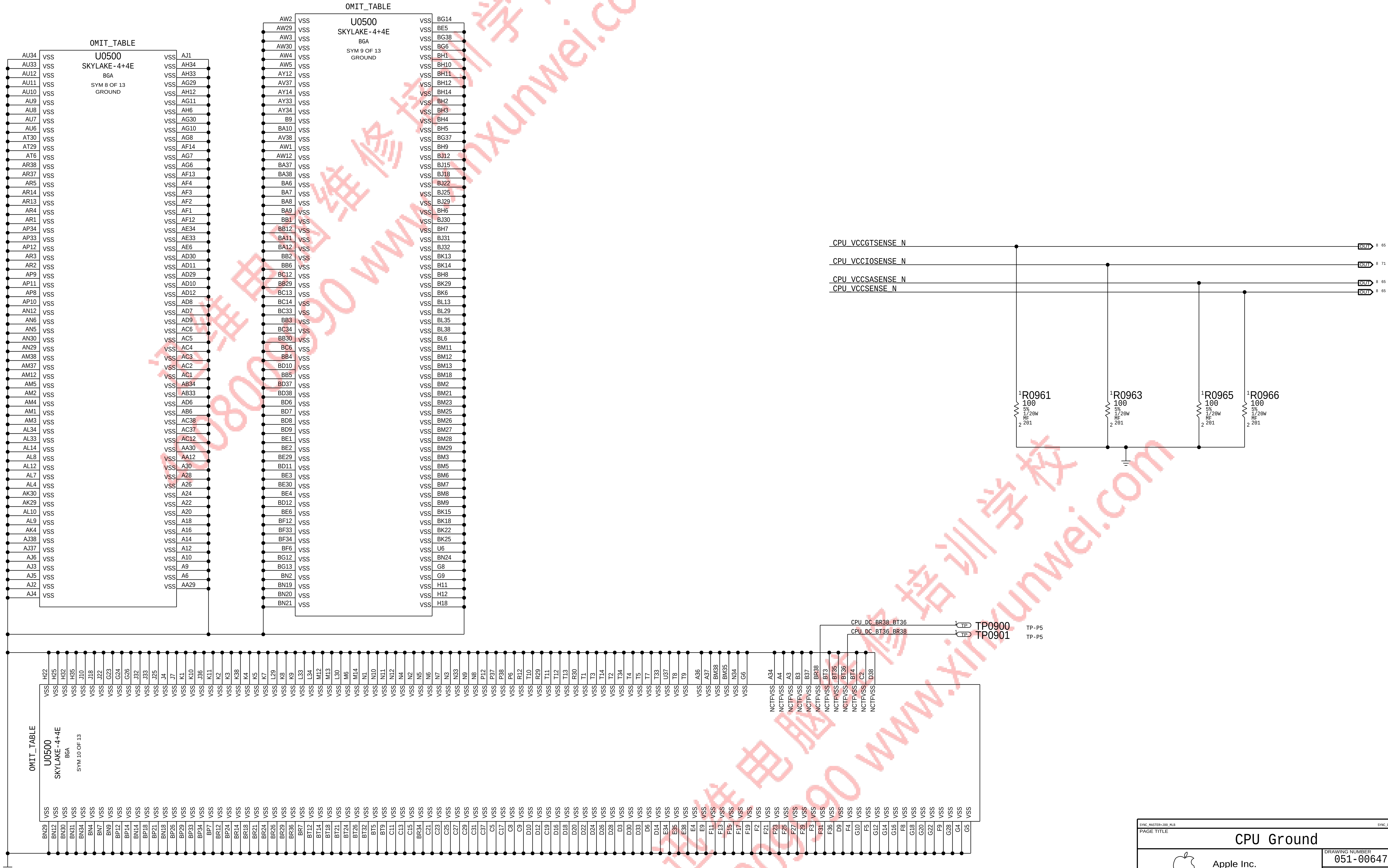

BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE

CPU Ground

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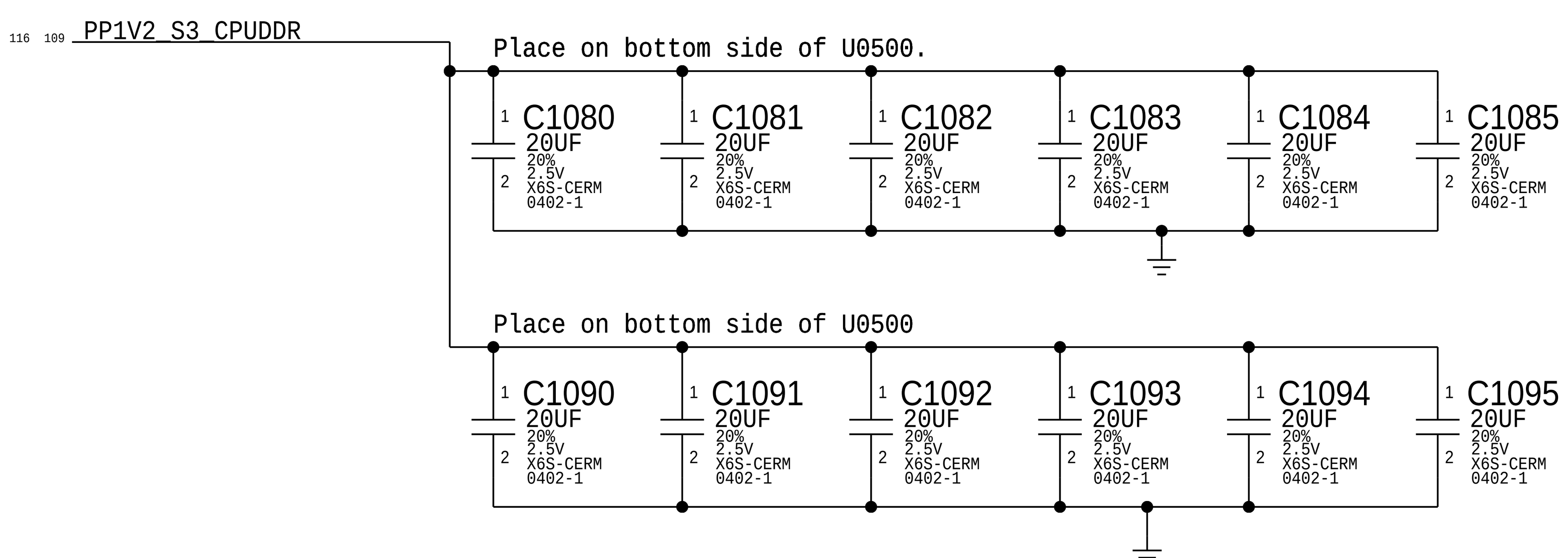
Vcc CPU Core Decoupling from 20140905 BOM

CPU VCORE Decoupling

Intel recommendation: 5x 220uF ESR 5m ohms ESL 1.9nH each,4x 47uF 0805 8x22uF 0603, 28x 10uF 0402, 3x 10uF 0402, 69x 1uF 0201   Board Edge: 2x 220uF, 4x 47uF rest on the back side
Apple Implementation:

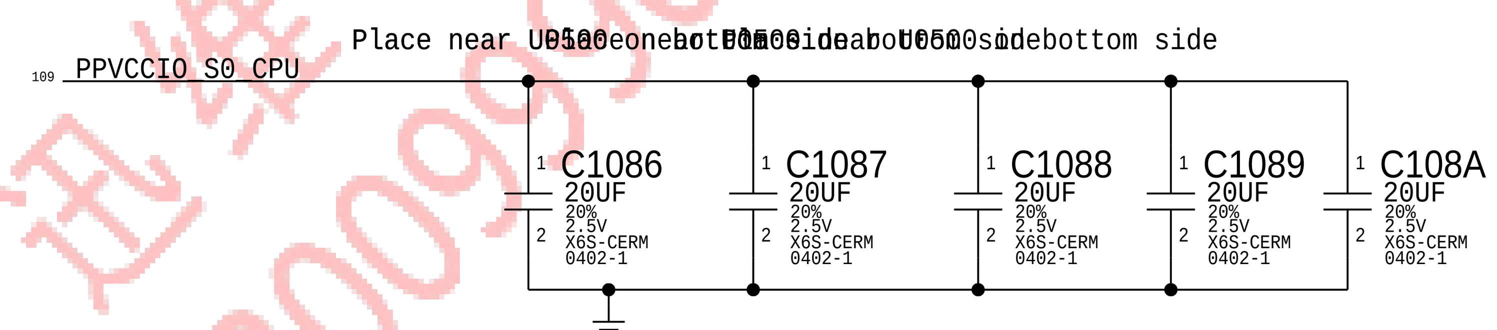
CPU VDDQ Decoupling

Intel recommendation: 10x 10uF 0402, 4x 22uF 0602
Apple Implementation:



CPU VCCIO Decoupling

Intel recommendation: 3x 10uF 0402 (opposite CPU)
Apple Implementation:



NOTE: Intel decoupling recommendations from CBR schematics for Skylake H doc#557227 and PDG section 48.1 (document# 546884)

BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE
CPU Decoupling 1 [10]

Apple Inc.

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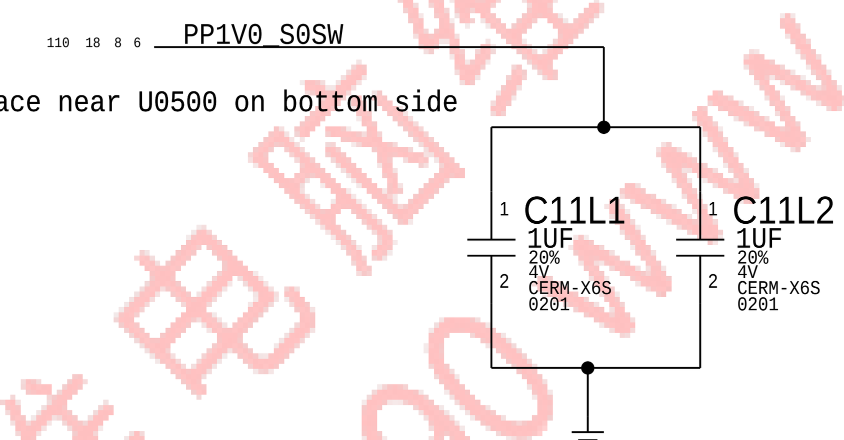
Vcc GT Slice Core Decoupling from 20140905 BOM

CPU VGTSlice Decoupling

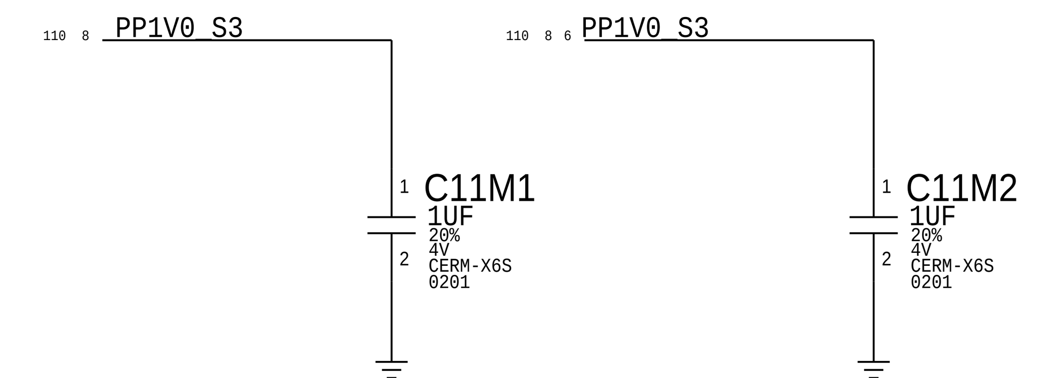
Intel recommendation: 7x 220uF, 6x 47uF 0805, 6x 22uF 0603, 35x 10uF 0402, 68 1uF 0201
Apple Implementation:

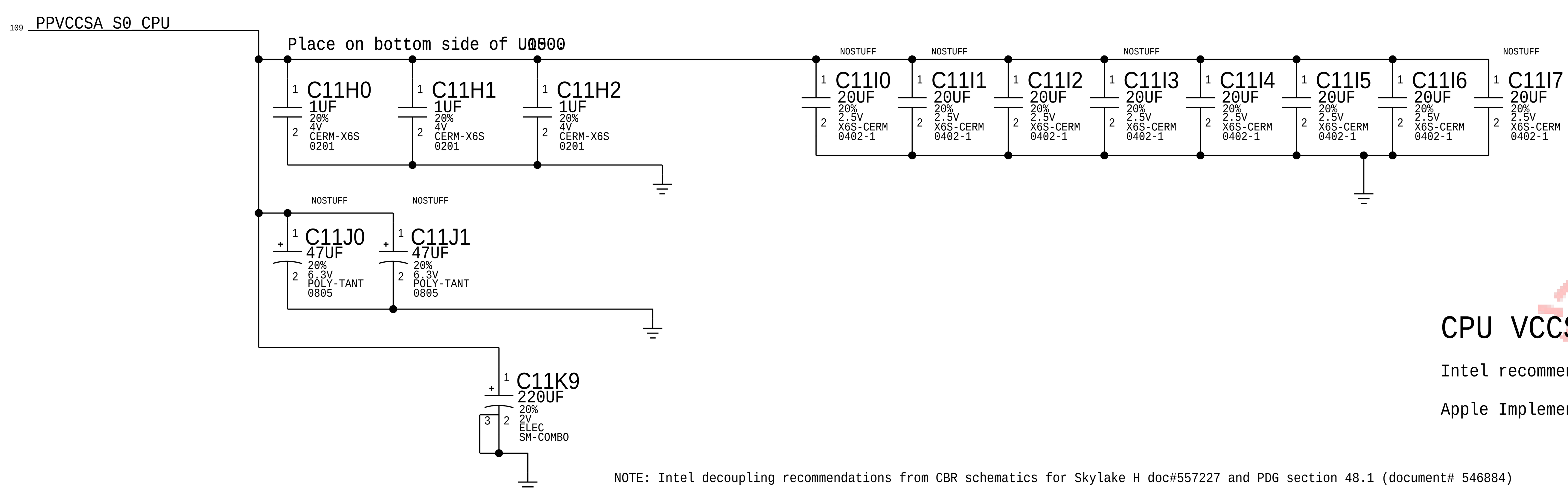
Board Edge: 4x220uF, 7x 47uF rest on back side

CPU VCCSTG Decoupling



CPU VCCPLL and VCCST Decoupling





CPU VCCSA Decoupling

Intel recommendation: 2x 220uF, 1x 47uF 0805. 1x 22uF. 7x 10uF 0402, 3x 1uF 0201_
2x 220uF, 1x 22uF on board edge, everything else on back side

Apple Implementation:

NOTE: Intel decoupling recommendations from CBR schematics for Skylake H doc#557227 and PDG section 48.1 (document# 546884)

BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE: CPU Decoupling 2 [11]

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PAGE TITLE
PCH RTC/HDA/JTAG/SATA/CLK

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BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE
PCH DMI/FDI/PM/GFX/PCI

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BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE

PCH PCI-E/USB

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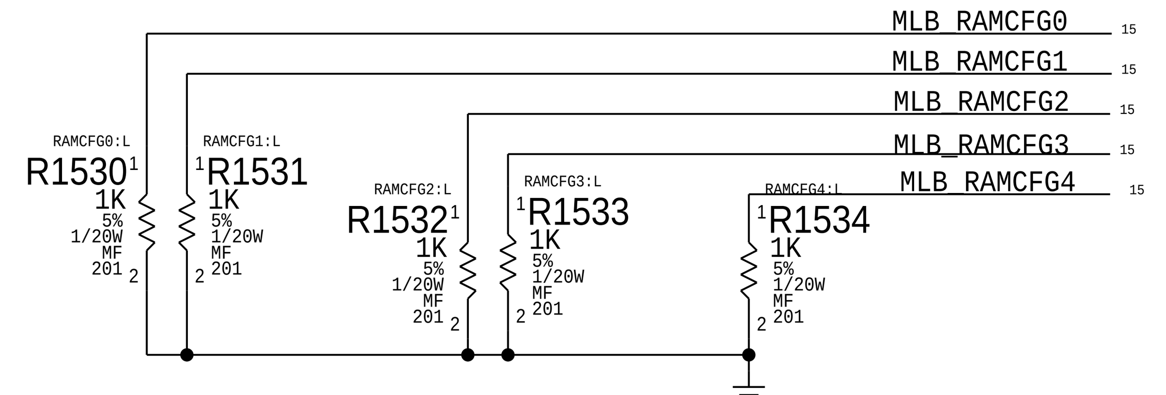
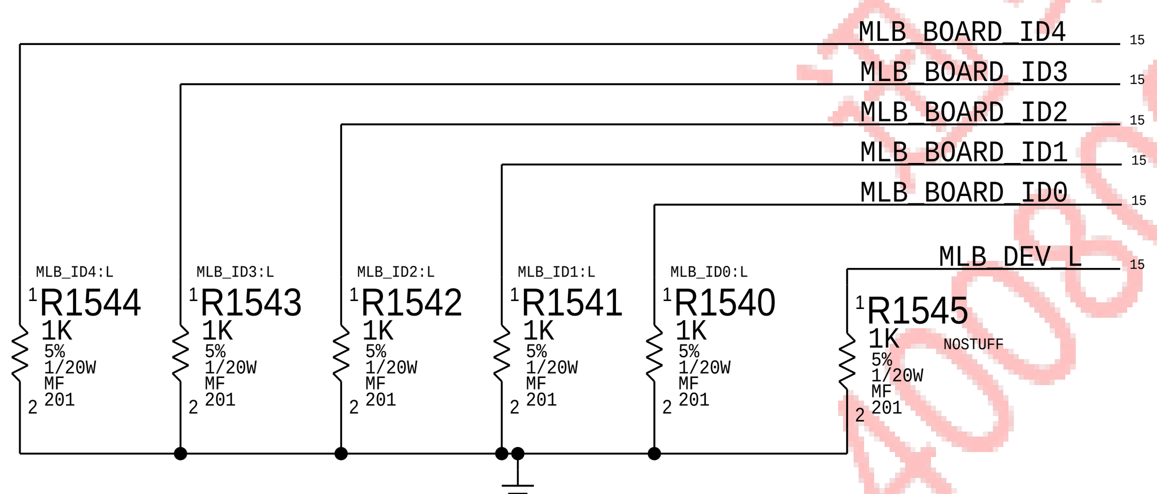


RAM Configuration Straps

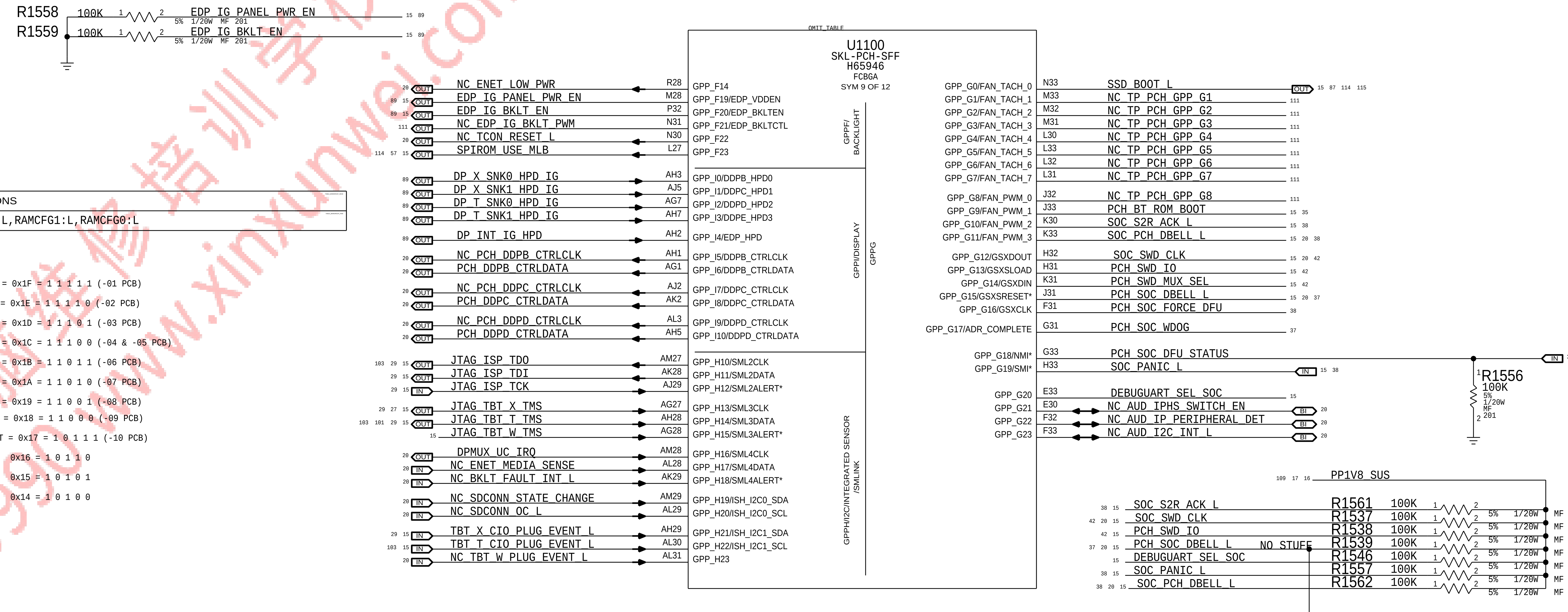
BOM GROUP	BOM OPTIONS
RAMCFG_SLOT	RAMCFG4:L,RAMCFG3:L,RAMCFG2:L,RAMCFG1:L,RAMCFG0:L

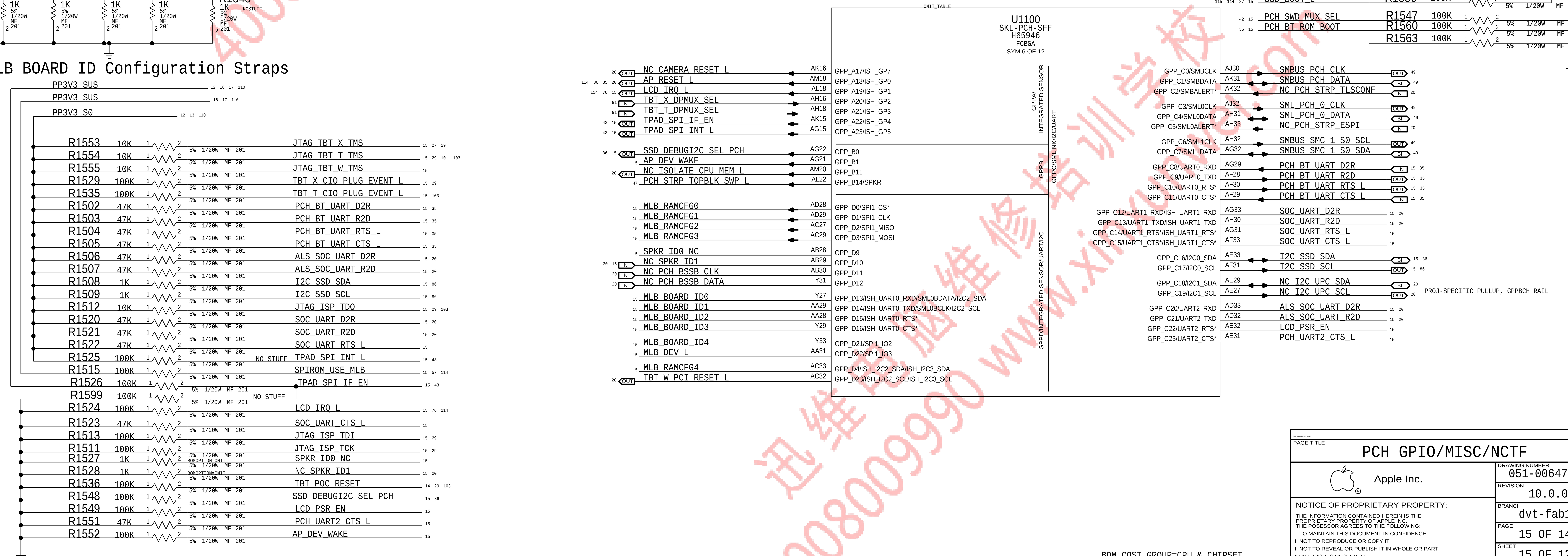
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
117S0006	0	RES,MF,1/20W,1KOHM,5%,0201,SMD		BOARD_ID:1F
117S0006	1	RES,MF,1/20W,1KOHM,5%,0201,SMD	R1540	BOARD_ID:1E
117S0006	1	RES,MF,1/20W,1KOHM,5%,0201,SMD	R1541	BOARD_ID:1D
117S0006	2	RES,MF,1/20W,1KOHM,5%,0201,SMD	R1541,R1540	BOARD_ID:1C
117S0006	1	RES,MF,1/20W,1KOHM,5%,0201,SMD	R1542	BOARD_ID:1B
117S0006	2	RES,MF,1/20W,1KOHM,5%,0201,SMD	R1542,R1540	BOARD_ID:1A
117S0006	2	RES,MF,1/20W,1KOHM,5%,0201,SMD	R1542,R1541	BOARD_ID:19
117S0006	3	RES,MF,1/20W,1KOHM,5%,0201,SMD	R1542,R1541,R1540	BOARD_ID:18
117S0006	1	RES,MF,1/20W,1KOHM,5%,0201,SMD	R1543	BOARD_ID:17
117S0006	2	RES,MF,1/20W,1KOHM,5%,0201,SMD	R1543,R1540	BOARD_ID:16
117S0006	2	RES,MF,1/20W,1KOHM,5%,0201,SMD	R1543,R1541	BOARD_ID:15
117S0006	3	RES,MF,1/20W,1KOHM,5%,0201,SMD	R1543,R1541,R1540	BOARD_ID:14

PROTO 0 = 0x1F = 1 1 1 1 1 (-01 PCB)
PROTO 0B = 0x1E = 1 1 1 1 0 (-02 PCB)
PROTO 1 = 0x1D = 1 1 1 0 1 (-03 PCB)
PROTO 2 = 0x1C = 1 1 1 0 0 (-04 & -05 PCB)
EVT1 = 0x1B = 1 1 0 1 1 (-06 PCB)
EVT2 = 0x1A = 1 1 0 1 0 (-07 PCB)
DVT = 0x19 = 1 1 0 0 1 (-08 PCB)
DVT1-1 = 0x18 = 1 1 0 0 0 (-09 PCB)
PVT = 0x17 = 1 0 1 1 1 (-10 PCB)
0x16 = 1 0 1 1 0
0x15 = 1 0 1 0 1
0x14 = 1 0 1 0 0



MLB BOARD ID Configuration Straps





BOM_COST_GROUP=CPU & CHIPSET

SYNC_MASTER=X363_SAKKOC SYNC_DATE=04/29/2016

PAGE TITLE
PCH GPIO/MISC/NCTF

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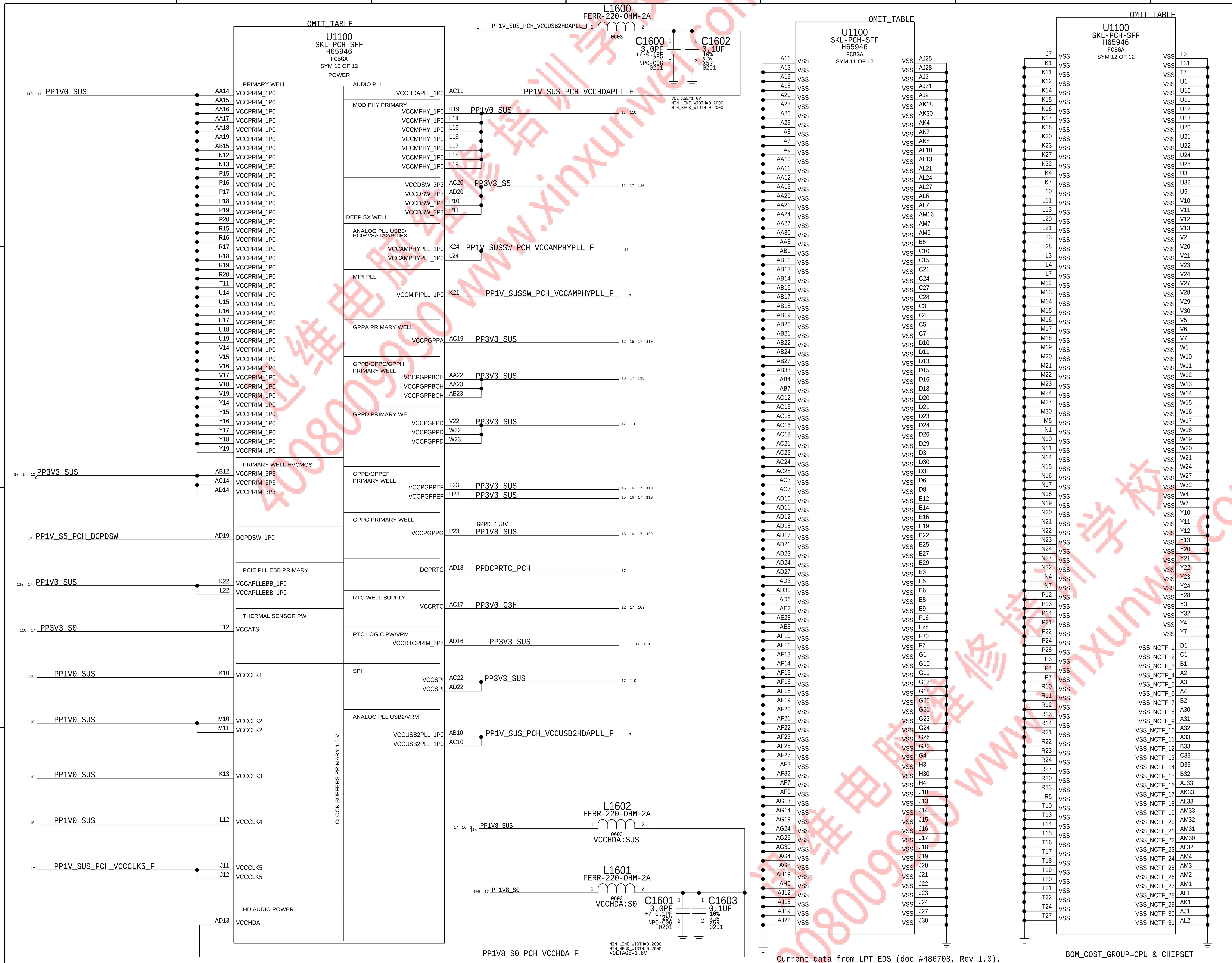

Current data from LPT EDS (doc #486708, Rev 1.0).

BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE

PCH Power

Apple Inc.

DRAWING NUMBER 051-00647 SIZE D

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SHEET 16 OF 121

1000PF CAPS ARE INTEL PLACEHOLDERS & ALL PLACENEAR NEED TO BE UPDATED

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
113S0022	3	RES,MF,1A MAX,0OHM,5%,0603	L1700,L1701,L1702		

PAGE TITLE
PCH DECOUPLING

Apple Inc.

DRAWING NUMBER 051-00647 SIZE D

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BRANCH dvt-fab10

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BOM_COST_GROUP=CPU & CHIPSET

Current data from LPT EDS (doc #486708, Rev 1.0).

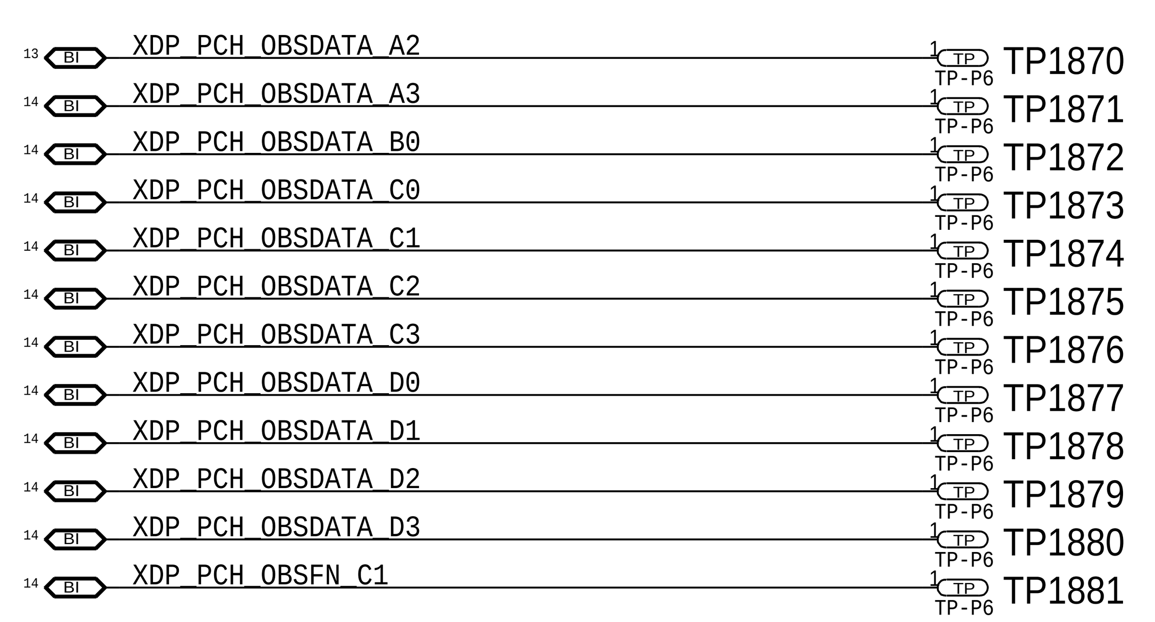
Extra BPM Testpoints

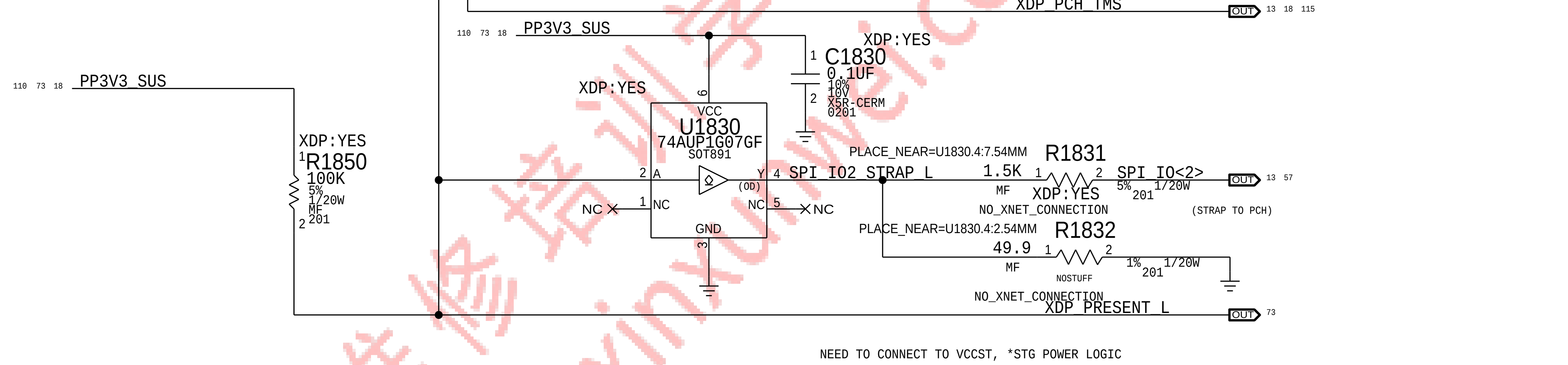
PCH XDP Signals

These signals do not connect to the Primary (Merged) XDP connector in this architecture.
The PDG puts them on a secondary XDP connector that is only needed in some PCH debugging situation.
They are listed here to show their secondary XDP functions and to provide test points for signals that are not used elsewhere.

PCH/XDP Signals

Non-XDP Signals





Unused GPIOs have TPs.

USB Overcurrents are aliased, do not cause USB OC# events during PCH debug.

JTAG_ISP (non-TMS) nets are aliased, do not attempt bit-banged JTAG during PCH debug.

BOM_COST_GROUP=DEBUG

DESIGN: X502/MLB

LAST CHANGE: Mon Jun 15 22:04:28 2015

PAGE TITLE

CPU/PCH Merged XDP

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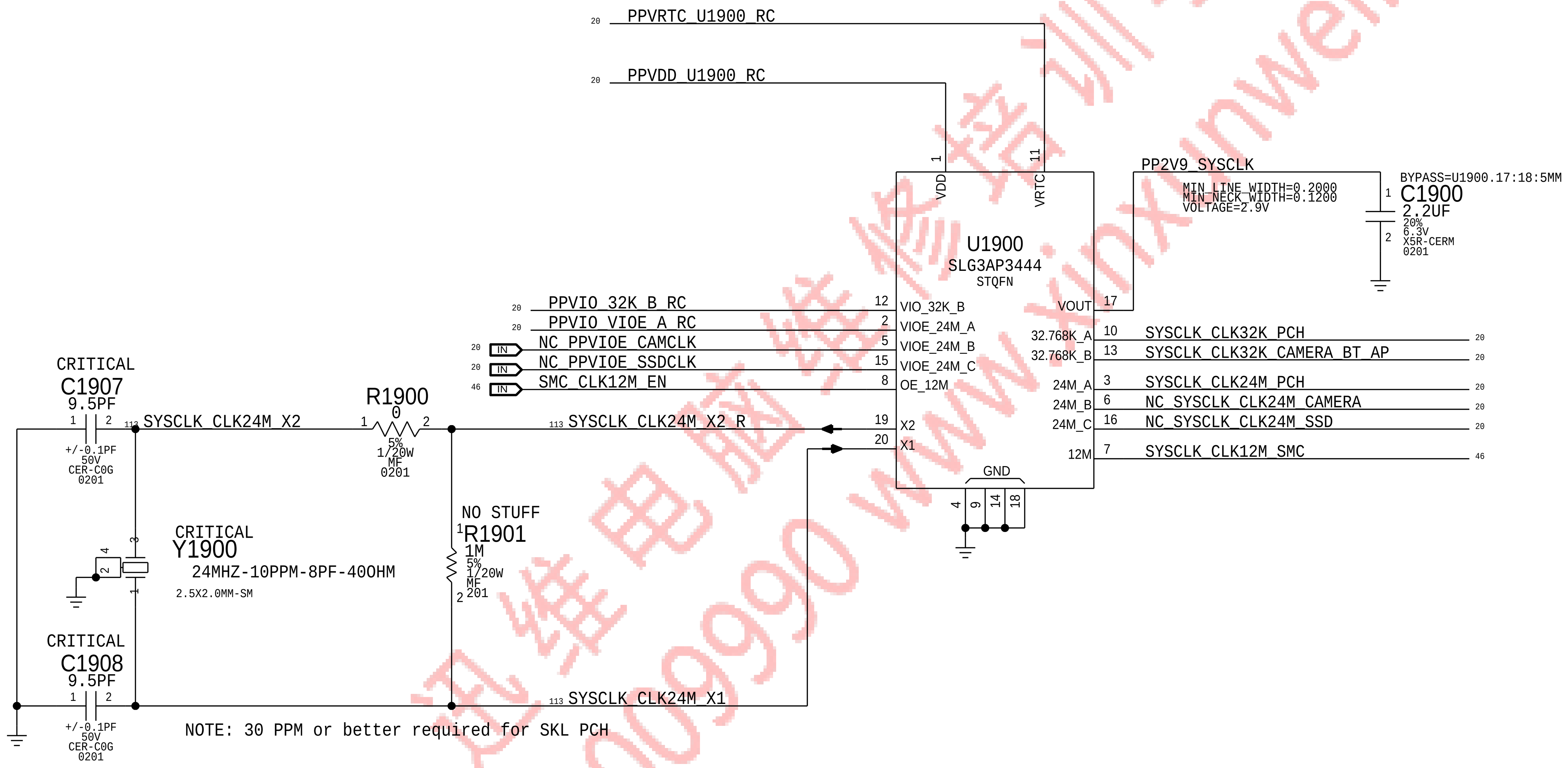
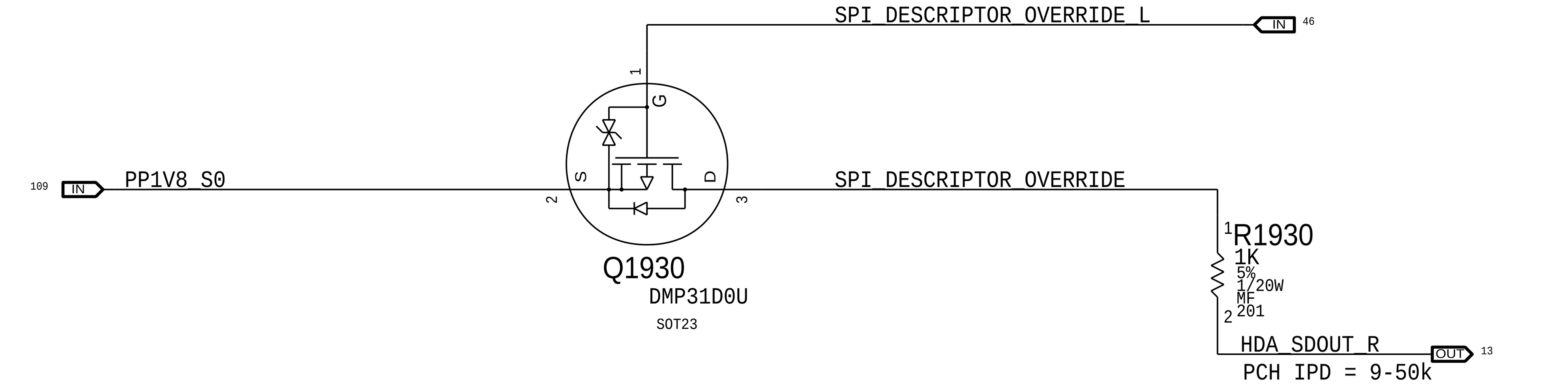
System 32kHz / 12MHz / 24MHz Clock Generator



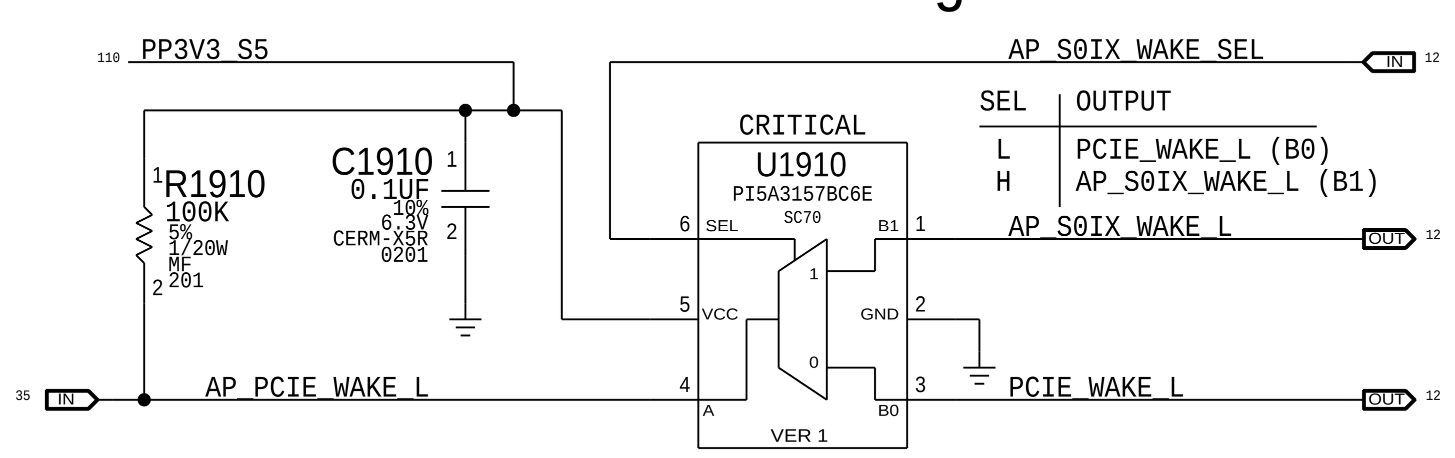
NOTE: 30 PPM or better required for SKL PCH

PCH ME Disable Strap



PCH uses HDA_SDO as a power-up strap. If low, ME functions normally.
If high, ME is disabled. This allows for full re-flashing of SPI ROM.
SMC controls strap enable to allow in-field control of strap setting.
***** Circuit does not support HDA voltage >3.3V.

PCIe Wake Muxing



SEL	OUTPUT
L	PCIE_WAKE_L (B0)
H	AP_S0IX_WAKE_L (B1)

BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE	Chipset Support 1
Apple Inc.	DRAWING NUMBER 051-00647 SIZE D
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PAGE TITLE
Chipset Support 2

Apple Inc.

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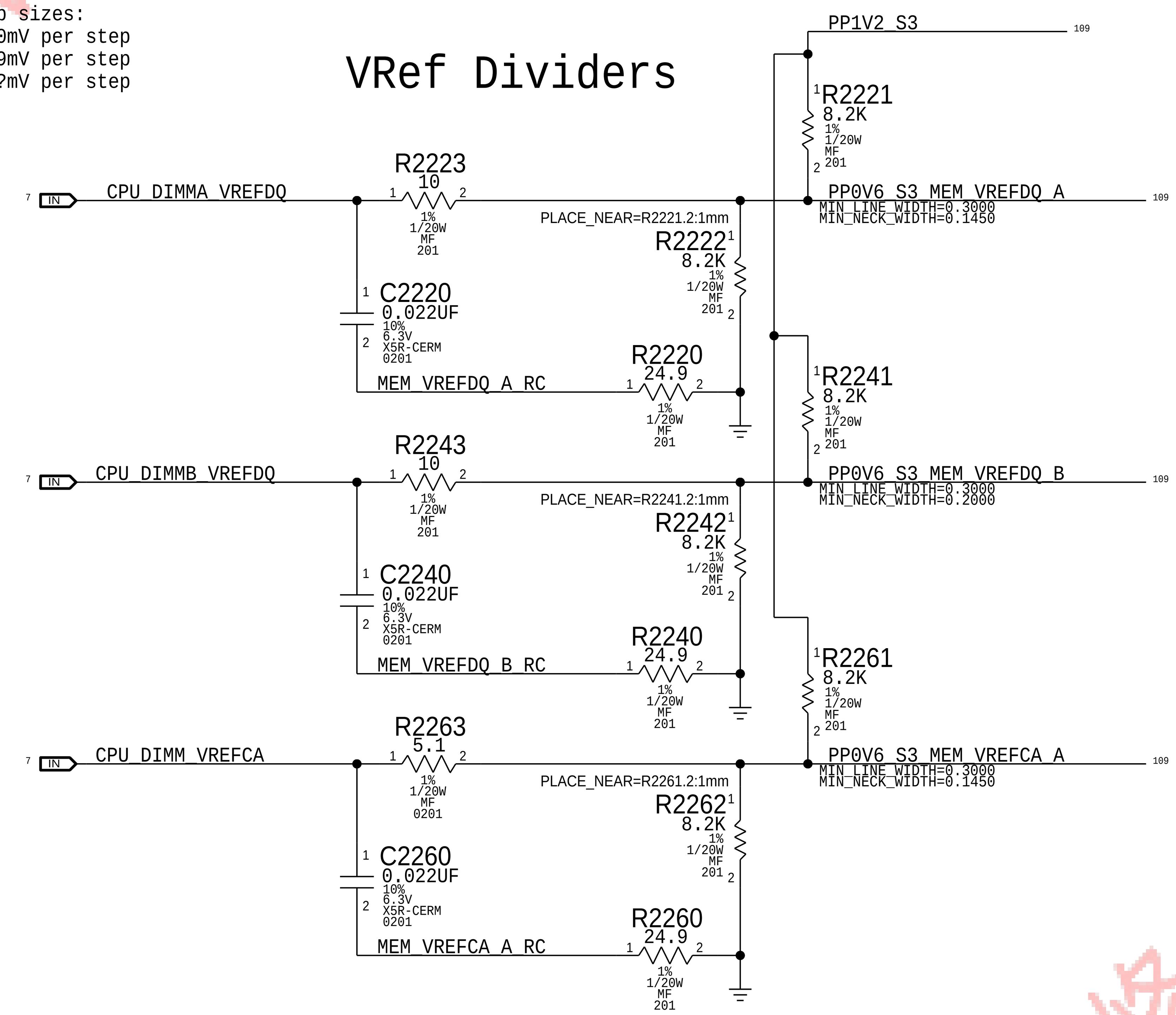
BRANCH dvt-fab10

CPU-Based Margining

NOTE: CPU DAC output step sizes:
DDR3 (1.5V) 7.70mV per step
DDR3L (1.35V) 6.99mV per step
LPDDR3 (1.2V) ?.??mV per step

VRef Dividers



BOM_COST_GROUP=DRAM

SYNC_MASTER=J80_MLB SYNC_DATE=11/06/2015

PAGE TITLE: LPDDR3 VREF MARGINING

Apple Inc.

DRAWING NUMBER: 051-00647 SIZE: D

REVISION: 10.0.0

BRANCH: dvt-fab10

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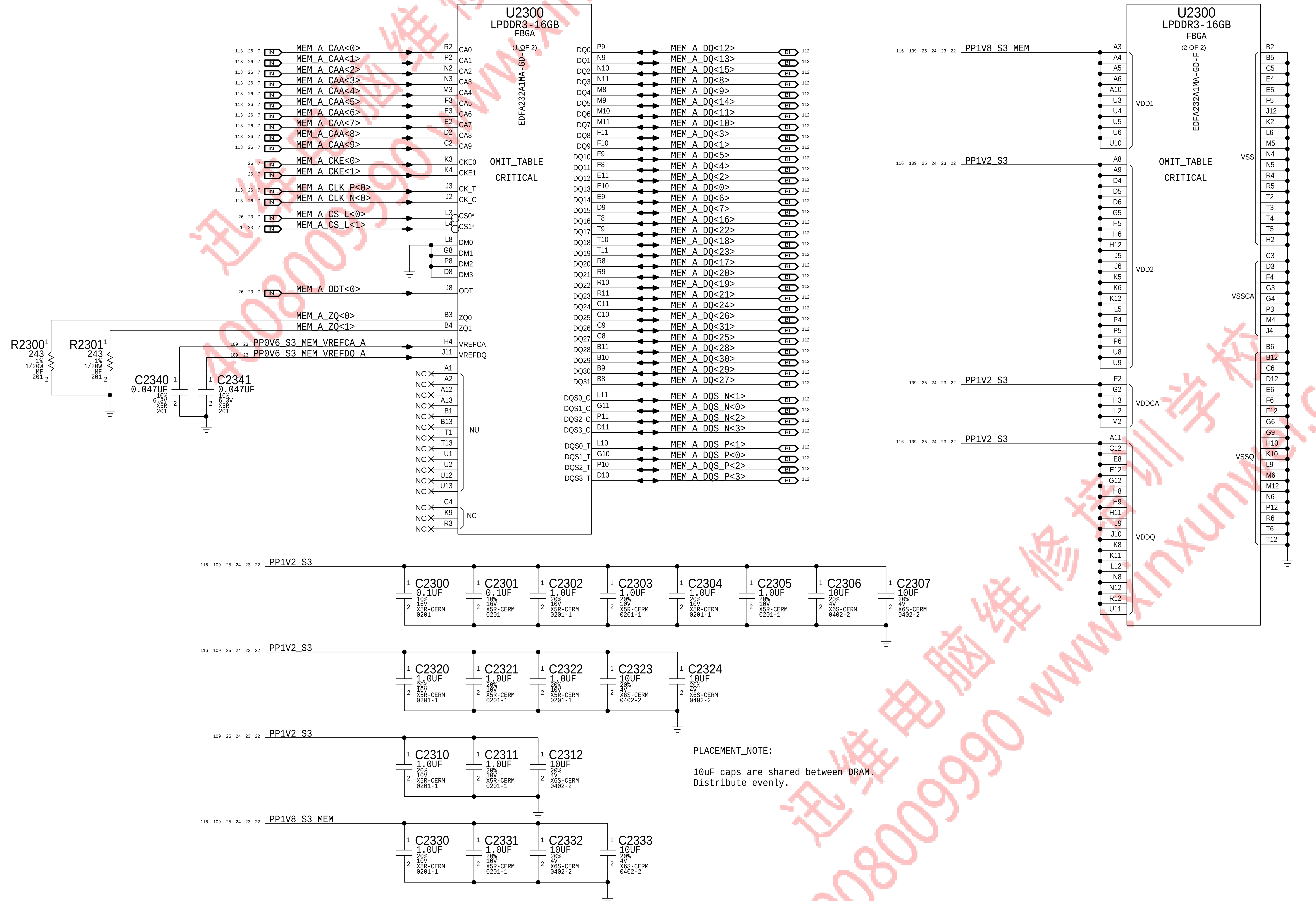
LPDDR3 CHANNEL A (0-31)



PLACEMENT_NOTE:

10uF caps are shared between DRAM.
Distribute evenly.

BOM_COST_GROUP=DRAM

SYNC_MASTER=J80_MLB SYNC_DATE=11/06/2015

PAGE TITLE: LPDDR3 DRAM Channel A (0-31)

Apple Inc.

DRAWING NUMBER: 051-00647 SIZE: D

REVISION: 10.0.0

BRANCH: dvt-fab10

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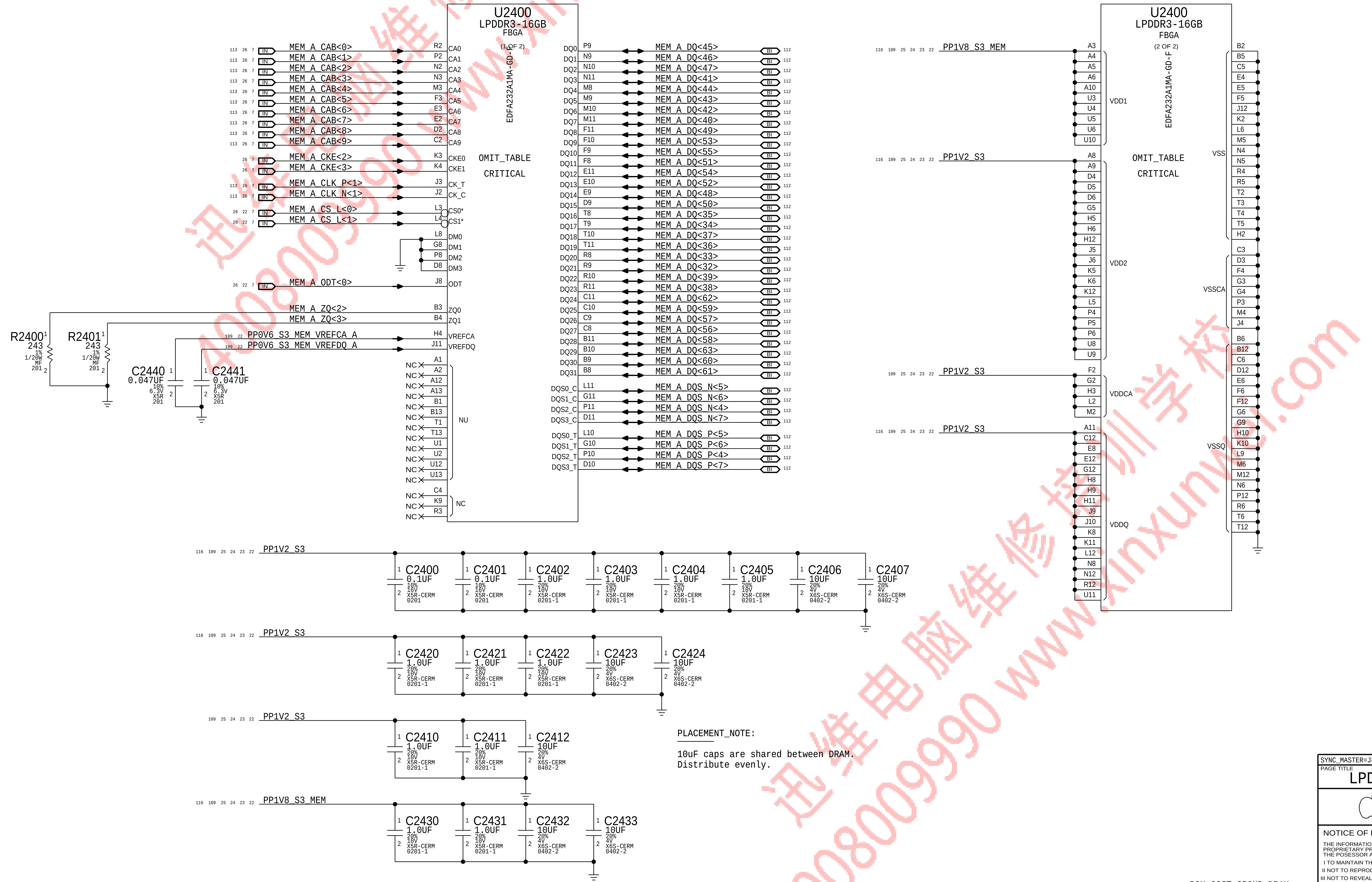
LPDDR3 CHANNEL A (32-63)



PLACEMENT_NOTE:

10uF caps are shared between DRAM.
Distribute evenly.

BOM_COST_GROUP=DRAM

SYNC_MASTER=J80_MLB SYNC_DATE=11/06/2015

PAGE TITLE: LPDDR3 DRAM Channel A (32-63)

Apple Inc.	DRAWING NUMBER 051-00647 SIZE D
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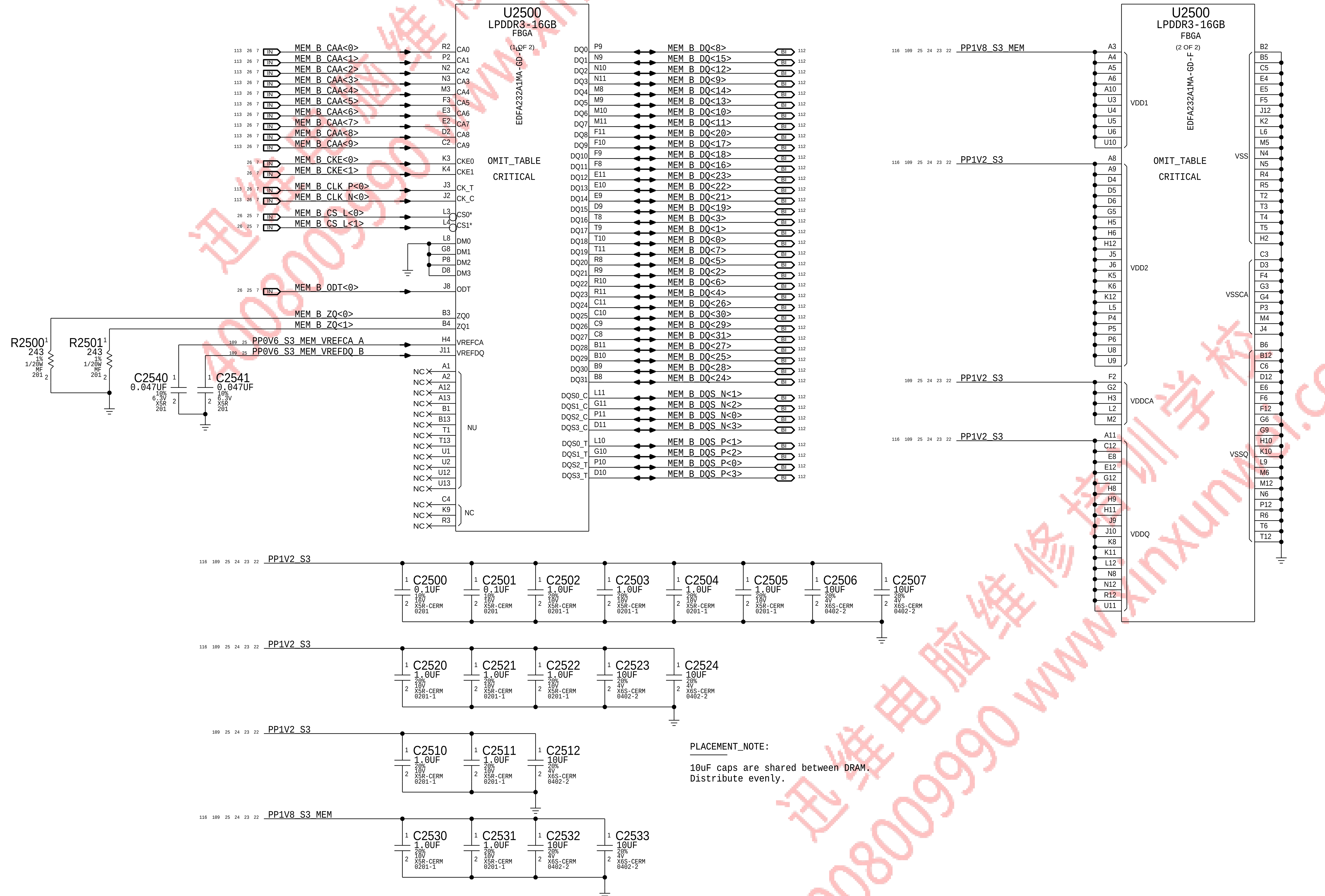
	BRANCH dvt-fab10
	PAGE 24 OF 145
	SHEET 23 OF 121

LPDDR3 CHANNEL B (0-31)



PLACEMENT_NOTE:

10uF caps are shared between DRAM.
Distribute evenly.

BOM_COST_GROUP=DRAM

SYNC_MASTER=J80_MLB SYNC_DATE=11/06/2015

PAGE TITLE: LPDDR3 DRAM Channel B (0-31)

Apple Inc.

DRAWING NUMBER: 051-00647 SIZE: D

REVISION: 10.0.0

BRANCH: dvt-fab10

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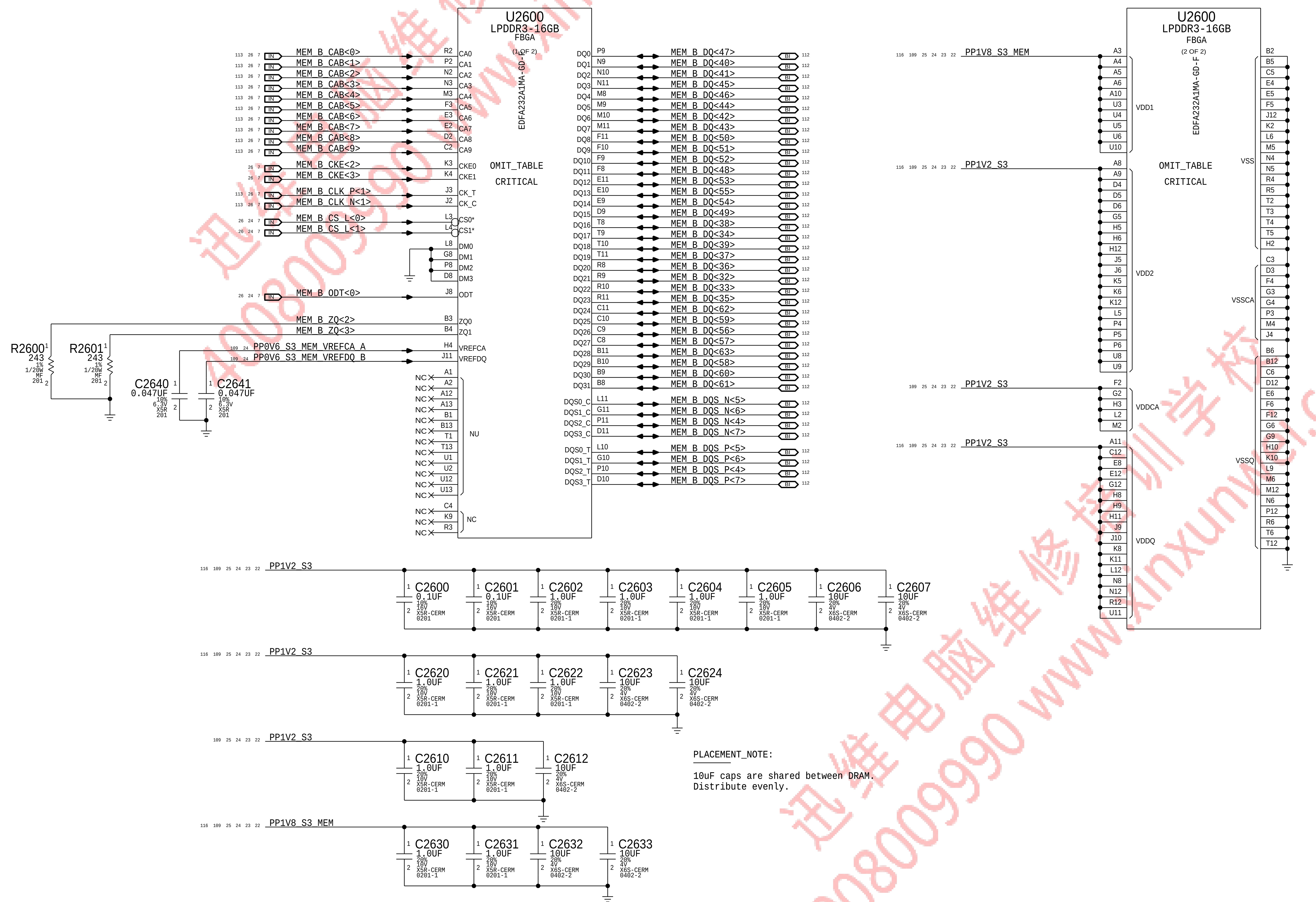
LPDDR3 CHANNEL B (32-63)



PLACEMENT_NOTE:

10uF caps are shared between DRAM.
Distribute evenly.

BOM_COST_GROUP=DRAM

SYNC_MASTER=J80_MLB SYNC_DATE=11/06/2015

PAGE TITLE: LPDDR3 DRAM Channel B (32-63)

Apple Inc.

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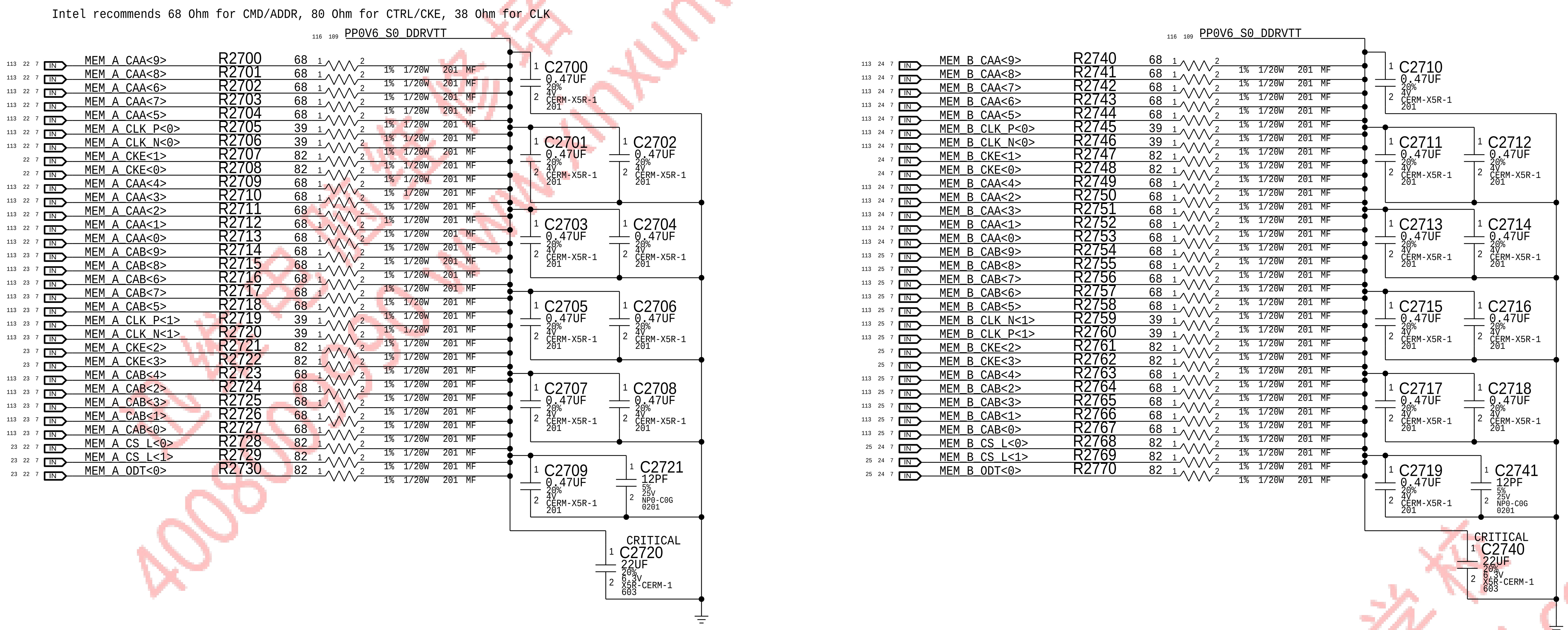

BOM_COST_GROUP=DRAM

SYNC_MASTER=J80_MLB SYNC_DATE=11/06/2015

PAGE TITLE

LPDDR3 DRAM Termination

Apple Inc.

DRAWING NUMBER 051-00647 SIZE D

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DRAWING LAST_MODIFIED=Wed Aug 24 09:57:49 2016

SYNC_MASTER=J80_MLB SYNC_DATE=11/06/2015

PAGE TITLE: USB-C HIGH SPEED 1

Apple Inc.

DRAWING NUMBER: 051-00647 SIZE: D

REVISION: 10.0.0

BRANCH: dvt-fab10

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USB-C HIGH SPEED 2

Apple Inc.

DRAWING NUMBER 051-00647 SIZE D

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BOM_COST_GROUP=TBT

PAGE TITLE
USB-C Support

Apple Inc.

DRAWING NUMBER 051-00647 SIZE D

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PRIMARY ACE USB-C PORT CONTROLLER (UPC)

BOM_COST_GROUP=USB-C

PAGE TITLE	USB-C PORT CONTROLLER A
DRAWING NUMBER	051-00647
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Apple Inc.

SECONDARY ACE USB-C PORT CONTROLLER (UPC)

BOM_COST_GROUP=USB-C

PAGE TITLE
USB-C PORT CONTROLLER B

Apple Inc.

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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0683	2	CAP,CER,X5R,1UF,10%,25V,0402	C3304, C3354	CRITICAL	NOSTUFF

BOM_COST_GROUP=USB-C

LAST CHANGE: Wed Apr 1 22:57:37 2015

PAGE TITLE

USB-C CONNECTOR A

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BRANCH dvt-fab10

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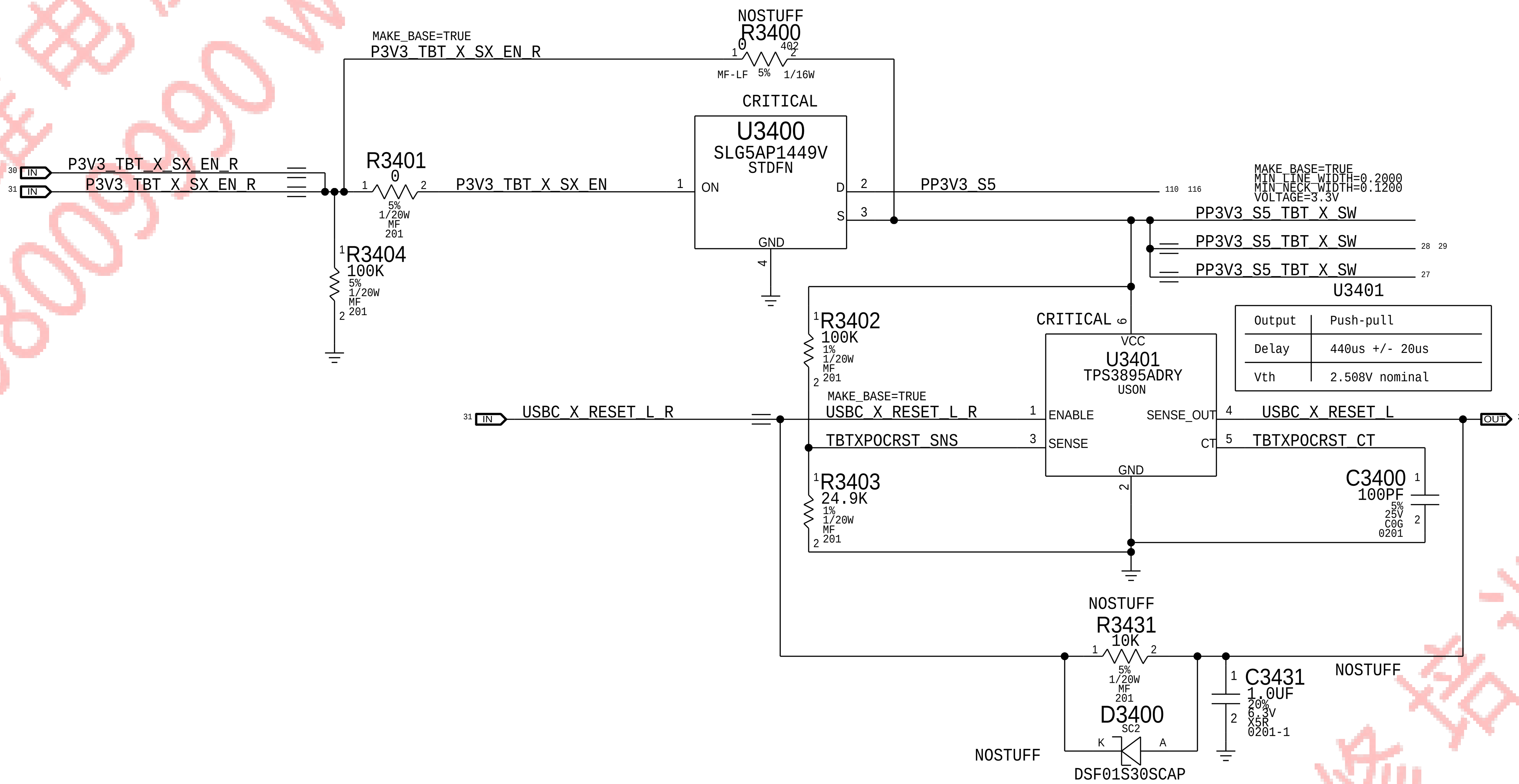
TBT X "POC" Power-up Reset



BOM_COST_GROUP=USB-C

DESIGN: X502/DEV_MLB_U

LAST CHANGE: Wed Feb 18 17:12:24 2015

PAGE TITLE

USB-C CONNECTOR B

DRAWING NUMBER 051-00647 SIZE D

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Vout = 5.23V
Freq = 500 kHz
Max OCP = 13.9A
Nom OCP = 11.6A
Min OCP = 9.37A

BOM_COST_GROUP=USB-C

PAGE TITLE
TBT 5V REGULATOR

Apple Inc.

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PAGE TITLE
WIFI/BT: MODULE 1

Apple Inc.

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BOM_COST_GROUP=WIRELESS

SYNC_MASTER=J80_MLB SYNC_DATE=11/06/2015

PAGE TITLE
WIFI/BT: MODULE 2

Apple Inc.

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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
343S00136	1	IC,M8+512MB 20NM DDR,A12,S,SCK,BGA700	U3900	CRITICAL	SOC:HYNIX

T208

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
338S00147	1	IC,RTM2,DEV,PN549A1,P61D0	U3905	CRITICAL	SE:DEV
338S00097	1	IC,RTM2,MP,PN549A1,P61D0	U3905	CRITICAL	SE:PROD

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
335S00203	1	IC,FLASH,SERIAL,SPI,4MX8,43MM,DFN8	U3906		

PAGE TITLE
Camera/DFR 1

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U3900
M8-LPDDR4-H-A-FUSE
UFBGA
(3 OF 7)
OMIT_TABLE
U3900
M8-LPDDR4-H-A-FUSE
UFBGA
(4 OF 7)
OMIT_TABLE
To/From PCH for logging/debug
To/From PCH for ALS
PLACE_NEAR=U3900.H4:2mm
PLACE_NEAR=U3900.K1:2mm
PLACE_NEAR=U3900.K2:3mm
L4010
120-OHM-0.2A-0.5-OHM
MIPIC FILTERING
L4002
L4003
L4004
FERR-120-OHM-1.5A
Y4000
24.000MHZ-30PPM-9.5PF-60OHM
R4010
499K
T208 SOC_S2R_ACK_L bi-directional Isolation
T208 SOC_S2R_ACK_L Isolation (NOSTUFF)
T208 FORCE_DFU Isolation
Q4000
DMN32D2LFB4
U4080
74LVC1G08FW5
U4070
74LVC1G08FW5
Q4000 to act as bi-directional islation for SOC_S2R_ACK_L
When in system S0 and M8 AWAKE, FET will be on & line will be pulled high
Below S0, SOC_S2R_ACK_L will be low and FET will be open (isolated from M8 power rails)
When M8 enters S2R, FET will be on, PP1V8_AWAKE_SW3C will turn off and SOC_S2R_ACK_L will be low
SOC_FORCE_DFU should be pulled up to S4
In the SOC_BOOT:DFU option
U4070 prevents it from leaking into PCH
BOM_COST_GROUP=T151
PAGE TITLE
Camera/DFR 2
Apple Inc.
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T208 BOARD REV/BOARD ID

S/W READ FLOW

1. SET GPIO AS INPUT
2. DISABLE PU AND ENABLE PD
3. READ
4. DISABLE PD AND ENABLE PU

BOARD REV[3:0]	MODE
1111	T208_Rev0 (X362 P0)
1110	T208_Rev1 (X363 P0)
1101	T208_Rev2 (X362 P1)
1100	T208_Rev3 (X362 P2)
1011	T208_Rev4 (X363 P2/X362 localEVT)
1010	T208_Rev5 (X363 EVT/X362 EVT1)

BOARD_ID[3:0]	MODE
1XXX	T202 DEV
0010	T208

BOM GROUP	BOM OPTIONS
T208_PROG:REV0	T208_CONFIG2_H,T208_CONFIG1_H,T208_CONFIG0_H
T208_PROG:REV1	T208_CONFIG2_H,T208_CONFIG1_H
T208_PROG:REV2	T208_CONFIG2_H,T208_CONFIG0_H
T208_PROG:REV3	T208_CONFIG2_H
T208_PROG:REV4	T208_CONFIG1_H, T208_CONFIG0_H
T208_PROG:REV5	T208_CONFIG1_H

BOM_COST_GROUP=T151

PAGE TITLE
Camera/DFR 3

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BOM_COST_GROUP=T151

PAGE TITLE
Berkelium - 1

Apple Inc.

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Berkelium - 2

BOM_COST_GROUP=T151

PAGE TITLE
Berkelium - 2

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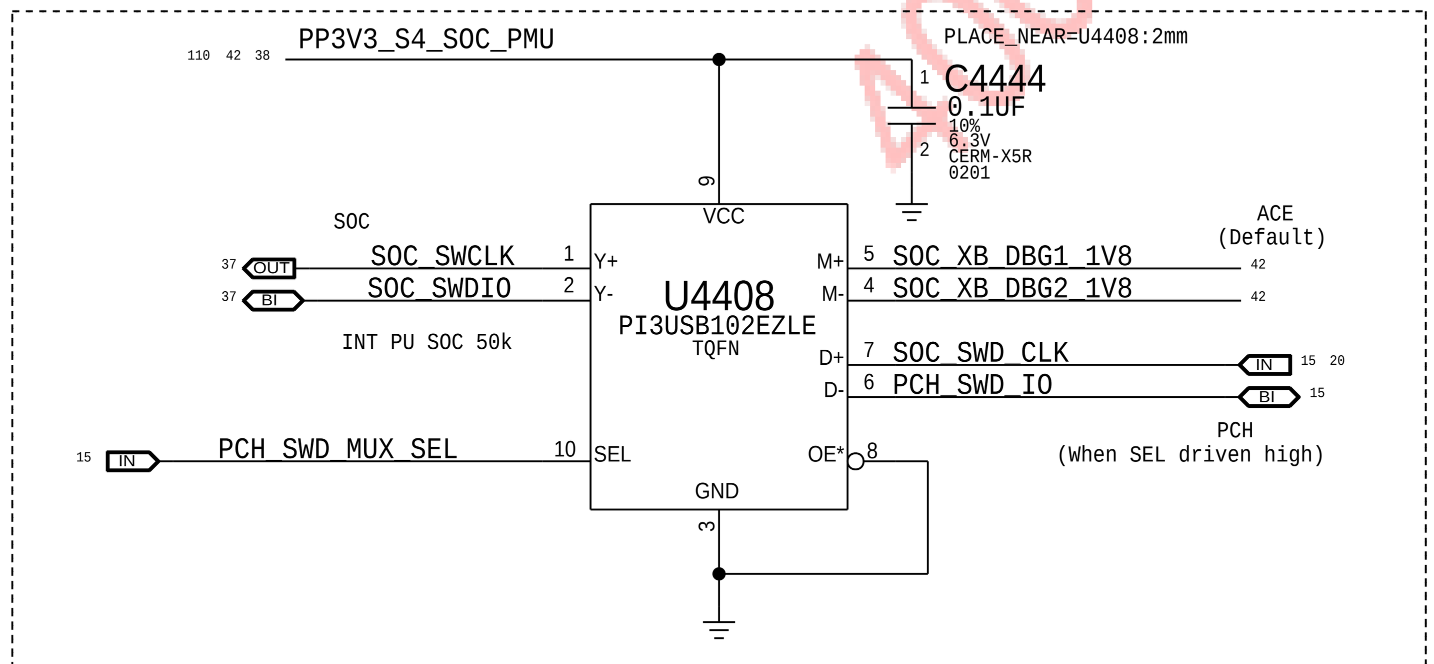
T208 Support

Part	SLG5AP1443V
Type	Load Switch
R(on) @ 3.3V	17 mOhm Typ 19 mOhm Max
Current	2.5A Max

U4405

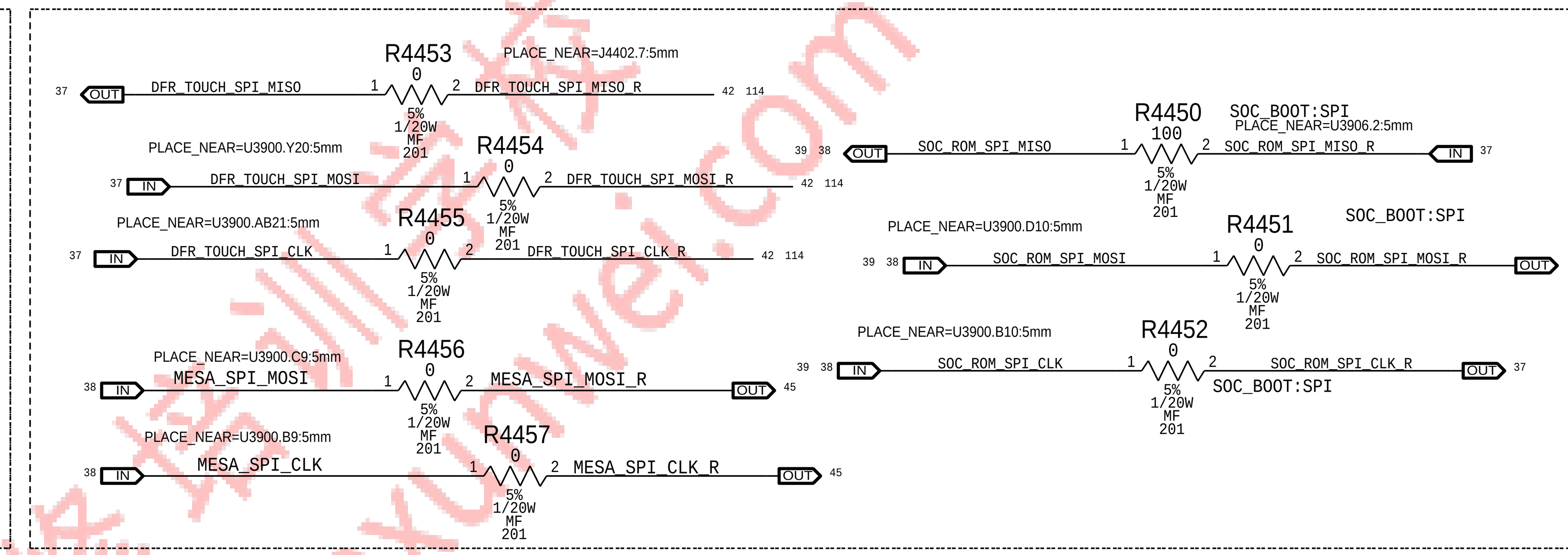
SWD DEBUG MUX



T208 I2C Mapping

Bus	Device	7-bit Address	8-bit Address Read	8-bit Address Write
AP0	PMU	0011110 (0x3C)	0x79	0x78
AP1	Touch EEPROM	1010000 (0x50)	0xA1	0xA0
AP2_0	Tesla	1010100 (0x4C)	0x99	0x98
AOP0	Mesa EEPROM	101000x (0x50/0x51)	0xA1/A3	0xA0/A2
AOP1	ALS	0111001 (0x39)	0x73	0x72
SEP	M34128 EEPROM	1010001 (0x51)	0xA3	0xA2

SPI TERM



T208 LEVEL SHIFTING

BOM_COST_GROUP=T151

PAGE TITLE: T208 Support

Apple Inc.

DRAWING NUMBER: 051-00647 SIZE: D

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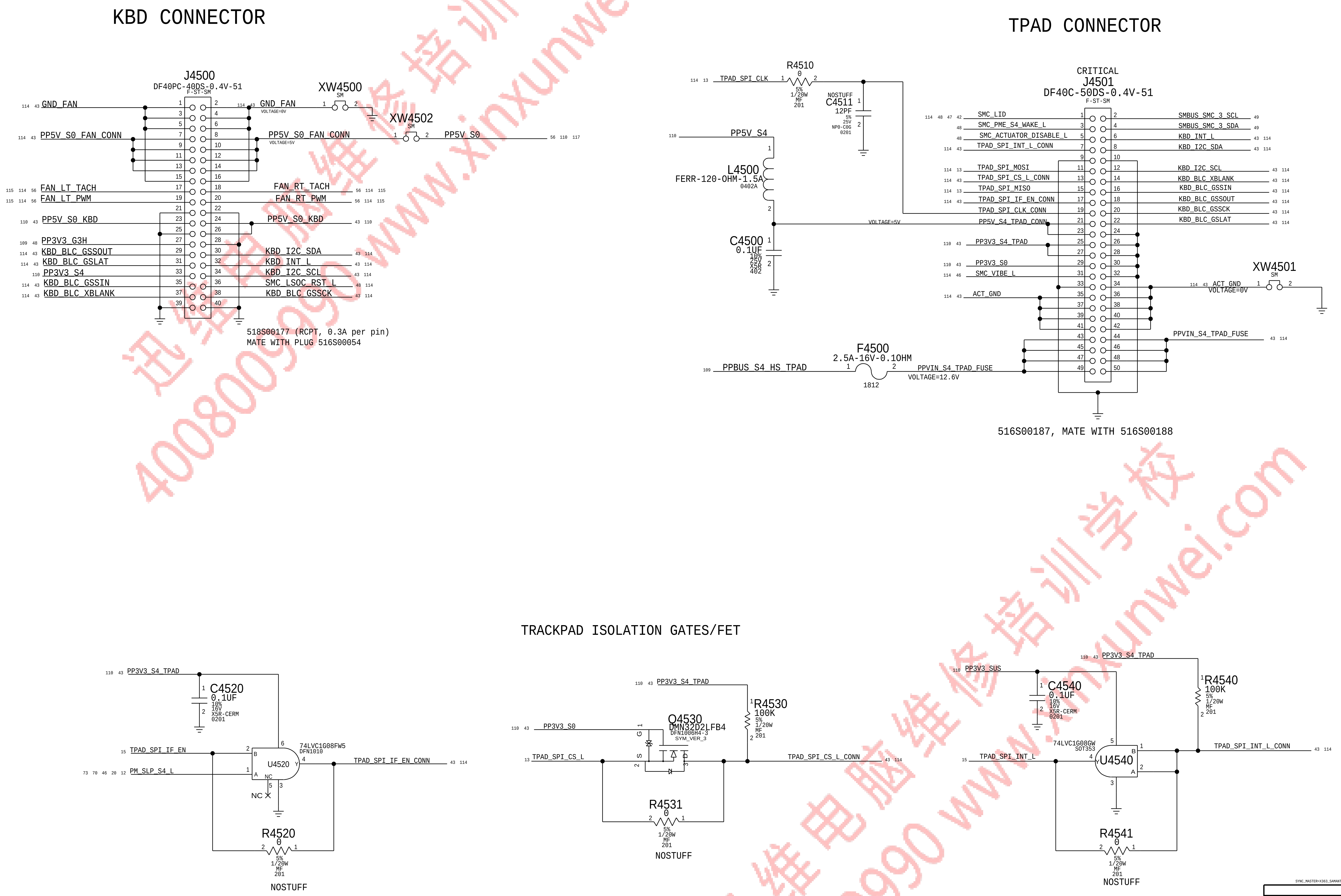

BOM_COST_GROUP=KEYBOARD

PAGE TITLE: Connectors&ESD

Apple Inc.

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Debug Stuff Was Here

BOM_COST_GROUP=DEBUG

SYNC_MASTER=J80_MLB
SYNC_DATE=08/26/2015

PAGE TITLE: External A USB3 Connector	
Apple Inc.	DRAWING NUMBER: 051-00647 — SIZE: D
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	BRANCH: dvt-fab10
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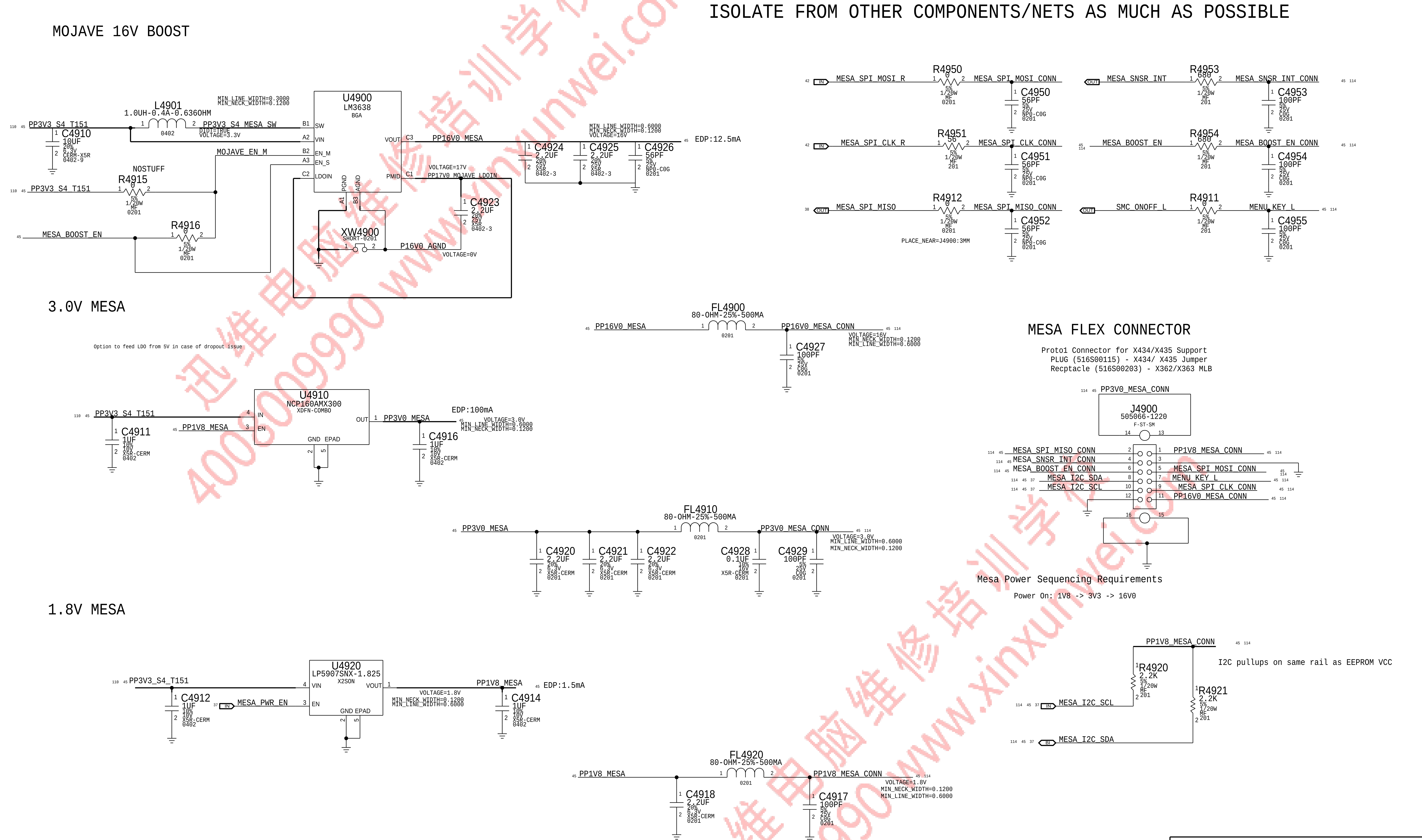

BOM_COST_GROUP=T151

PAGE TITLE
MESA

Apple Inc.

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NOTE:
SMS Interrupt can be active high or low, rename net accordingly.
If SMS interrupt is not used, pull up to SMC rail.

NOTE:
Unused pins have "SMC_Pxx" names. Unused
pins designed as outputs can be left floating,
those designated as inputs require pull-ups.

PAGE TITLE
SMC
Apple Inc.

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PAGE TITLE

SMC Shared Support

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Specify one of these BOM GROUPs.

BOM GROUP	BOM OPTIONS
CPUTHRM:BOTH	CPUTHRM_THRM:SMC,CPUTHRM_ALRT:SMC
CPUTHRM:THRM	CPUTHRM_THRM:SMC
CPUTHRM:ALRT	CPUTHRM_ALRT:SMC

Specify one of these BOM GROUPs.

BOM GROUP	BOM OPTIONS
TBTTHRM:BOTH	TBTTHRM_THRM:SMC,TBTTHRM_ALRT:SMC
TBTTHRM:THRM	TBTTHRM_THRM:SMC,TBTTHRM_ALRT:PU
TBTTHRM:ALRT	TBTTHRM_THRM:PU,TBTTHRM_ALRT:SMC
TBTTHRM:NONE	TBTTHRM_THRM:PU,TBTTHRM_ALRT:PU

PAGE TITLE
SMC Project Support

Apple Inc.

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LYNX POINT LP S0 "SMBus 0" Connections
PP3V3_S0
LYNX POINT LP
U0500
(MASTER)
SMBUS_PCH_CLK
SMBUS_PCH_DATA
MAKE_BASE=TRUE
R5300 1K
R5301 1K
SMC SMBus "0" S0 Connections
PP3V3_S0
SMC
U5000
(MASTER)
SMBUS_SMC_0_S0_SCL
SMBUS_SMC_0_S0_SDA
R5350 1.5K
R5351 1.5K
Internal DP
J8500
16 addresses
(Write: 0x20 Read: 0x21)
thru
(Write: 0x3E Read: 0x3F)
(when VRR_FLAG = 0)
SMC SMBus "5" G3H Connections
PP3V3_G3H
SMC
U5000
(MASTER)
SMBUS_SMC_5_G3_SCL
SMBUS_SMC_5_G3_SDA
R5380 2.0K
R5381 2.0K
Battery Charger
ISL6259 - U7000
(Write: 0x12 Read: 0x13)
Banjo
U7800
(Write: 0x68 Read: 0x69)
Battery
J6950
(Write: 0x16 Read: 0x17)
DPMUX Connections
DPMUX IC
U9800
(MASTER)
NC_I2C_DPMUX_A_SCL
NC_I2C_DPMUX_A_SDA
DPMUX Connections
PP3V3_S0
DPMUX IC
U9800
(MASTER)
I2C_DPMUX_UC_SCL
I2C_DPMUX_UC_SDA
R5305 2.0K
R5306 2.0K
SMC SMBus "2" S4 Connections
PP3V3_S4
SMC
U5000
(MASTER)
SMBUS_SMC_2_S4_SCL
SMBUS_SMC_2_S4_SDA
R5370 1K
R5371 1K
Berkelium
U4200
(Write: 0x78 Read: 0x79)
SMBUS_SOC_PMU_SCL
SMBUS_SOC_PMU_SDA
EADC1
U5700
(Write: 0x10 Read: 0x11)
EADC2
U5710
(Write: 0x14 Read: 0x15)
R5373 0
UGLY HACK
R5372 0
SMC SMBus "3" S0 Connections
PP3V3_S0
SMC
U5000
(MASTER)
SMBUS_SMC_3_SCL
SMBUS_SMC_3_SDA
R5390 2.0K
R5391 2.0K
X100 Temp
TMP105: J9510
(Write: 0x92 Read: 0x93)
Trackpad
J4501
(Write: 0x99 Read: 0x98)
TBT & Airflow Left
TMP461: U5850
(Write: 0x90 Read: 0x91)
CARBON
U6000
(Write: 0xD5 Read: 0xD4)
TBT & Airflow Right
TMP461: U5800
(Write: 0x96 Read: 0x97)
LYNX POINT LP S0 "SMLink 0" Connections
PP3V3_S0
LYNX POINT LP
U0500
(MASTER)
SML_PCH_0_CLK
SML_PCH_0_DATA
R5310 8.2K
R5311 8.2K
SMC SMBus "1" S0 Connections
PP3V3_S0
SMC
U5000
(MASTER)
SMBUS_SMC_1_S0_SCL
SMBUS_SMC_1_S0_SDA
R5360 2.0K
R5361 2.0K
CPU, Mem, Airflow,
Fixstack Prox
TMP513AISAR:U5870
(Write: 0xB8 Read: 0xB9)
GPU THERM
TMP442A: UA960
(Write: 0x98 Read: 0x99)
GPU
UA000
(Write: 0x82 Read: 0x83)
SMC SMBUS "4" G3H CONNECTIONS
PP3V3_G3H
SMC
U5000
(MASTER)
SMBUS_SMC_4_G3H_SCL
SMBUS_SMC_4_G3H_SDA
R5320 1.5K
R5321 1.5K
USB-C PORT CONTROLLER XA
CD3215A (ACE) - U3100
(WRITE: 0x70 READ: 0x71)
USB-C PORT CONTROLLER TA
CD3215A (ACE) - UB300
(WRITE: 0x40 READ: 0x41)
USB-C PORT CONTROLLER XB
CD3215A (ACE) - U3200
(WRITE: 0x7E READ: 0x7F)
USB-C PORT CONTROLLER TB
CD3215A (ACE) - UB400
(WRITE: 0x4E READ: 0x4F)
LYNX POINT LP S0 "SMLink 1" Connections
LYNX POINT LP
U0500
(Write: 0x88 Read: 0x89)
SMBUS_SMC_1_S0_SCL
SMBUS_SMC_1_S0_SDA
SMLink 1 is slave port to
access PCH.
BOM_COST_GROUP=SMC
PAGE TITLE
SMBus Connections
Apple Inc.
DRAWING NUMBER 051-00647
SIZE D
REVISION 10.0.0
BRANCH dvt-fab10
PAGE 53 OF 145
SHEET 49 OF 121
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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	2	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5419,C5449		OTHERRC:NO
117S0008	1	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5459		LOADRC:NO
117S0008	1	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5469		TPADRC:NO

PAGE TITLE
Power Sensors: High Side

Apple Inc.

DRAWING NUMBER 051-00647 SIZE D
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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	6	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5519,C5529,C5539,C5549,C5569,C5599		LOADRC:NO
117S0008	1	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5579		DDRRC:NO

BOM_COST_GROUP=SENSORS

PAGE TITLE: Power Sensors: Load Side

Apple Inc.

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REVISION: 10.0.0

BRANCH: dvt-fab10

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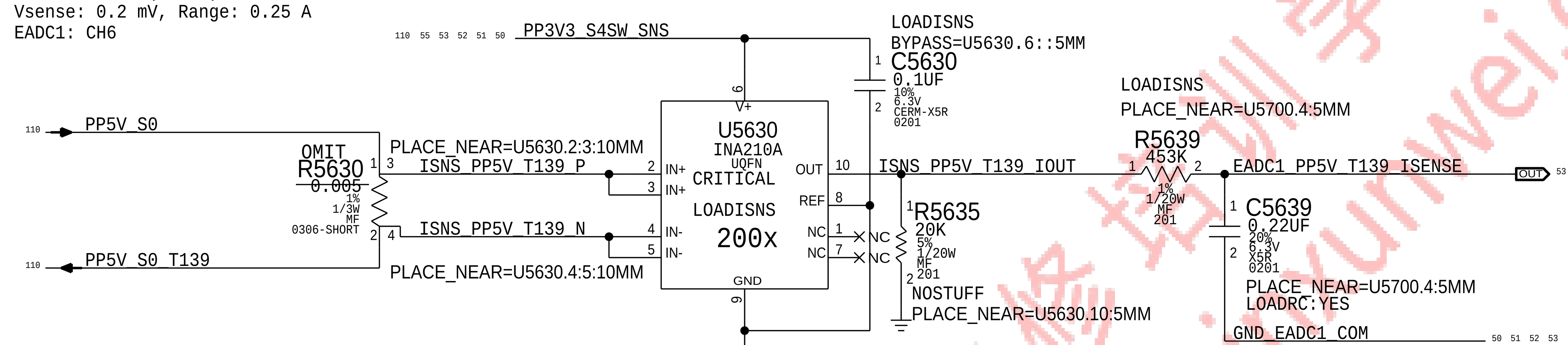
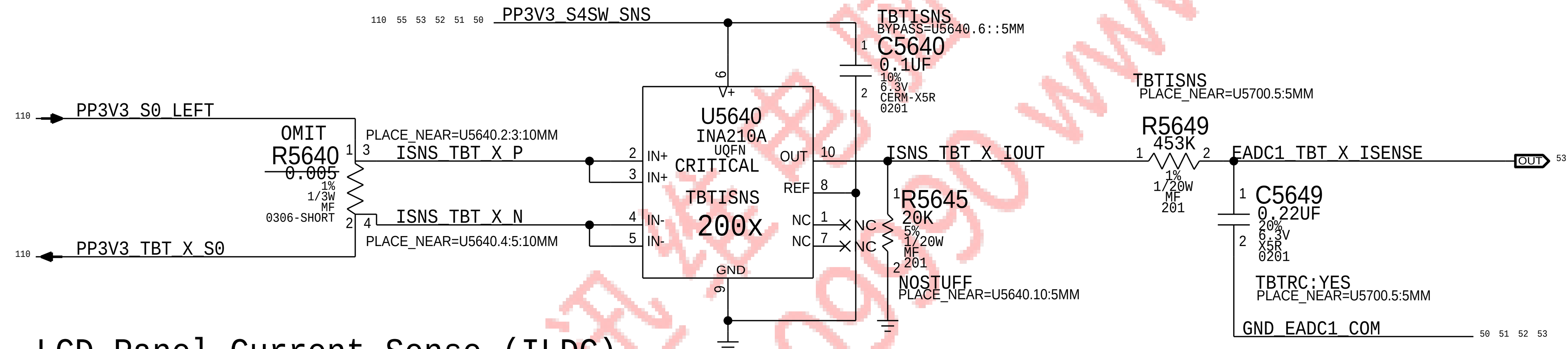
T139 5V Current Sense (IF5C)

Gain: 200x, EDP: 0.004 A
Rsense: 0.05 (R5630) or Rsense SHORT
Vsense: 0.2 mV, Range: 0.25 A
EADC1: CH6



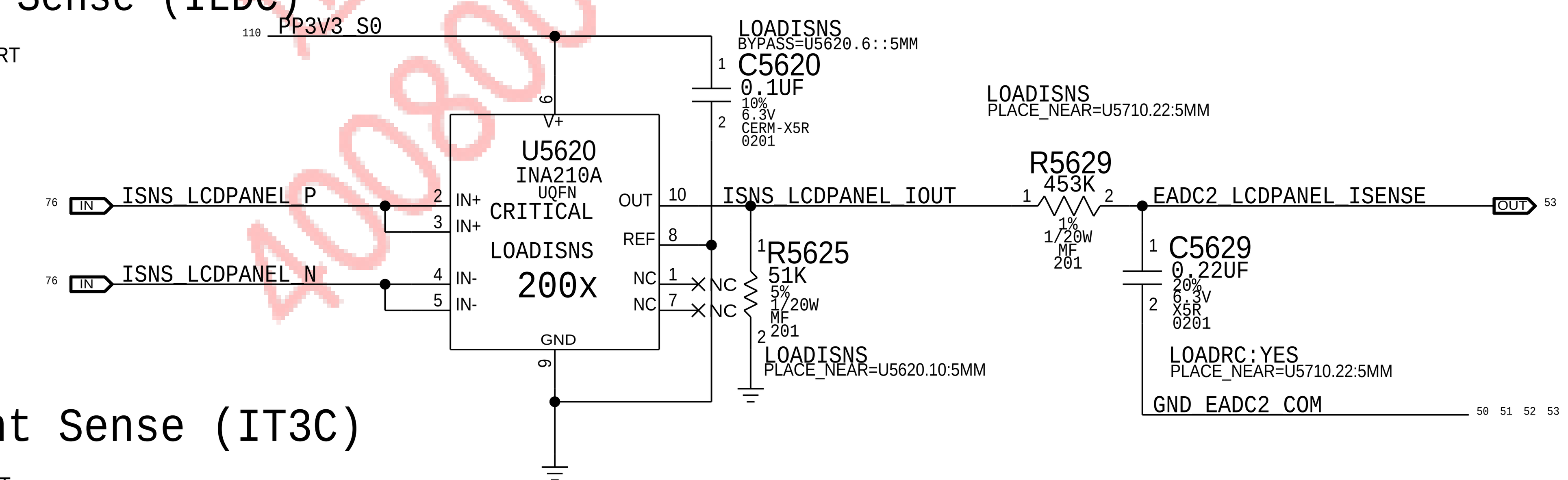
Thunderbolt TBT LEFT Current Sense (IULC)

Gain: 200x. EDP: 0.5 A
Rsense: 0.025 (R5640) or Rsense SHORT
Vsense: 12.5 mV, Range: 0.5 A
EADC1: CH7



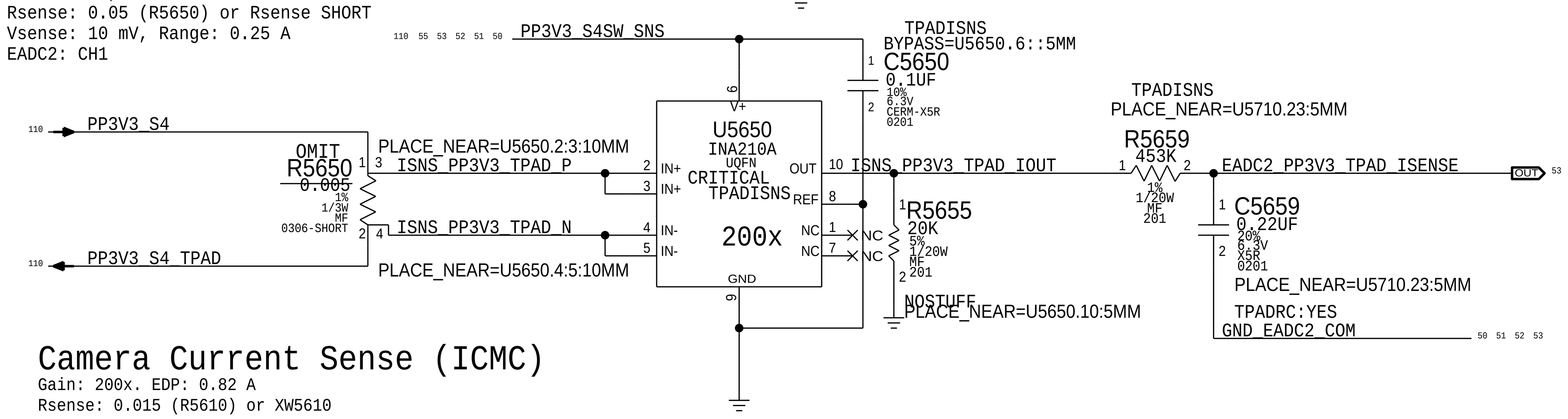
LCD Panel Current Sense (ILDC)

Gain: 200x. EDP: 1 A
RSENSE: 0.01 (R8520) or Rsense SHORT
Vsense: 5 mV, Range: 1.25 A
EADC2: CH0



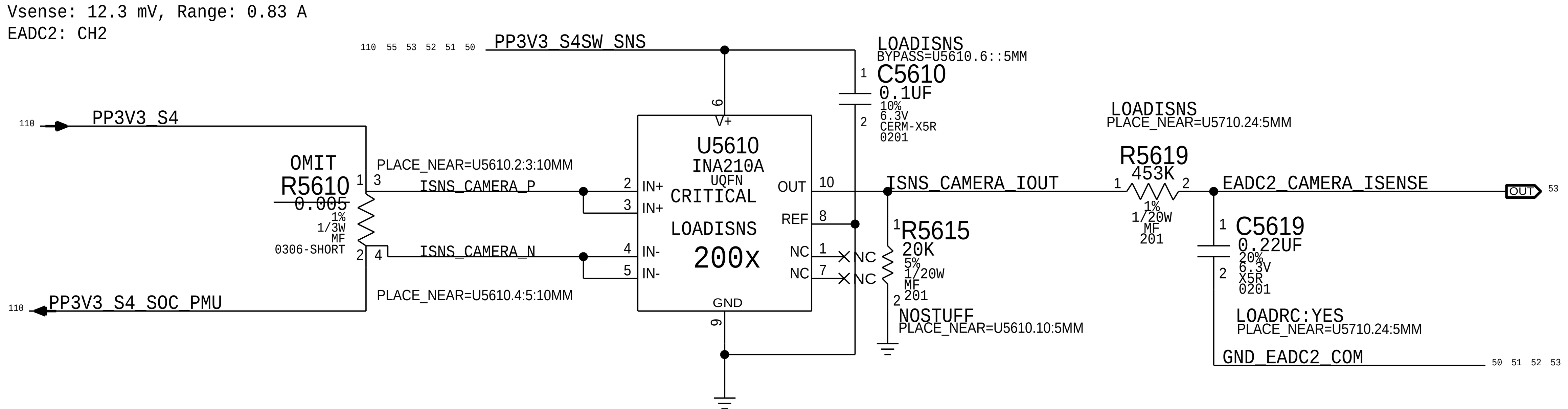
Trackpad 3V Current Sense (IT3C)

Gain: 200x, EDP: 0.2 A
Rsense: 0.05 (R5650) or Rsense SHORT
Vsense: 10 mV, Range: 0.25 A
EADC2: CH1

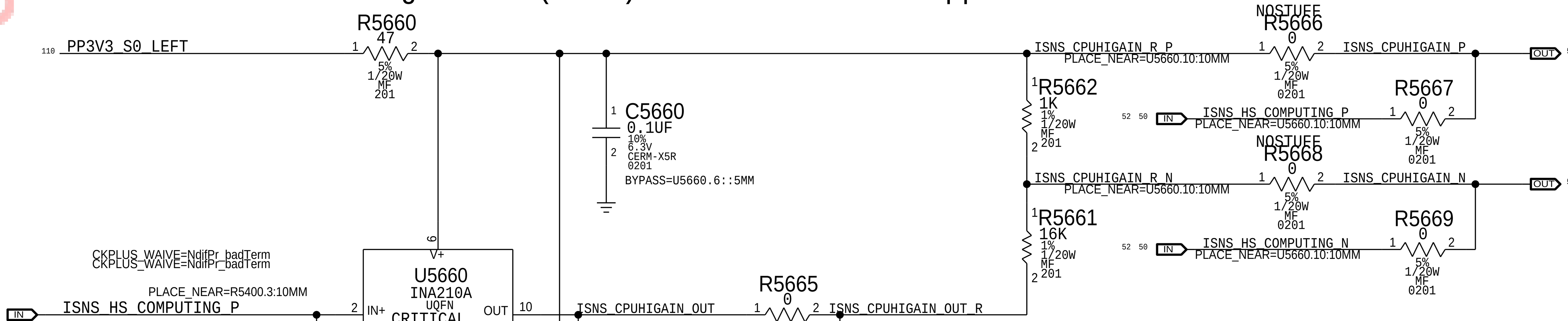


Camera Current Sense (ICMC)

Gain: 200x. EDP: 0.82 A
Rsense: 0.015 (R5610) or XW5610
Vsense: 12.3 mV, Range: 0.83 A
EADC2: CH2



CPU High Side (IC0R) Peak Detection Support

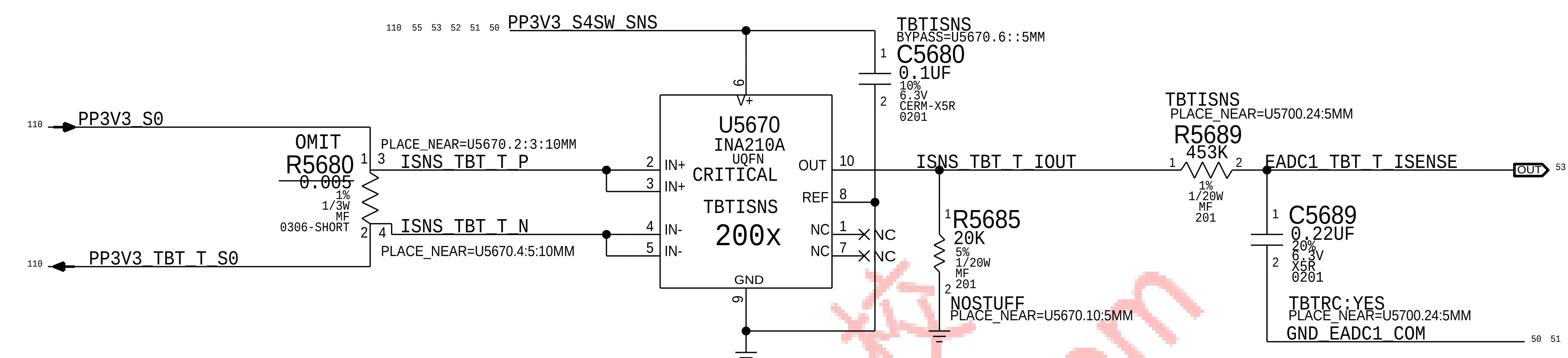


In battery discharge scenario negative voltage will be present on IN+/- pins with INA output voltage decreasing from 3.3V with increasing discharge current.

SENSE+ pins of EMC1704 sink 10-20uA current.
This deviation has been designed in our Peak Detection circuit.
With 10uA sink: 0.125A - 2.1A -> 13mV - 83 mV
With 20uA sink: 0.125A - 2.1A -> 23mV - 92 mV

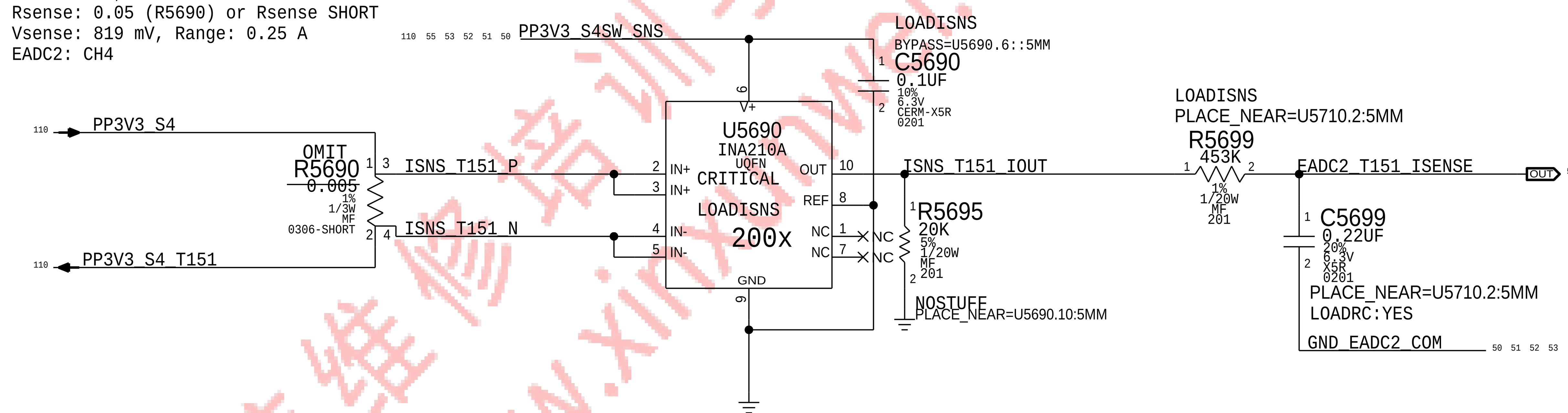
Thunderbolt TBT RIGHT Current Sense (IURC)

Gain: 200x. EDP: 0.5 A
Rsense: 0.025 (R5680) or Rsense SHORT
Vsense: 12.5 mV, Range: 0.5 A
EADC1: CH2



T151 Current Sense (IIDC)

Gain: 200x, EDP: 0.1638 A
Rsense: 0.05 (R5690) or Rsense SHORT
Vsense: 819 mV, Range: 0.25 A
EADC2: CH4



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	4	RES,MTL FILM,100K,1/16W,0201,SMD,LF	C5619,C5629,C5639,C5659		LOADRC:NO
117S0008	2	RES,MTL FILM,100K,1/16W,0201,SMD,LF	C5649,C5689		TBTRC:NO
117S0008	1	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5659		TPADRC:NO

BOM_COST_GROUP=SENSORS

PAGE TITLE
Power Sensors: Extended

Apple Inc.

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BRANCH dvt-fab10

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	4	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5739,C5749,C5779,C5789		LOADRC:NO

PAGE TITLE
Power Sensors: Extended 2

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PAGE TITLE: Thermal Sensors

DRAWING NUMBER: 051-00647 SIZE: D

Apple Inc.

REVISION: 10.0.0

BRANCH: dvt-fab10

SENSORS: EXTENDER 3

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	5	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5921,C5941,C5951,C5961,C5971		LOADRC:NO

BOM_COST_GROUP=SENSORS

PAGE TITLE
Sensor Extended 3
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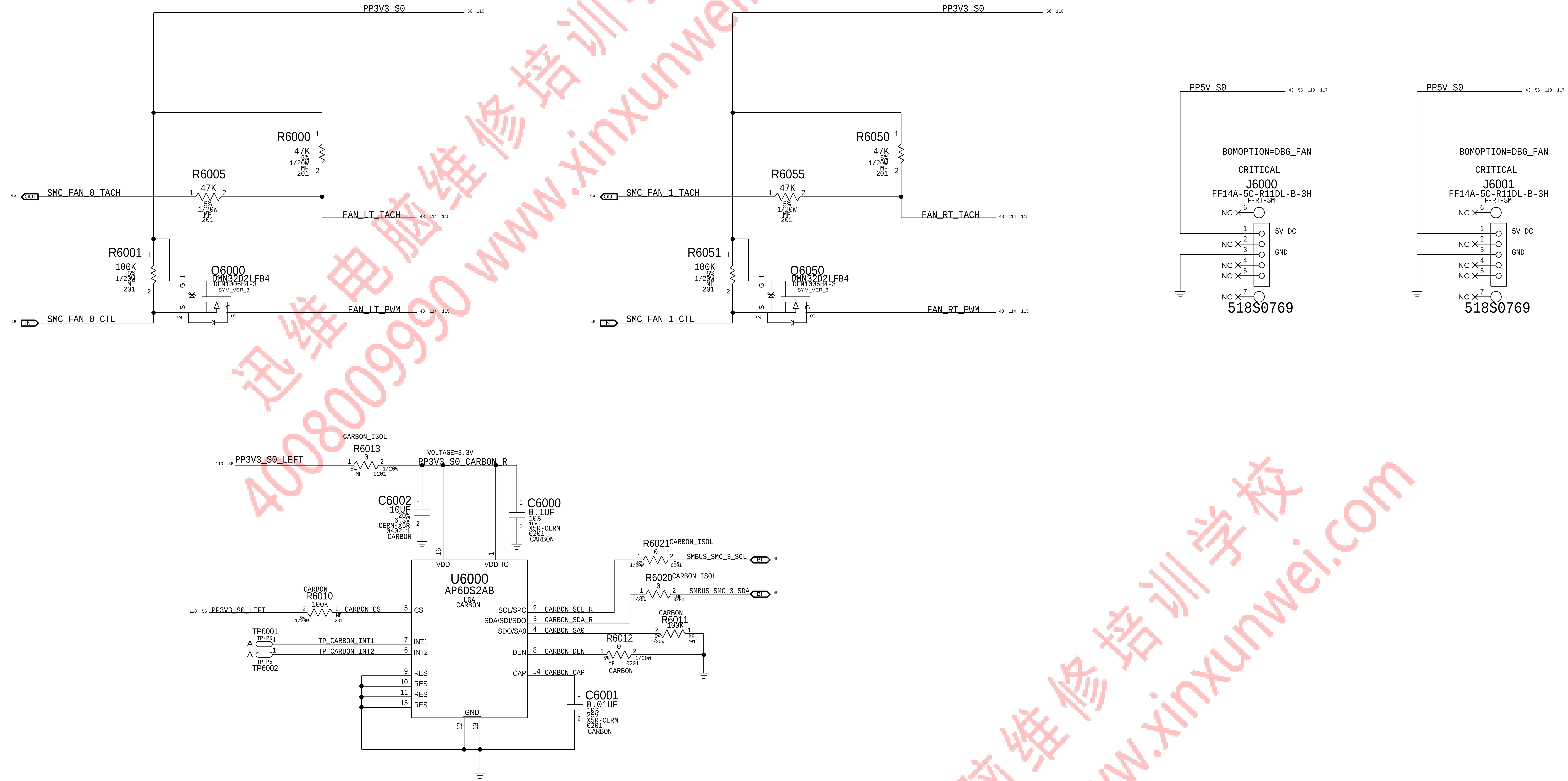
FAN CONNECTOR

KEEP THE 5 PIN CONNECTOR FROM D1



BOM_COST_GROUP=FAN

PAGE TITLE	Fans
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SYNC_MASTER=J80_MLB SYNC_DATE=11/06/2015

PAGE TITLE

SPI Debug Connector

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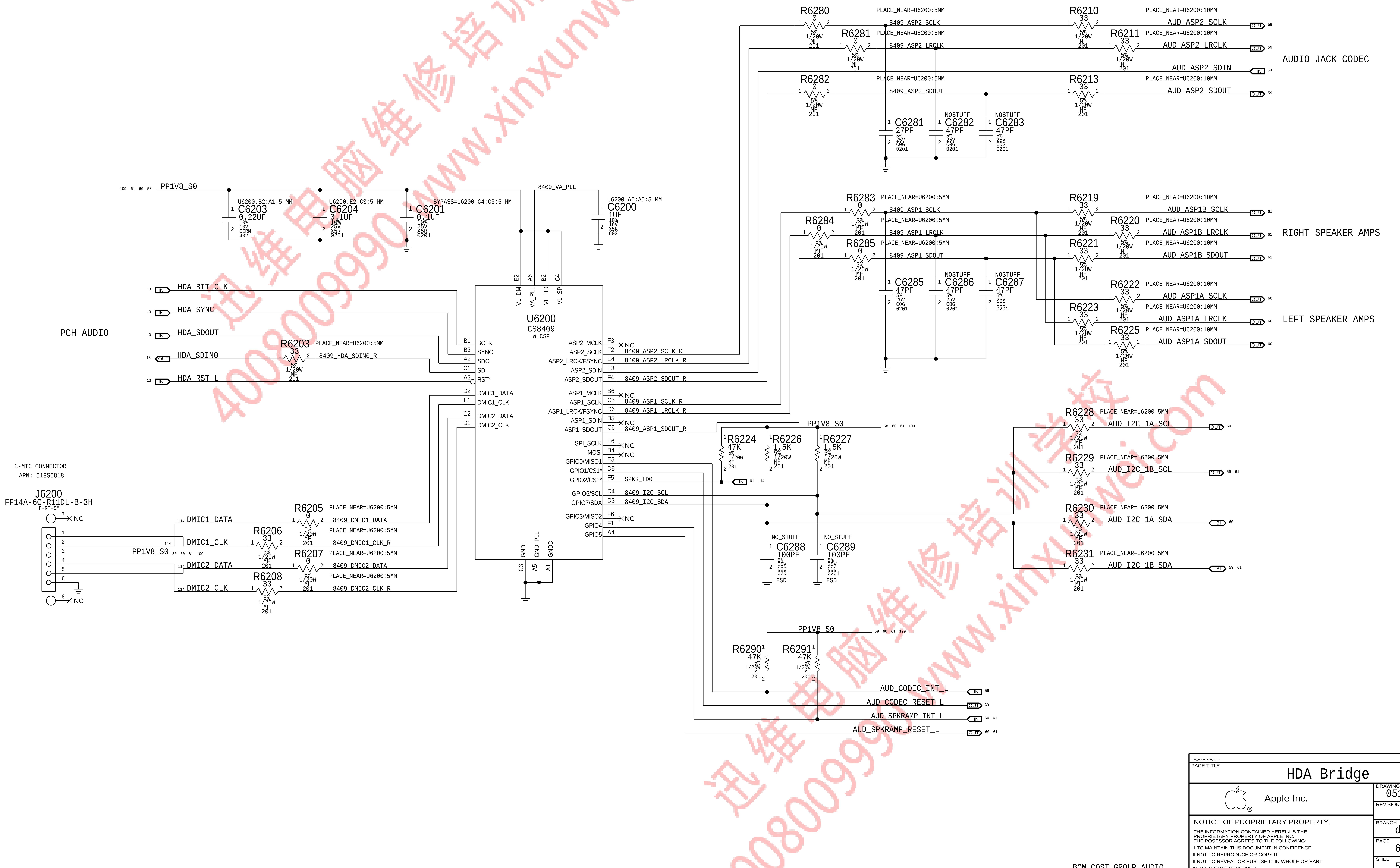

PAGE TITLE
HDA Bridge

Apple Inc.

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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0719	2	CAP,CER,4.7UF,20%,10V,X5R,0402,MURATA	C6305,C6306	CRITICAL	

BOM_COST_GROUP=AUDIO

PAGE TITLE
AUDIO JACK CODEC

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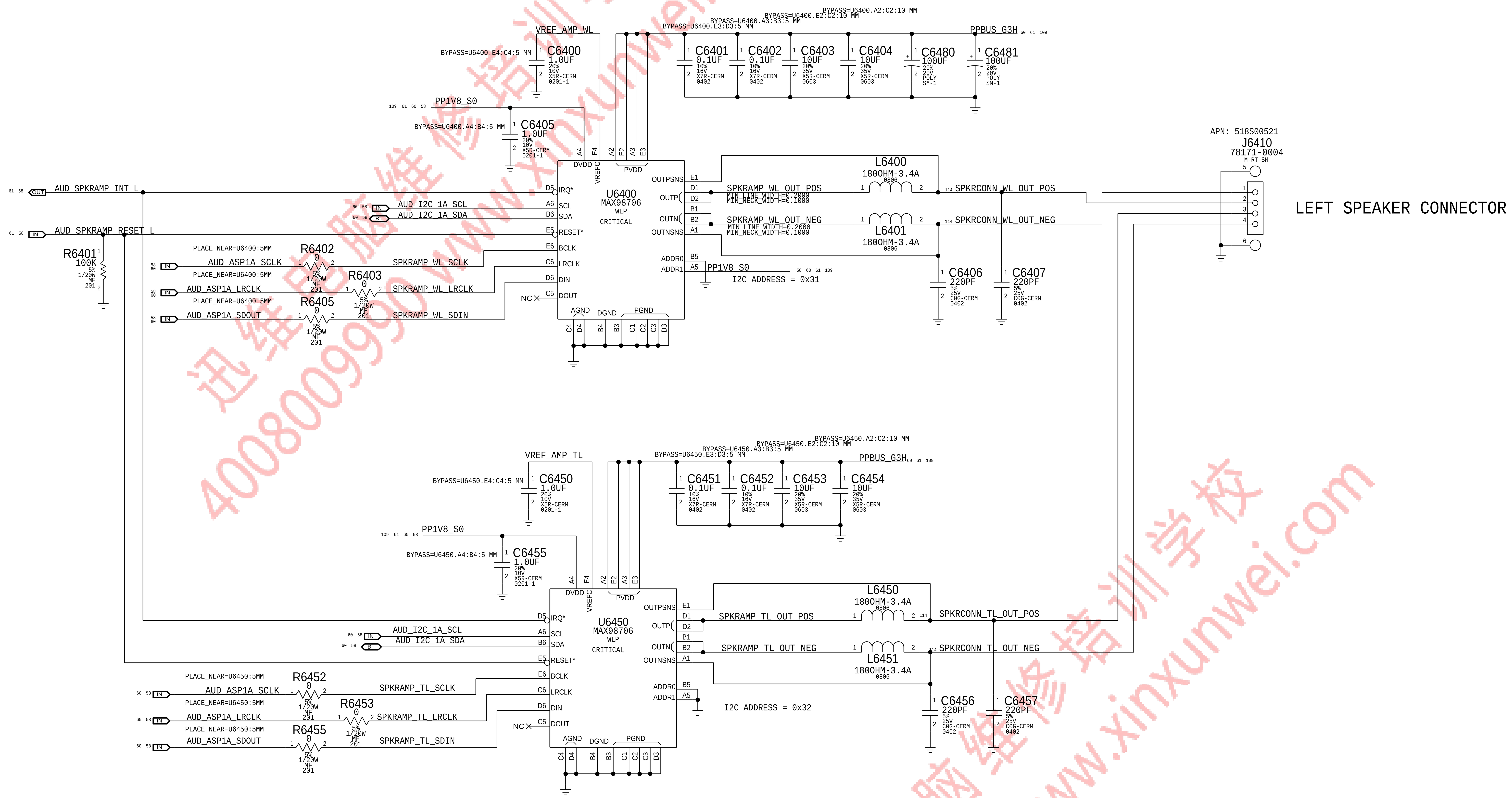
2X MONO SPEAKER AMPLIFIER
APN: 353S00604
0dBFS = 9VPK



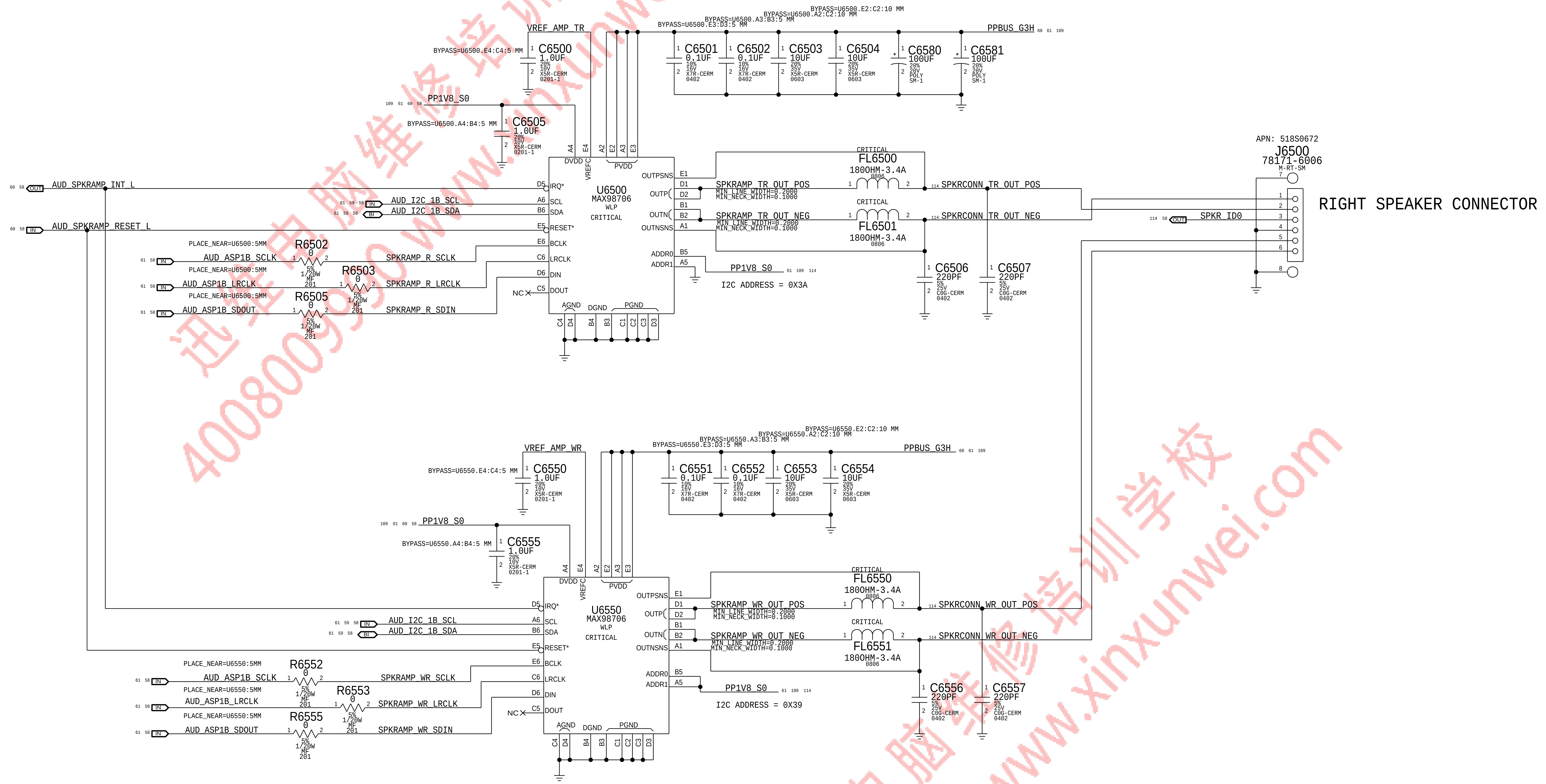
BOM_COST_GROUP=AUDIO

PAGE TITLE
AUDIO Speaker Amps & Conn

Apple Inc.

DRAWING NUMBER 051-00647 SIZE D
REVISION 10.0.0
BRANCH dvt-fab10

2X MONO SPEAKER AMPLIFIER
APN: 353S00604
0dBFS = 9VPK



BOM_COST_GROUP=AUDIO

PAGE TITLE
AUDIO Speaker Amps & Conn
Apple Inc.
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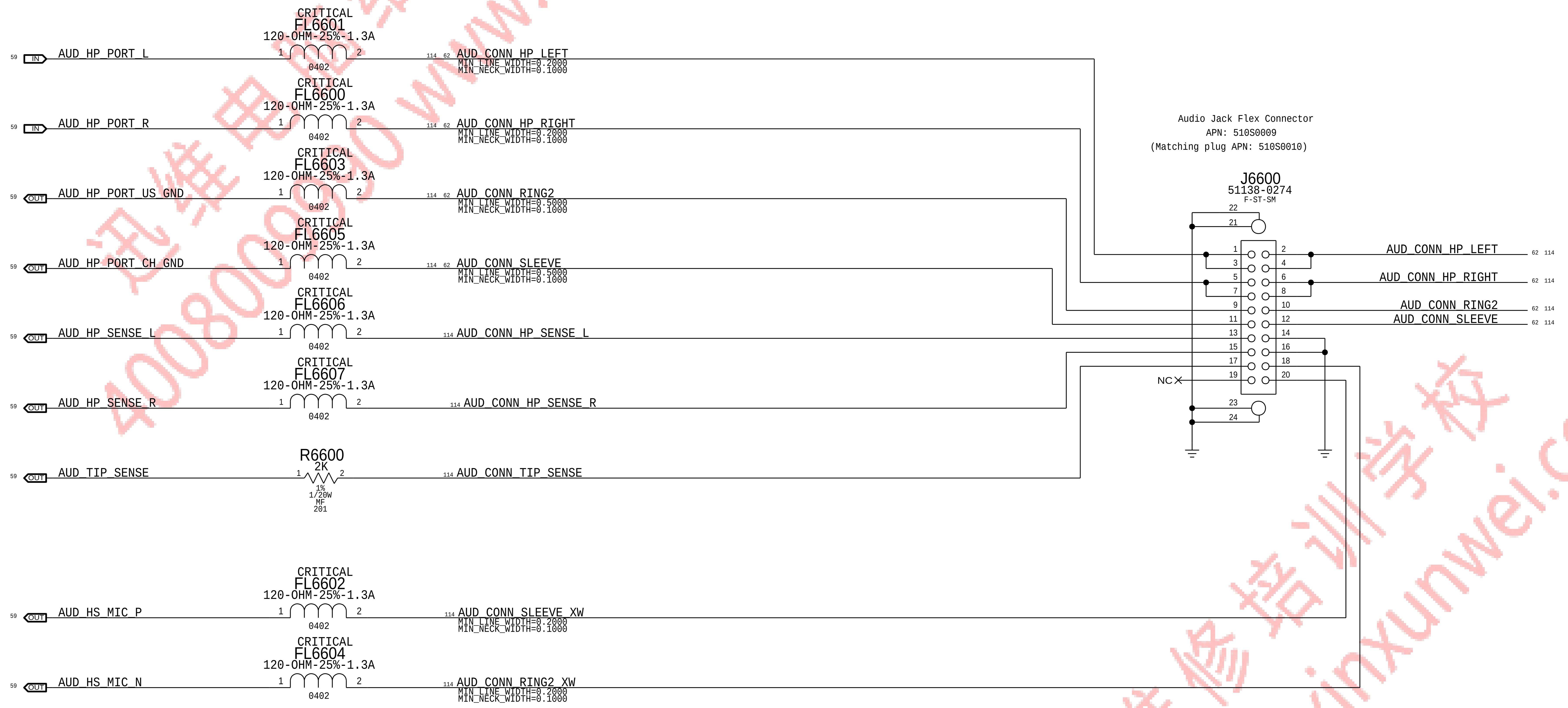
D C B A



BOM_COST_GROUP=AUDIO

PAGE TITLE
AUDIO JACK CONNECTOR

Apple Inc.

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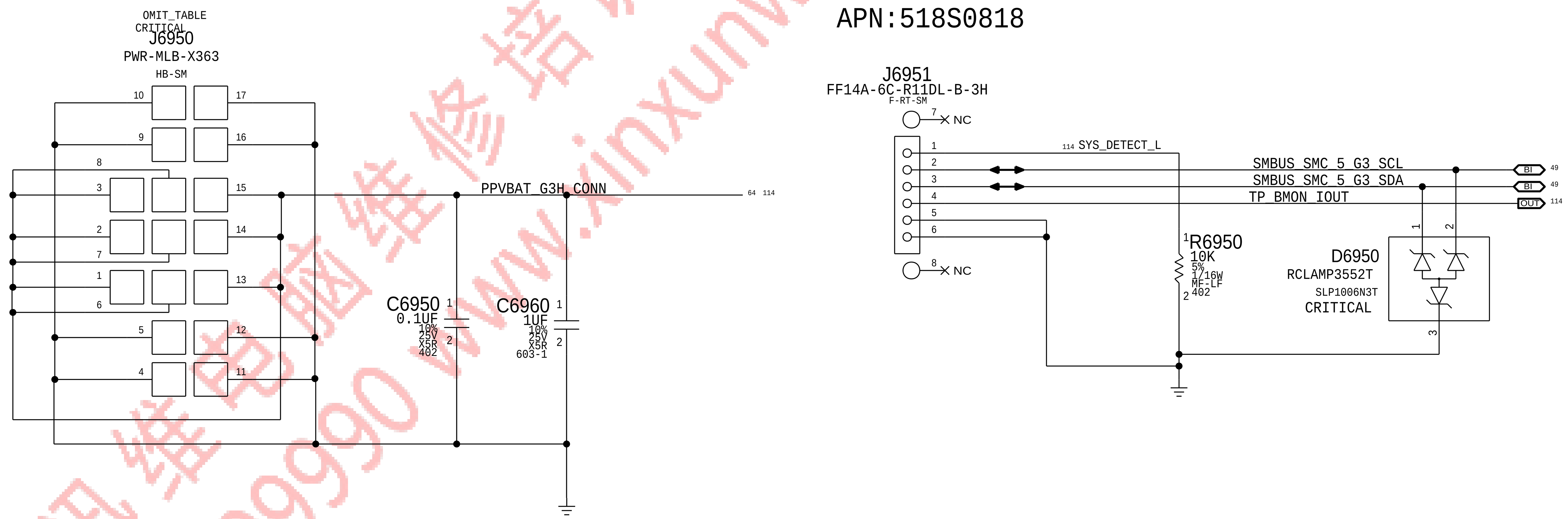
J80 Battery Hotbar Flex Pads 998-03902

Flex Pad TO MLB 998-03780.

APN:518S0818



BMU POWER FLEX HOTBAR'd TO THE MLB:

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
632-00862	1	PCBA,FLEX,BMU PWR,X363	J6950	CRITICAL	

BOM_COST_GROUP=PLATFORM POWER

PAGE TITLE

DC-In & Battery Connectors

Apple Inc.

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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
152S00199	1	IND,MLD,2.7UH,16A,14.8MO,10.9x10.0x3.00	L7030	CRITICAL	

BOM_COST_GROUP=PLATFORM POWER

PAGE TITLE
PBUS Supply & Battery Charger

Apple Inc.

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CORE & SA IMVP IC

Apple Inc.

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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S00497	3	IC,SIC635,DRMOS,IMVP-8,40A,PQFN31,5X5	U7210,U7220,U7230	CRITICAL	

BOM_COST_GROUP=CPU & CHIPSET

SYNC_MASTER=J80_DTUZMAN_MLB_BAFFIN SYNC_DATE=09/03/2015

PAGE TITLE

CORE IMVP POWER BLOCK

Apple Inc.

DRAWING NUMBER 051-00647 SIZE D

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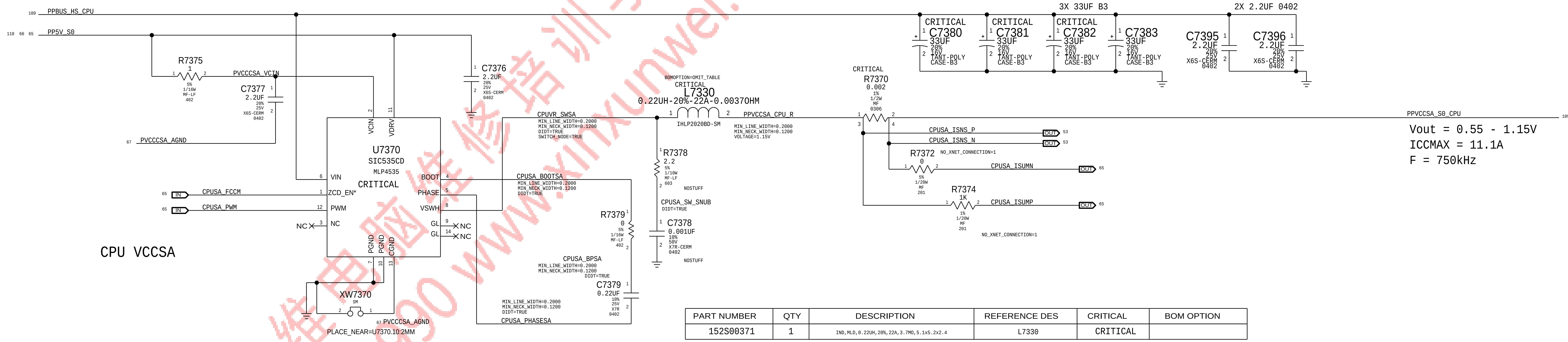
BRANCH dvt-fab10



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
152S00371	1	IND,MLD,0.22UH,20%,22A,3.7MO,5.1x5.2x2.4	L7330	CRITICAL	

BOM_COST_GROUP=CPU & CHIPSET

SYNC_MASTER=J80_DTUZMAN_MLB_BAFFIN SYNC_DATE=11/18/2015

PAGE TITLE
SA IMVP IC

Apple Inc.

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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S00497	2	IC,SIC635,DRMOS,IMVP-8,40A,PQFN31,5X5	U7410,U7420	CRITICAL	

BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE
GT & GTX IMVP POWER BLOCK

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5V S0 - V5

3.3V DSW - V6

BOM_COST_GROUP=PLATFORM POWER

PAGE TITLE

Power - 5V 3.3V Supply

Apple Inc.

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BANJO - PMIC Control

PAGE TITLE
PMIC-1 & Power Control

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SYNC_MASTER=J80_MLB SYNC_DATE=11/06/2015

PAGE TITLE

PMIC-1 1.2V 0.6V VCCIO

Apple Inc.

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1V - V11 (Banjo#1 VR1)

1.8V - V8 (Banjo#1 VR2)

PAGE TITLE

PMIC-1 1V 1.8V VCCPCH

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PAGE TITLE
PMIC-1 Aliases & TPs

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U8200

Part	SLG5AP1569V
Type	Load Switch
R(on) @ 3.6V	34 mOhm Typ 46 mOhm Max
Current	1A Max

U8295

Part	SLG5AP1445V
Type	Load Switch
R(on) @ 3.6V	7.8 MOHM TYP 8.5 MOHM MAX
Current	4A MAX

U8203

Part	TPS22969
Type	Load Switch
R(on) @ 3.6V	4.4 MOHM TYP 6.6 MOHM MAX
Current	6A MAX

U8215

Part	SLG5AP1453V
Type	Load Switch
R(on) @ 5.3A	7.8 mOhm Typ 9.6 mOhm Max
Current	5.3A Max

U8205

Part	SLG5AP1445V
Type	Load Switch
R(on) @ 3.6V	7.8 MOHM TYP 8.5 MOHM MAX
Current	4A MAX

U8283

Part	SLG5AP1635V
Type	Load Switch
R(on) @ 25C	20 mOhm Typ 28 mOhm Max
Current	2.5A Max
Ton Total	39us max @ 1V

U8209

Part	SLG5AP1453V
Type	Load Switch
R(on)	7.8 mOhm Typ 9.6 mOhm Max
Current	5.3A

U8218

Part	SLG5AP1453V
Type	Load Switch
R(on) @ 5.3A	7.8 mOhm Typ 9.6 mOhm Max
Current	5.3A Max

U8285

Part	SLG5AP1635V
Type	Load Switch
R(on) @ 25C	20 mOhm Typ 28 mOhm Max
Current	2.5A Max
Ton Total	39us max @ 1V

U8210

Part	SLGAP1569V
Type	Load Switch
R(on) @ 3.6V	34 mOhm Typ 46 mOhm Max
Current	1A Max

U8220

Part	SLG5AP1453V
Type	Load Switch
R(on) @ 5.3A	7.8 mOhm Typ 9.6 mOhm Max
Current	5.3A Max

BOM_COST_GROUP=PLATFORM POWER

PAGE TITLE
Power FETs

Apple Inc.

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Page Notes

Power aliases required by this page:

- =PPVIN_S0SW_LCDBKLTFET (9-12.6V LCD BACKLIGHT INPUT)
- =PP5V_S0_BKLT (5V BACKLIGHT DRIVER INPUT)

PAGE TITLE

LCD Backlight Driver

Apple Inc.

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LCD PANEL INTERFACE (eDP) + Camera (MIPI)

PAGE TITLE
eDP Display Connector

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POLARIS_CONTROLLER

Apple Inc.

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POLARIS POWER

Apple Inc.

DRAWING NUMBER 051-00647 SIZE D

REVISION 10.0.0

BRANCH dvt-fab10

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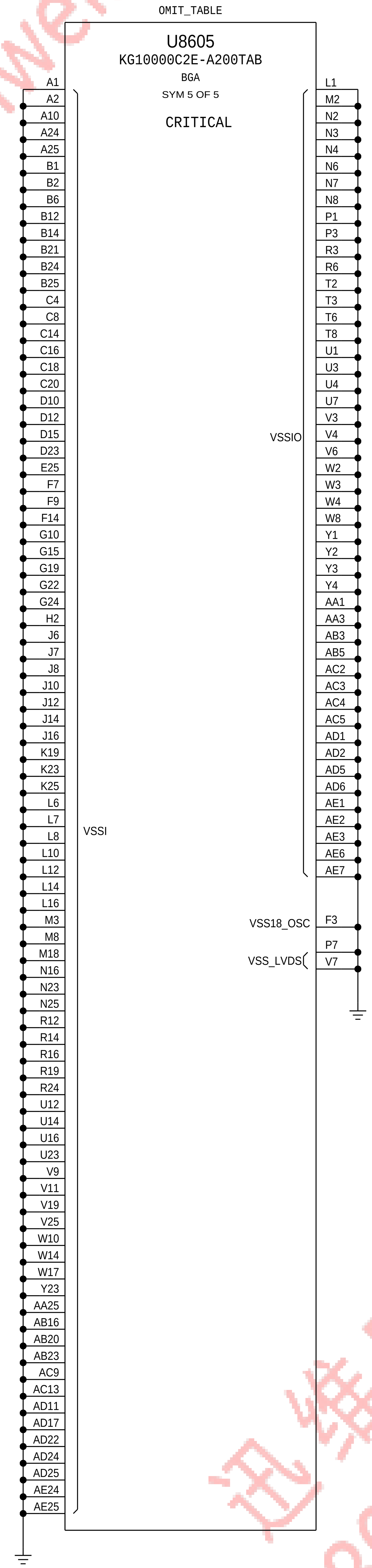
D

C

B

A



BOM_COST_GROUP=SSD

PAGE TITLE
PAGE_TITLE=POLARIS GND

Apple Inc.

DRAWING NUMBER	051-00647
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SSD RELATED BOM Groups

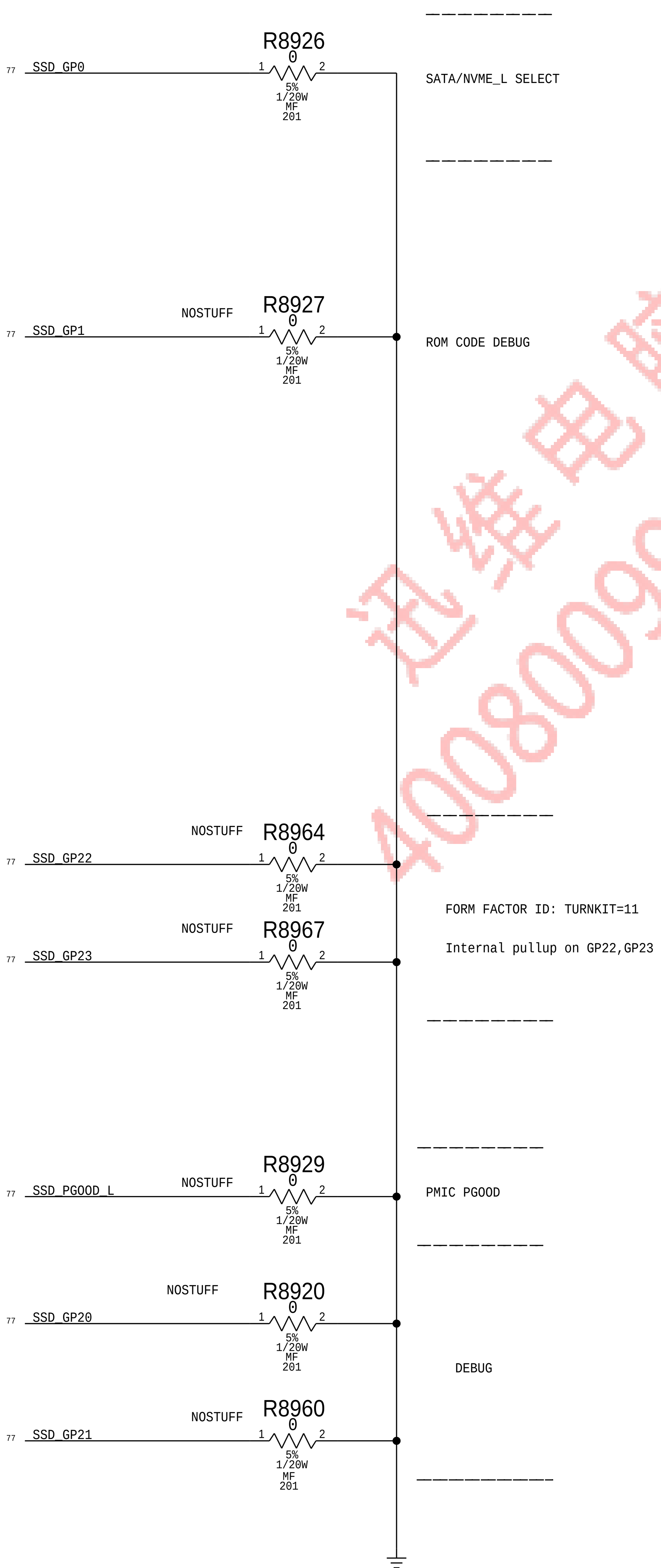
BOM GROUP	BOM OPTIONS
SSD_CONFIG:256GB	SSD_GPIO9,NAND_TYPE:256GB,SSD_CTRL_TYPE:4GBIT
SSD_CONFIG:512GB	SSD_GPIO8,NAND_TYPE:512GB,SSD_CTRL_TYPE:4GBIT
SSD_CONFIG:1TB	NAND_TYPE:1TB,SSD_CTRL_TYPE:8GBIT
SSD_CONFIG:2TB	SSD_GPIO10,NAND_TYPE:2TB,SSD_CTRL_TYPE:8GBIT

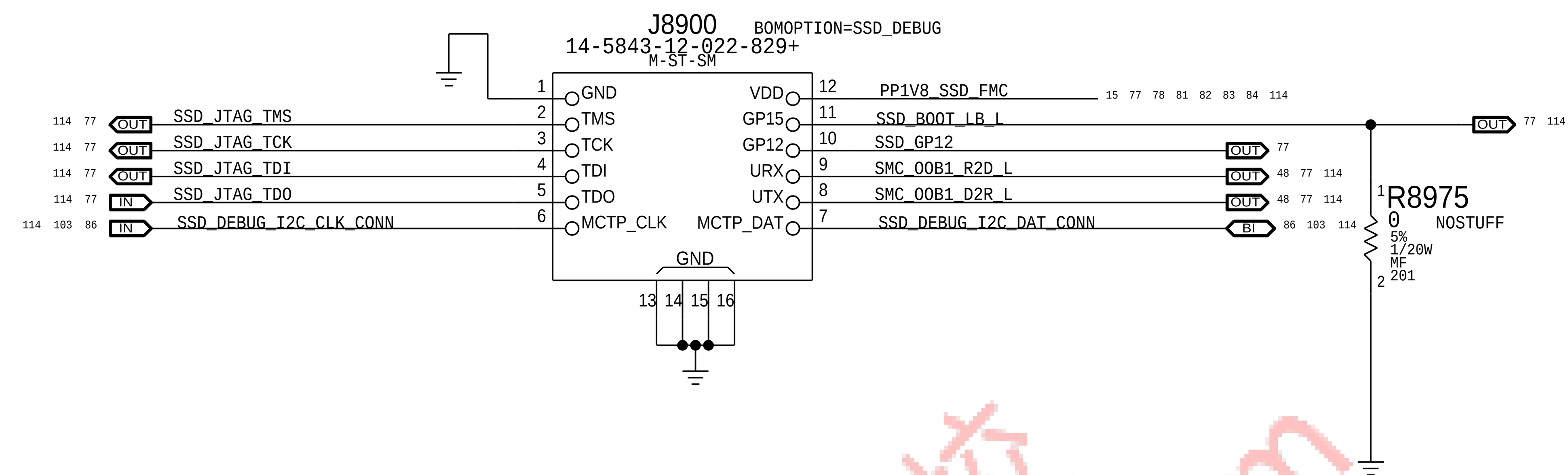
NAND Parts

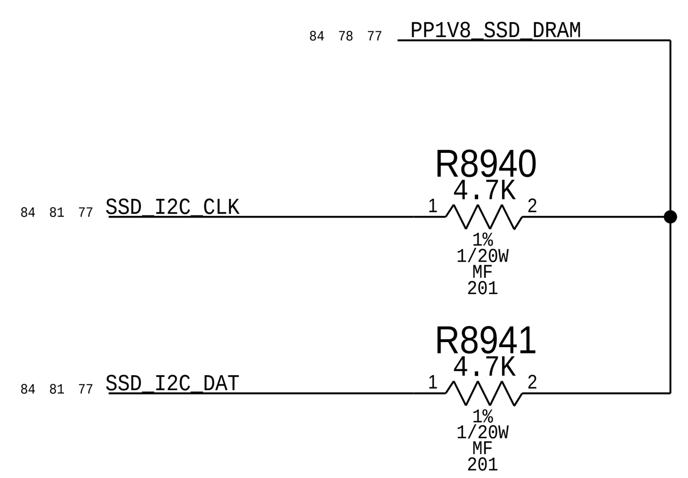
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
335S00149	4	NAND,1Z,64GBM,TOGG DDR2,64G,SS,BGA 168	U9100,U9110,U9200,U9210	CRITICAL	NAND_TYPE:256GB
335S00204	4	NAND,V3,128GBM,TOGG DDR2,256G,SSBGA 168	U9100,U9110,U9200,U9210	CRITICAL	NAND_TYPE:512GB
335S00205	4	NAND,V3,256GBM,TOGG DDR2,256G,SS,BGA 168	U9100,U9110,U9200,U9210	CRITICAL	NAND_TYPE:1TB
335S00219	4	NAND,V3,512GBM,TOGG DDR2,256G,SS,BGA 168	U9100,U9110,U9200,U9210	CRITICAL	NAND_TYPE:2TB

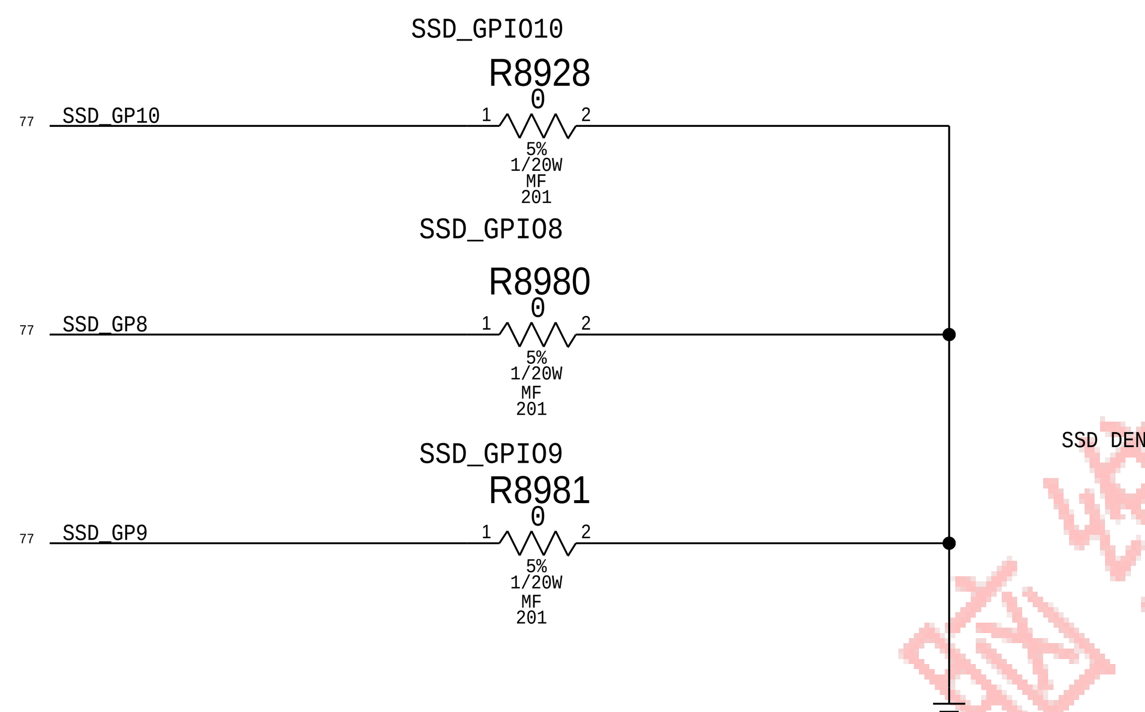
SSD Controller Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
339S00154	1	POP,POLARIS+4GBIT,SSD CTRL,A2,BGA516	U8605	CRITICAL	SSD_CTRL_TYPE:4GBIT
339S00155	1	POP,POLARIS+8GBIT,SSD CTRL,A2,BGA516	U8605	CRITICAL	SSD_CTRL_TYPE:8GBIT









SSD CONFIGURATIONS

CAPACITY	NAND APN	CONTROLLER APN	R8980/GPIO8	R8981/GPIO9	R8928/GPIO10
128 GB	-	-	STUFF-0	STUFF-0	STUFF-0
256 GB	335S00149	339S00154 (A2 4GB DRAM)	NOSTUFF-1	STUFF-0	NOSTUFF-1
512 GB	335S00204	339S00154 (A2 4GB DRAM)	STUFF-0	NOSTUFF-1	NOSTUFF-1
1 TB	335S00205	339S00155 (A2 8GB DRAM)	NOSTUFF-1	NOSTUFF-1	NOSTUFF-1
2 TB	335S00219	339S00155 (A2 8GB DRAM)	NOSTUFF-1	NOSTUFF-1	STUFF-0

BOM_COST_GROUP=SSD

PAGE TITLE
Connector

Apple Inc.

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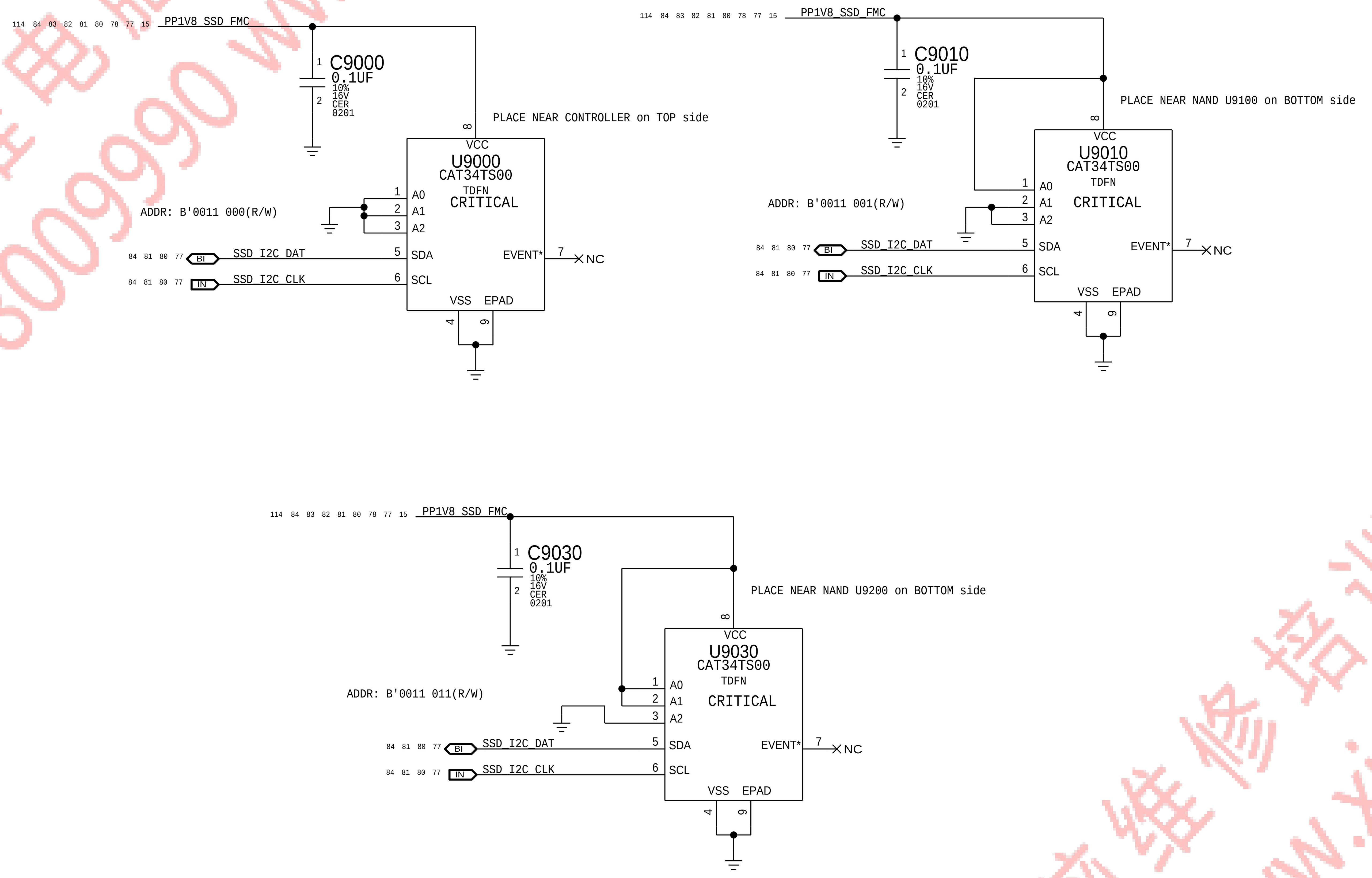

BOM_COST_GROUP=SSD

PAGE TITLE

PAGE_TITLE=TEMP SENSORS

Apple Inc.

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PAGE TITLE

NAND 1/2

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BOM_COST_GROUP=SSD

PAGE TITLE
NAND 2/2

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SSD PMIC Alternate Parts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00457	152S00309		ALL	
152S1721	152S00310		ALL	

PAGE TITLE
PAGE_TITLE=POLARIS PMIC

Apple Inc.

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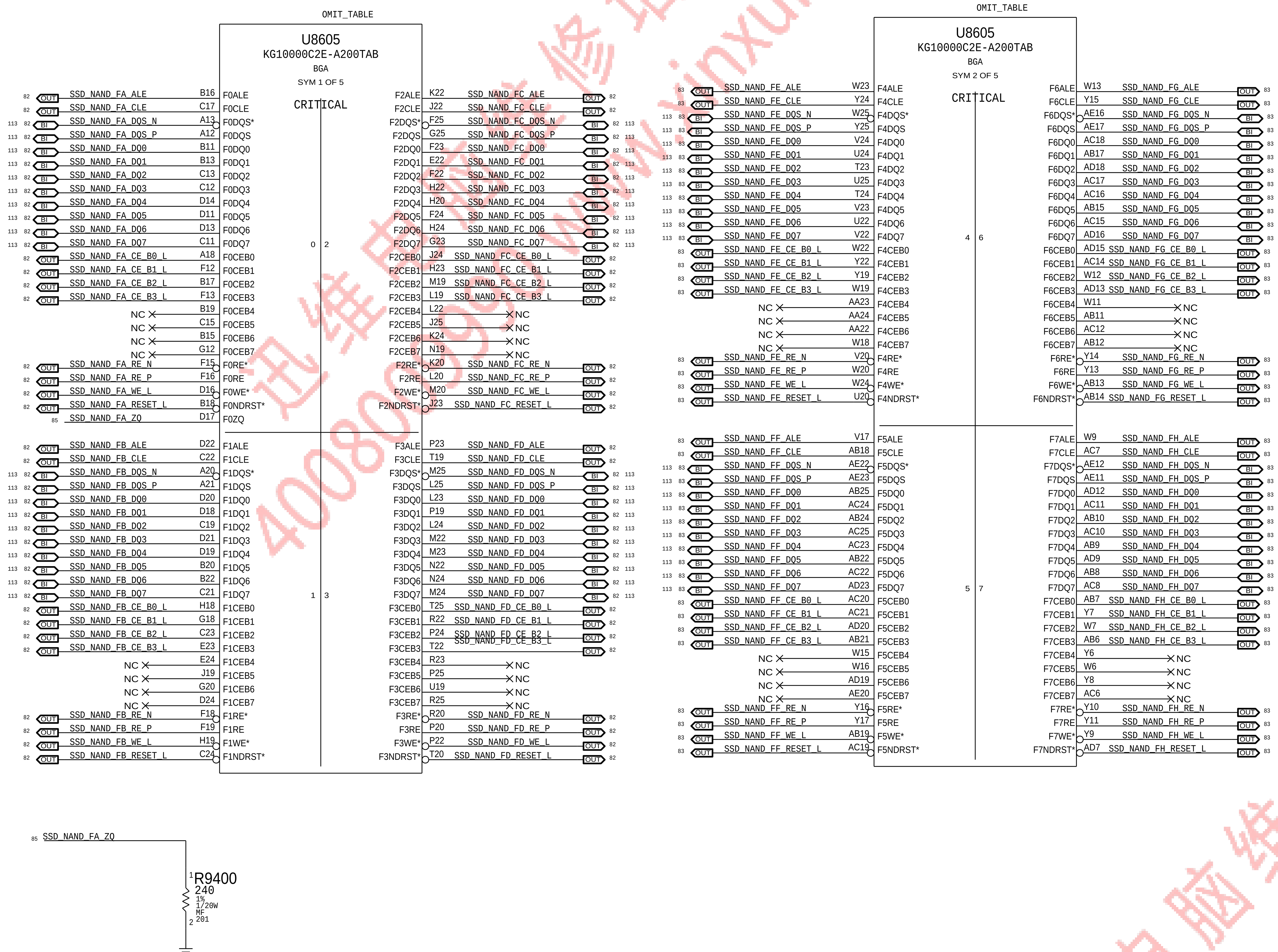

BOM_COST_GROUP=SSD

PAGE TITLE	SSD NAND VR
DRAWING NUMBER	051-00647
SIZE	D
REVISION	10.0.0
BRANCH	dvt-fab10
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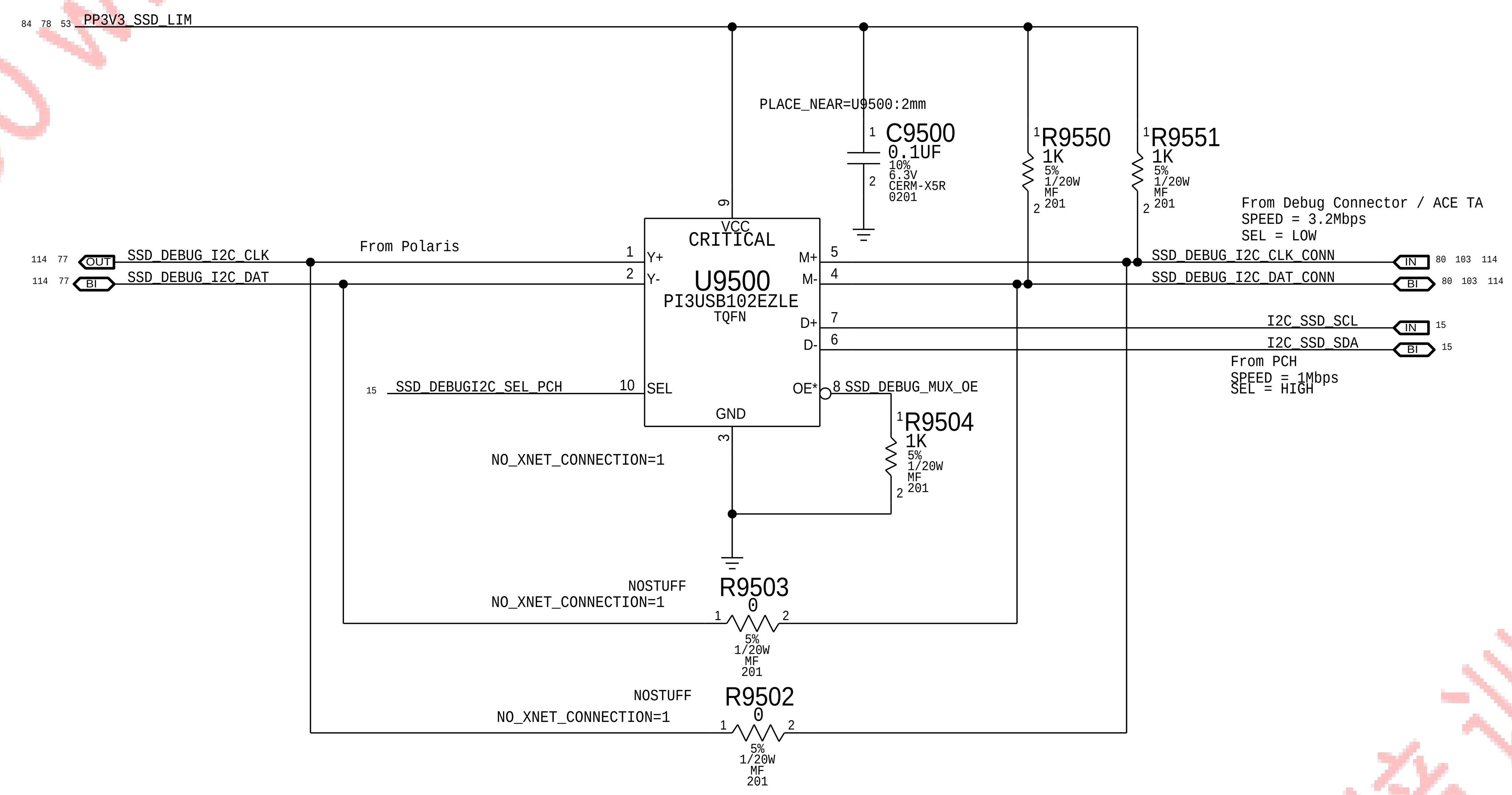
Apple Inc.



BOM_COST_GROUP=SSD

PAGE TITLE

SSD SUPPORT

Apple Inc.

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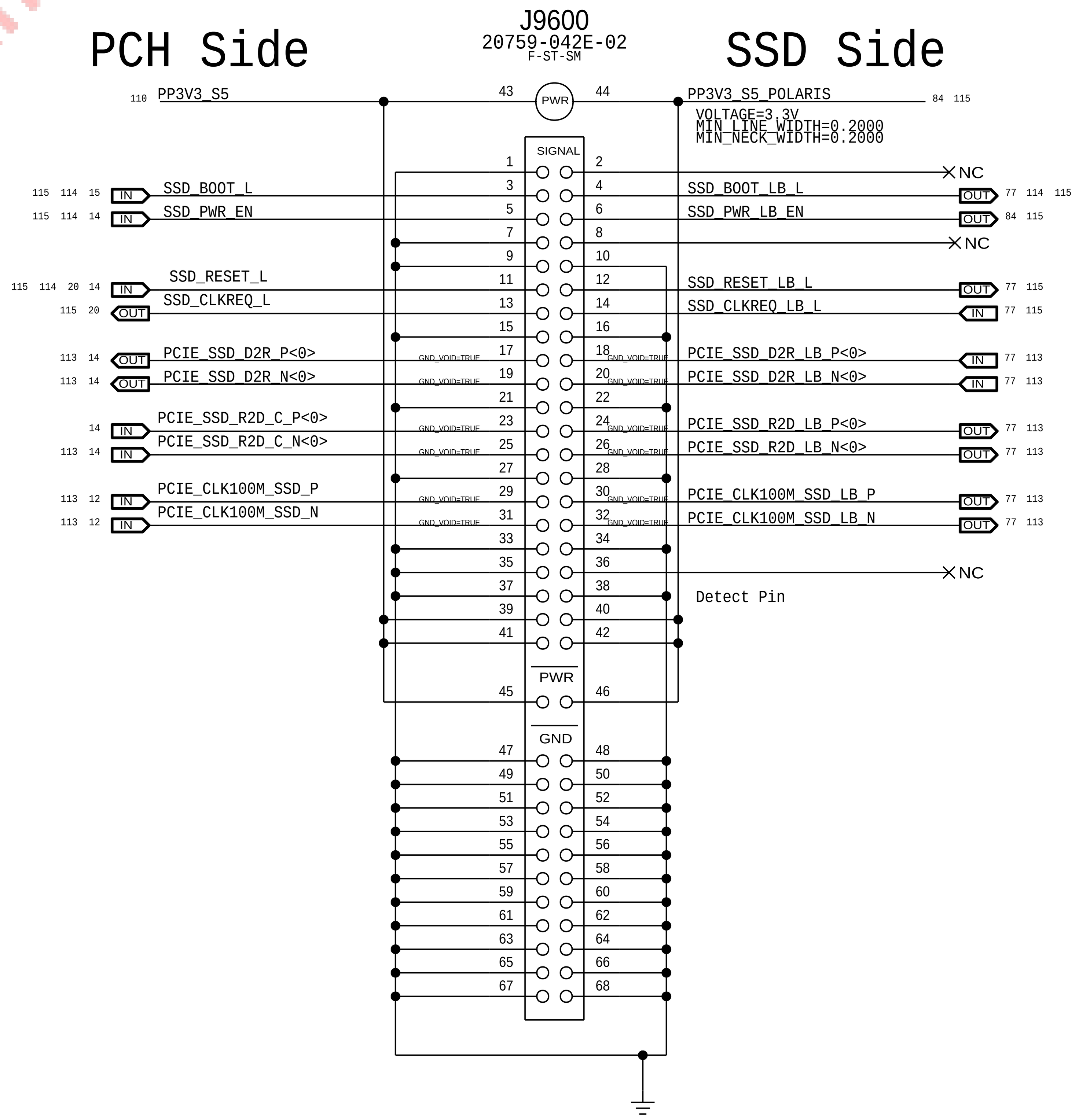
LIFEBOAT



BOM_COST_GROUP=SSD

PAGE TITLE	Lifeboat
DRAWING NUMBER	051-00647
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Apple Inc.

8 7 6 5 4 3 2 1

D C B A

PAGE TITLE

Constraints

Apple Inc.

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SYNC_MASTER=dpmux

SYNC_DATE=08/22/2015

PAGE TITLE

eDP Mux

Apple Inc.

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PAGE TITLE

GPU PCC

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PAGE TITLE

BAFFIN PCI-E

Apple Inc.

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PAGE TITLE

Baffin CORE/FB POWER

Apple Inc.

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PAGE TITLE
Baffin FRAME BUFFER I/F

Apple Inc.

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BRANCH dvt-fab10

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GPU FB SUPPLY
VOUT = 1.5V / 1.35V
10.7A MAX OUTPUT
F = 500 KHZ
VID1 VID0 FBVDD
0 0 1.5V
1 0 1.35V
GPU VDDCI SUPPLY
VOUT = 0.8V-0.9V
?9.7A? MAX OUTPUT
F = 450 KHZ
Note:
Regulator requires
a minimum load to
prevent noise in the
audio frequencies
UA300
ISL95870AH
UTQFN
CRITICAL
QA300
SI7710DT
CRITICAL
LA300
0.68UH-20%-14A
RA300
0.002
QA350
CSD58873Q3D
LA350
0.2UH-20%-28A-0.0011OHM
RA368
0.002
PP3V3_S0_GPU
PP5V_S0
PPBUS_HS_GPU
PP1V5R1V35_S0_GPU_MEM
PP1V5R1V35_GPU_REG_R
PPVDDCI_S0_GPU
FBVDD_ALTVO
P1V5R1V35_GPU_FB_SNS_P
P1V5R1V35_GPU_FB_SNS_N
PVDDCI_GPU_FB_SNS_P
PVDDCI_GPU_FB_SNS_N
PVCORE_GPU_FB_SNS_N
REG_GPU_VDDCI_VSEN
REG_GPU_VDDCI_FB
REG_GPU_VDDCI_COMP
REG_GPU_VDDCI_FCCM
REG_GPU_VDDCI_IMON
REG_GPU_VDDCI_ISEN1
REG_BOOT_GPU_VDDCI
REG_UGATE_GPU_VDDCI_R
REG_PHASE_GPU_VDDCI
REG_LGATE_GPU_VDDCI
REG_GPU_VDDCI_ISUMP
REG_GPU_VDDCI_ISUMN
VDDCIS0_CS_P
VDDCIS0_CS_N
BOM_COST_GROUP=GRAPHICS
Baffin 1V05 GPU / 1V35 FB Power Supply
Apple Inc.
DRAWING NUMBER 051-00647
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BRANCH dvt-fab10
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SYNC_MASTER=J88_SEAN SYNC_DATE=04/29/2015

PAGE TITLE

GDDR5 Frame Buffer A

Apple Inc.

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SYNC_MASTER=J80_SEAN SYNC_DATE=04/29/2015

PAGE TITLE

GDDR5 Frame Buffer B

Apple Inc.

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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
376S00174	2	MOSFET,CTRL+SYNC,25V,4.1/1.4MO,QFN32,6x6	QA640,QA650	CRITICAL	

TABLE 6. PRE-PWROK METAL VID CODES

SVC	SVD	OUTPUT VOLTAGE (V)
0	0	1.1
0	1	1.0
1	0	0.9
1	1	0.8

ISL6277A datasheet (December 19,2012)

BOM_COST_GROUP=GRAPHICS

PAGE TITLE
GFX IMVP VCore Regulator [106]

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former site of the SPEED MLPS STRAPS
PCIe Gen3 enabled, Half-Swing, TX De-emp enabled
VBIOS disabled, Boot from EFI
VGA enabled
All power audio-capable display output
256MB FB Aperture Size
PS_0: 01001 82nF 8.45k 2k
PS_1: 10001 10nF 8.45k 2k
PS_2: 10000 10nF NC 4.75k
PS_3: 00000 680nF NC 4.75k

BAFFIN STRAPS
PCIe Gen3 enabled, FULL-Swing, TX De-emp enabled
VBIOS disabled, Boot from EFI
VGA enabled, CLKREQ_L enabled
All power audio-capable display output
Audio for DP or HDMI dongle
256MB FB Aperture Size
SMBUS address 41

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
197S0499	1	XTAL,27.000MHZ,30PPM,12PF,2.5x2.0MM	YA740	CRITICAL	GPUCLK:XTAL
197S0056	1	OSC,MEMS,100MHZ,+/-20PPM,1.8V,2520	YA740	CRITICAL	GPUCLK:OSC
117S0201	1	RES,0 OHM,5%,0201	RA743	CRITICAL	GPUCLK:XTAL
117S0080	1	RES,33 OHM,5%,0201	RA743	CRITICAL	GPUCLK:OSC
155S0387	1	FERRITE BEAD, 470OHM,0.1A,1.5MOHM DCR,060	RA740	CRITICAL	GPUCLK:OSC
117S0201	1	RES,0 OHM,5%,0201	RA742	CRITICAL	GPUCLK:XTAL
132S0444	1	CAP,CER,X5R,0.1UF,10%,6.3V,0201	RA742	CRITICAL	GPUCLK:OSC

BOM_COST_GROUP=GRAPHICS

PAGE TITLE
Baffin GPIOs,CLK & Straps

Apple Inc.

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PAGE TITLE
Baffin DP/GPIO

Apple Inc.

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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S00897	1	IC,TPS62130B-S,3A BUCK CNVTRT,QFN16,3X3M	UA900	CRITICAL	

Placement Note:
Place UA960 Bottom side, under the VRAM shield can, but not within 2mm of any IC

I2C Write: 0x98, I2C Read: 0x99

Placement Note:
Place QA960 TOP side, under the GPU

UA950

Part	SLG5AP1453V
Type	Load Switch
R(on) @ 5.3A	7.8 mOhm Typ 9.6 mOhm Max
Current	5.3A Max

UA951

Part	SLG5AP1453V
Type	Load Switch
R(on) @ 5.3A	7.8 mOhm Typ 9.6 mOhm Max
Current	5.3A Max

BOM_COST_GROUP=GRAPHICS

PAGE TITLE
Baffin VSS & MISC
Apple Inc.
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LAST_MODIFIED=Wed Aug 24 09:58:00 2016

USB-C HIGH SPEED 1

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BOM_COST_GROUP=TBT

USB-C HIGH SPEED 2

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TMU CLKs
DP SRC OPTIONS
IF DP SRC NOT USED
FUSES FOR UPC
ACE Debug Support
USBC 5V EN PD
ACE A/B RPD STRAPPING
Ridge PDs
ACE PDs
ACE FET DCIN Bypass Caps
I2C SERIES R'S
ACE DEBUG CONN
Place on bottom
RIDGE 0.9V SVR XW
SIGNAL ALIASES
25MHz xtal
to/from Ridge
TBT to ACE
TBT
Alpine Ridge U2800
(MASTER)
Pri ACE
U3100
(Write: 0x70 Read: 0x71)
Sec ACE
U3200
(Write: 0x7E Read: 0x7F)
POWER ALIASES
AR/ACE SPI BUS SERIES R'S
ROM
Ace
AR
Ridge PCIE Caps
R2D
D2R
NO TEST
GND ALIASES
BOM_COST_GROUP=TBT
PAGE TITLE
USB-C Support
Apple Inc.
DRAWING NUMBER
051-00647
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BRANCH
dvt-fab10
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PRIMARY ACE USB-C PORT CONTROLLER (UPC)

BOM_COST_GROUP=USB-C

PAGE TITLE: USB-C PORT CONTROLLER A

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DRAWING NUMBER: 051-00647 | SIZE: D

REVISION: 10.0.0

BRANCH: dvt-fab10

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SECONDARY ACE USB-C PORT CONTROLLER (UPC)

BOM_COST_GROUP=USB-C

PAGE TITLE
USB-C PORT CONTROLLER B

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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0683	2	CAP,CER,X5R,1UF,10%,25V,0402	CB504, CB554	CRITICAL	NOSTUFF

BOM_COST_GROUP=USB-C

LAST CHANGE: Wed Apr 1 22:57:37 2015

PAGE TITLE

USB-C CONNECTOR A

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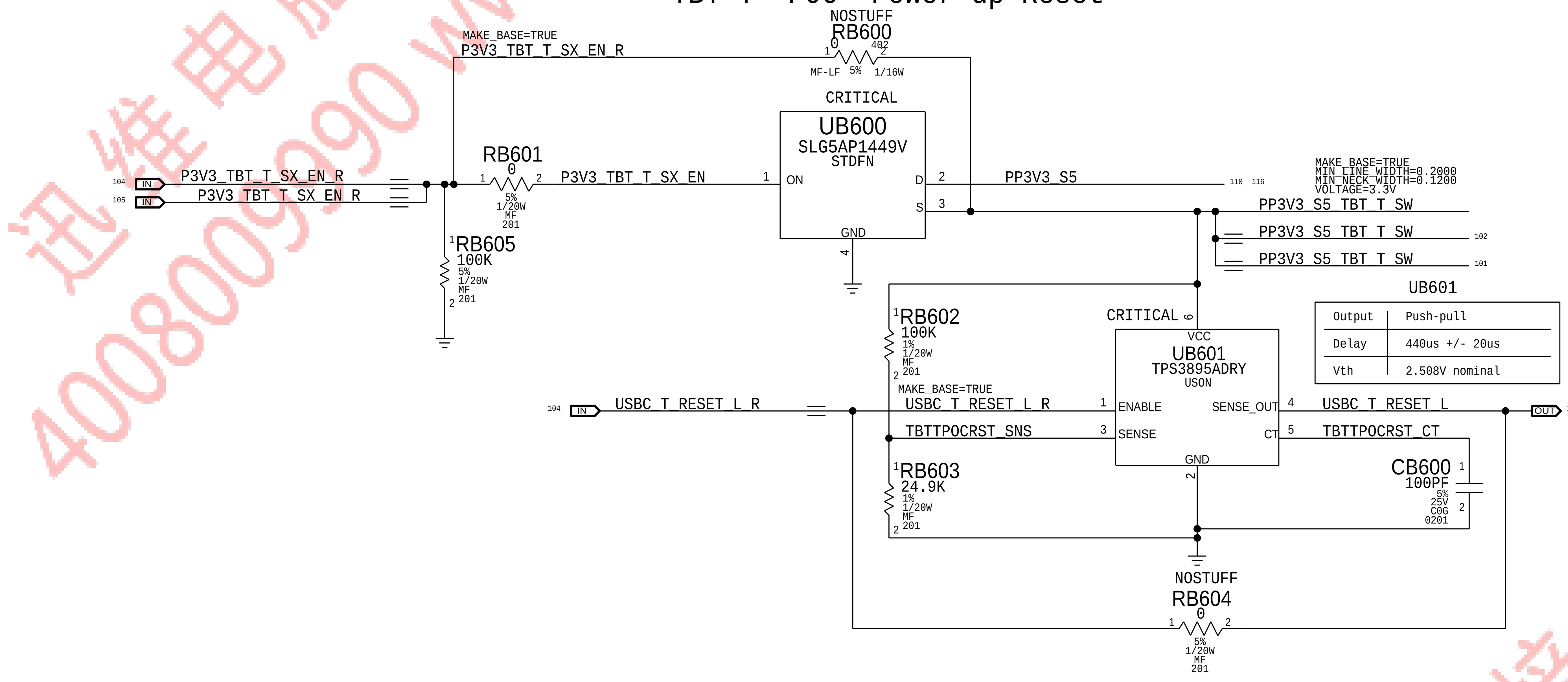
TBT T "POC" Power-up Reset



BOM_COST_GROUP=USB-C

DESIGN: j130/dev_mlb_u
LAST CHANGE: Wed Apr 1 22:57:37 2015

PAGE TITLE
USB-C CONNECTOR B

Apple Inc.

DRAWING NUMBER	051-00647
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Vout = 5.23V
Freq = 500 kHz
Max OCP = 13.9A
Nom OCP = 11.6A
Min OCP = 9.37A

BOM_COST_GROUP=USB-C

PAGE TITLE
TBT 5V REGULATOR

Apple Inc.

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PAGE TITLE
Power Aliases - 1

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1V0 Rails
PP1V0_SUS
PP1V0_S3
PP1V0_S0SW
T208 Rails
PP3V3_S5_T139
PP3V3_S4_SOC_PMU
PP5V_S0_T139
5V Rails
PP5V_S5
PP5V_S4
PP5V_S0
PP5V_S0_KBD
3V3 Rails
PP3V3_S5
PP3V3_SUS
PP3V3_S4
PP3V3_S0
PP3V3_S0_LEFT
PP3V3_TBT_X_S0
PP3V3_TBT_T_S0
PP3V3_S4_T151
PP3V3_S4_BT
PP3V3_S4_WLAN
PP3V3_S4_TPAD
PP3V3_S4SW_SNS
GPU Rails
PPVCORE_S0_GPU
PPVDDCI_S0_GPU
PP1V5R1V35_S0_GPU_MEM
PP1V5R1V35_S0_GPU_IC
PP1V8_GPU
PP1V8_S0_GPU
PP3V3_S0_GPU
Digital Ground
GND
MIN_NECK_WIDTH=0.1000
MIN_LINE_WIDTH=0.1270
PAGE TITLE
Power Aliases - 2
Apple Inc.
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PAGE TITLE
Signal Aliases

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Memory Bit/Byte Swizzle

LPDDR3 COMMAND/ADDRESS

	BDW-H LPDDR3 NET (MAKE_BASE=TRUE)	BIT SWIZZLE	
113 7	MEM_A_DQ<0>	MEM_A_DQ<0>	22
113 7	MEM_A_DQ<1>	MEM_A_DQ<1>	22
113 7	MEM_A_DQ<2>	MEM_A_DQ<2>	22
113 7	MEM_A_DQ<3>	MEM_A_DQ<3>	22
113 7	MEM_A_DQ<4>	MEM_A_DQ<4>	22
113 7	MEM_A_DQ<5>	MEM_A_DQ<5>	22
113 7	MEM_A_DQ<6>	MEM_A_DQ<6>	22
113 7	MEM_A_DQ<7>	MEM_A_DQ<7>	22
113 7	MEM_A_DQ<8>	MEM_A_DQ<8>	22
113 7	MEM_A_DQ<9>	MEM_A_DQ<9>	22
113 7	MEM_A_DQ<10>	MEM_A_DQ<10>	22
113 7	MEM_A_DQ<11>	MEM_A_DQ<11>	22
113 7	MEM_A_DQ<12>	MEM_A_DQ<12>	22
113 7	MEM_A_DQ<13>	MEM_A_DQ<13>	22
113 7	MEM_A_DQ<14>	MEM_A_DQ<14>	22
113 7	MEM_A_DQ<15>	MEM_A_DQ<15>	22
113 7	MEM_A_DQ<16>	MEM_A_DQ<16>	22
113 7	MEM_A_DQ<17>	MEM_A_DQ<17>	22
113 7	MEM_A_DQ<18>	MEM_A_DQ<18>	22
113 7	MEM_A_DQ<19>	MEM_A_DQ<19>	22
113 7	MEM_A_DQ<20>	MEM_A_DQ<20>	22
113 7	MEM_A_DQ<21>	MEM_A_DQ<21>	22
113 7	MEM_A_DQ<22>	MEM_A_DQ<22>	22
113 7	MEM_A_DQ<23>	MEM_A_DQ<23>	22
113 7	MEM_A_DQ<24>	MEM_A_DQ<24>	22
113 7	MEM_A_DQ<25>	MEM_A_DQ<25>	22
113 7	MEM_A_DQ<26>	MEM_A_DQ<26>	22
113 7	MEM_A_DQ<27>	MEM_A_DQ<27>	22
113 7	MEM_A_DQ<28>	MEM_A_DQ<28>	22
113 7	MEM_A_DQ<29>	MEM_A_DQ<29>	22
113 7	MEM_A_DQ<30>	MEM_A_DQ<30>	22
113 7	MEM_A_DQ<31>	MEM_A_DQ<31>	22
113 7	MEM_A_DQ<32>	MEM_A_DQ<32>	23
113 7	MEM_A_DQ<33>	MEM_A_DQ<33>	23
113 7	MEM_A_DQ<34>	MEM_A_DQ<34>	23
113 7	MEM_A_DQ<35>	MEM_A_DQ<35>	23
113 7	MEM_A_DQ<36>	MEM_A_DQ<36>	23
113 7	MEM_A_DQ<37>	MEM_A_DQ<37>	23
113 7	MEM_A_DQ<38>	MEM_A_DQ<38>	23
113 7	MEM_A_DQ<39>	MEM_A_DQ<39>	23
113 7	MEM_A_DQ<40>	MEM_A_DQ<40>	23
113 7	MEM_A_DQ<41>	MEM_A_DQ<41>	23
113 7	MEM_A_DQ<42>	MEM_A_DQ<42>	23
113 7	MEM_A_DQ<43>	MEM_A_DQ<43>	23
113 7	MEM_A_DQ<44>	MEM_A_DQ<44>	23
113 7	MEM_A_DQ<45>	MEM_A_DQ<45>	23
113 7	MEM_A_DQ<46>	MEM_A_DQ<46>	23
113 7	MEM_A_DQ<47>	MEM_A_DQ<47>	23
113 7	MEM_A_DQ<48>	MEM_A_DQ<48>	23
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113 7	MEM_A_DQ<53>	MEM_A_DQ<53>	23
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113 7	MEM_A_DQ<55>	MEM_A_DQ<55>	23
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113 7	MEM_A_DQ<59>	MEM_A_DQ<59>	23
113 7	MEM_A_DQ<60>	MEM_A_DQ<60>	23
113 7	MEM_A_DQ<61>	MEM_A_DQ<61>	23
113 7	MEM_A_DQ<62>	MEM_A_DQ<62>	23
113 7	MEM_A_DQ<63>	MEM_A_DQ<63>	23

	BDW-H LPDDR3 NET (MAKE_BASE=TRUE)	BIT SWIZZLE	
113 7	MEM_B_DQ<0>	MEM_B_DQ<0>	24
113 7	MEM_B_DQ<1>	MEM_B_DQ<1>	24
113 7	MEM_B_DQ<2>	MEM_B_DQ<2>	24
113 7	MEM_B_DQ<3>	MEM_B_DQ<3>	24
113 7	MEM_B_DQ<4>	MEM_B_DQ<4>	24
113 7	MEM_B_DQ<5>	MEM_B_DQ<5>	24
113 7	MEM_B_DQ<6>	MEM_B_DQ<6>	24
113 7	MEM_B_DQ<7>	MEM_B_DQ<7>	24
113 7	MEM_B_DQ<8>	MEM_B_DQ<8>	24
113 7	MEM_B_DQ<9>	MEM_B_DQ<9>	24
113 7	MEM_B_DQ<10>	MEM_B_DQ<10>	24
113 7	MEM_B_DQ<11>	MEM_B_DQ<11>	24
113 7	MEM_B_DQ<12>	MEM_B_DQ<12>	24
113 7	MEM_B_DQ<13>	MEM_B_DQ<13>	24
113 7	MEM_B_DQ<14>	MEM_B_DQ<14>	24
113 7	MEM_B_DQ<15>	MEM_B_DQ<15>	24
113 7	MEM_B_DQ<16>	MEM_B_DQ<16>	24
113 7	MEM_B_DQ<17>	MEM_B_DQ<17>	24
113 7	MEM_B_DQ<18>	MEM_B_DQ<18>	24
113 7	MEM_B_DQ<19>	MEM_B_DQ<19>	24
113 7	MEM_B_DQ<20>	MEM_B_DQ<20>	24
113 7	MEM_B_DQ<21>	MEM_B_DQ<21>	24
113 7	MEM_B_DQ<22>	MEM_B_DQ<22>	24
113 7	MEM_B_DQ<23>	MEM_B_DQ<23>	24
113 7	MEM_B_DQ<24>	MEM_B_DQ<24>	24
113 7	MEM_B_DQ<25>	MEM_B_DQ<25>	24
113 7	MEM_B_DQ<26>	MEM_B_DQ<26>	24
113 7	MEM_B_DQ<27>	MEM_B_DQ<27>	24
113 7	MEM_B_DQ<28>	MEM_B_DQ<28>	24
113 7	MEM_B_DQ<29>	MEM_B_DQ<29>	24
113 7	MEM_B_DQ<30>	MEM_B_DQ<30>	24
113 7	MEM_B_DQ<31>	MEM_B_DQ<31>	24
113 7	MEM_B_DQ<32>	MEM_B_DQ<32>	25
113 7	MEM_B_DQ<33>	MEM_B_DQ<33>	25
113 7	MEM_B_DQ<34>	MEM_B_DQ<34>	25
113 7	MEM_B_DQ<35>	MEM_B_DQ<35>	25
113 7	MEM_B_DQ<36>	MEM_B_DQ<36>	25
113 7	MEM_B_DQ<37>	MEM_B_DQ<37>	25
113 7	MEM_B_DQ<38>	MEM_B_DQ<38>	25
113 7	MEM_B_DQ<39>	MEM_B_DQ<39>	25
113 7	MEM_B_DQ<40>	MEM_B_DQ<40>	25
113 7	MEM_B_DQ<41>	MEM_B_DQ<41>	25
113 7	MEM_B_DQ<42>	MEM_B_DQ<42>	25
113 7	MEM_B_DQ<43>	MEM_B_DQ<43>	25
113 7	MEM_B_DQ<44>	MEM_B_DQ<44>	25
113 7	MEM_B_DQ<45>	MEM_B_DQ<45>	25
113 7	MEM_B_DQ<46>	MEM_B_DQ<46>	25
113 7	MEM_B_DQ<47>	MEM_B_DQ<47>	25
113 7	MEM_B_DQ<48>	MEM_B_DQ<48>	25
113 7	MEM_B_DQ<49>	MEM_B_DQ<49>	25
113 7	MEM_B_DQ<50>	MEM_B_DQ<50>	25
113 7	MEM_B_DQ<51>	MEM_B_DQ<51>	25
113 7	MEM_B_DQ<52>	MEM_B_DQ<52>	25
113 7	MEM_B_DQ<53>	MEM_B_DQ<53>	25
113 7	MEM_B_DQ<54>	MEM_B_DQ<54>	25
113 7	MEM_B_DQ<55>	MEM_B_DQ<55>	25
113 7	MEM_B_DQ<56>	MEM_B_DQ<56>	25
113 7	MEM_B_DQ<57>	MEM_B_DQ<57>	25
113 7	MEM_B_DQ<58>	MEM_B_DQ<58>	25
113 7	MEM_B_DQ<59>	MEM_B_DQ<59>	25
113 7	MEM_B_DQ<60>	MEM_B_DQ<60>	25
113 7	MEM_B_DQ<61>	MEM_B_DQ<61>	25
113 7	MEM_B_DQ<62>	MEM_B_DQ<62>	25
113 7	MEM_B_DQ<63>	MEM_B_DQ<63>	25

	MAKE_BASE			
113 7	TRUE	MEM_A_DQS_P<0>	MEM_A_DQS_P<0>	22
113 7	TRUE	MEM_A_DQS_N<0>	MEM_A_DQS_N<0>	22
113 7	TRUE	MEM_A_DQS_P<1>	MEM_A_DQS_P<1>	22
113 7	TRUE	MEM_A_DQS_N<1>	MEM_A_DQS_N<1>	22
113 7	TRUE	MEM_A_DQS_P<2>	MEM_A_DQS_P<2>	22
113 7	TRUE	MEM_A_DQS_N<2>	MEM_A_DQS_N<2>	22
113 7	TRUE	MEM_A_DQS_P<3>	MEM_A_DQS_P<3>	22
113 7	TRUE	MEM_A_DQS_N<3>	MEM_A_DQS_N<3>	22
113 7	TRUE	MEM_A_DQS_P<4>	MEM_A_DQS_P<4>	23
113 7	TRUE	MEM_A_DQS_N<4>	MEM_A_DQS_N<4>	23
113 7	TRUE	MEM_A_DQS_P<5>	MEM_A_DQS_P<5>	23
113 7	TRUE	MEM_A_DQS_N<5>	MEM_A_DQS_N<5>	23
113 7	TRUE	MEM_A_DQS_P<6>	MEM_A_DQS_P<6>	23
113 7	TRUE	MEM_A_DQS_N<6>	MEM_A_DQS_N<6>	23
113 7	TRUE	MEM_A_DQS_P<7>	MEM_A_DQS_P<7>	23
113 7	TRUE	MEM_A_DQS_N<7>	MEM_A_DQS_N<7>	23

	MAKE_BASE			
113 7	TRUE	MEM_B_DQS_P<0>	MEM_B_DQS_P<0>	24
113 7	TRUE	MEM_B_DQS_N<0>	MEM_B_DQS_N<0>	24
113 7	TRUE	MEM_B_DQS_P<1>	MEM_B_DQS_P<1>	24
113 7	TRUE	MEM_B_DQS_N<1>	MEM_B_DQS_N<1>	24
113 7	TRUE	MEM_B_DQS_P<2>	MEM_B_DQS_P<2>	24
113 7	TRUE	MEM_B_DQS_N<2>	MEM_B_DQS_N<2>	24
113 7	TRUE	MEM_B_DQS_P<3>	MEM_B_DQS_P<3>	24
113 7	TRUE	MEM_B_DQS_N<3>	MEM_B_DQS_N<3>	24
113 7	TRUE	MEM_B_DQS_P<4>	MEM_B_DQS_P<4>	25
113 7	TRUE	MEM_B_DQS_N<4>	MEM_B_DQS_N<4>	25
113 7	TRUE	MEM_B_DQS_P<5>	MEM_B_DQS_P<5>	25
113 7	TRUE	MEM_B_DQS_N<5>	MEM_B_DQS_N<5>	25
113 7	TRUE	MEM_B_DQS_P<6>	MEM_B_DQS_P<6>	25
113 7	TRUE	MEM_B_DQS_N<6>	MEM_B_DQS_N<6>	25
113 7	TRUE	MEM_B_DQS_P<7>	MEM_B_DQS_P<7>	25
113 7	TRUE	MEM_B_DQS_N<7>	MEM_B_DQS_N<7>	25

SYNC_MASTER=J80_MLB SYNC_DATE=11/06/2015

PAGE TITLE: Memory Bit/Byte Swizzle

Apple Inc.

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High Speed NO_TEST

DMI
DMI_S2N_P<3..0> NO_TEST=1
DMI_S2N_N<3..0> NO_TEST=1
DMI_N2S_P<3..0> NO_TEST=1
DMI_N2S_N<3..0> NO_TEST=1

DP - CPU/ACE
DP_X_SNK0_ML_C_N<3..0> NO_TEST=1
DP_X_SNK0_ML_C_P<3..0> NO_TEST=1
DP_X_SNK0_ML_N<3..0> NO_TEST=1
DP_X_SNK0_ML_P<3..0> NO_TEST=1
DP_X_SNK1_ML_C_N<3..0> NO_TEST=1
DP_X_SNK1_ML_C_P<3..0> NO_TEST=1
DP_X_SNK1_ML_N<3..0> NO_TEST=1
DP_X_SNK1_ML_P<3..0> NO_TEST=1
DP_T_SNK0_ML_C_N<3..0> NO_TEST=1
DP_T_SNK0_ML_C_P<3..0> NO_TEST=1
DP_T_SNK0_ML_N<3..0> NO_TEST=1
DP_T_SNK0_ML_P<3..0> NO_TEST=1
DP_T_SNK1_ML_C_N<3..0> NO_TEST=1
DP_T_SNK1_ML_C_P<3..0> NO_TEST=1
DP_T_SNK1_ML_N<3..0> NO_TEST=1
DP_T_SNK1_ML_P<3..0> NO_TEST=1
DP_X_SNK0_AUXCH_C_P NO_TEST=1
DP_X_SNK0_AUXCH_C_N NO_TEST=1
DP_T_SNK0_AUXCH_C_P NO_TEST=1
DP_T_SNK0_AUXCH_C_N NO_TEST=1
DP_X_SNK0_AUXCH_P NO_TEST=1
DP_X_SNK0_AUXCH_N NO_TEST=1
DP_T_SNK0_AUXCH_P NO_TEST=1
DP_T_SNK0_AUXCH_N NO_TEST=1
DP_X_SNK1_AUXCH_C_P NO_TEST=1
DP_X_SNK1_AUXCH_C_N NO_TEST=1
DP_T_SNK1_AUXCH_C_P NO_TEST=1
DP_T_SNK1_AUXCH_C_N NO_TEST=1
DP_X_SNK1_AUXCH_P NO_TEST=1
DP_X_SNK1_AUXCH_N NO_TEST=1
DP_T_SNK1_AUXCH_P NO_TEST=1
DP_T_SNK1_AUXCH_N NO_TEST=1

CPU/EDP
EDP_AUXCH_C_N
EDP_AUXCH_C_P

PCH/AR
PCIE_AP_D2R_P NO_TEST=1
PCIE_AP_D2R_N NO_TEST=1
PCIE_AP_R2D_P NO_TEST=1
PCIE_AP_R2D_N NO_TEST=1
PCIE_AP_R2D_C_P NO_TEST=1
PCIE_AP_R2D_C_N NO_TEST=1

PCH/SSD
PCIE_SSD_D2R_N<3..0> NO_TEST=1
PCIE_SSD_D2R_P<3..0> NO_TEST=1
PCIE_SSD_R2D_N<3..0> NO_TEST=1
PCIE_SSD_R2D_P<3..0> NO_TEST=1
PCIE_SSD_D2R_C_N<3..0> NO_TEST=1
PCIE_SSD_R2D_C_N<3..0> NO_TEST=1
PCIE_SSD_D2R_N<0> NO_TEST=1
PCIE_SSD_D2R_P<0> NO_TEST=1
PCIE_SSD_D2R_C_P<3..0> NO_TEST=1
PCIE_SSD_R2D_C_N<3..0> NO_TEST=1
PCIE_SSD_D2R_LB_P<0>
PCIE_SSD_R2D_LB_N<0>
PCIE_SSD_D2R_P<3> NO_TEST=1
PCIE_SSD_D2R_LB_N<0>
PCIE_SSD_D2R_N<3> NO_TEST=1
PCIE_SSD_R2D_LB_P<0>

USB-C X
USBC_XB_D2R_N<2..1> NO_TEST=1
USBC_XB_D2R_P<2..1> NO_TEST=1
USBC_XA_D2R_P<2..1> NO_TEST=1
USBC_XB_R2D_C_P<2..1> NO_TEST=1
DP_XA_AUXCH_P NO_TEST=1
USBC_XA_D2R_N<2..1> NO_TEST=1
USBC_XB_R2D_C_N<2..1> NO_TEST=1
USBC_XA_R2D_C_P<2..1> NO_TEST=1
DP_XB_AUXCH_P NO_TEST=1
USBC_XA_R2D_C_N<2..1> NO_TEST=1
DP_XB_AUXCH_N NO_TEST=1
DP_XA_AUXCH_N NO_TEST=1
DP_XA_AUXCH_C_N NO_TEST=1
DP_XA_AUXCH_C_P NO_TEST=1
DP_XB_AUXCH_C_P NO_TEST=1
DP_XB_AUXCH_C_N NO_TEST=1

PEG
PEG_GPU_D2R_N<7..0> NO_TEST=1
PEG_GPU_D2R_P<7..0> NO_TEST=1
PEG_GPU_R2D_C_N<7..0> NO_TEST=1
PEG_GPU_R2D_C_P<7..0> NO_TEST=1
PCIE_TBT_X_D2R_P<3..0> NO_TEST=1
PCIE_TBT_X_D2R_N<3..0> NO_TEST=1
PCIE_TBT_X_R2D_C_P<3..0> NO_TEST=1
PCIE_TBT_X_R2D_C_N<3..0> NO_TEST=1
PCIE_TBT_T_D2R_P<3..0> NO_TEST=1
PCIE_TBT_T_D2R_N<3..0> NO_TEST=1
PCIE_TBT_T_D2R_C_P<3..0> NO_TEST=1
PCIE_TBT_T_D2R_C_N<3..0> NO_TEST=1
PCIE_TBT_X_D2R_C_P<3..0> NO_TEST=1
PCIE_TBT_X_R2D_P<3..0> NO_TEST=1
PCIE_TBT_X_R2D_N<3..0> NO_TEST=1
PCIE_TBT_T_R2D_P<3..0> NO_TEST=1
PCIE_TBT_T_R2D_N<3..0> NO_TEST=1
PCIE_TBT_T_R2D_C_P<3..0> NO_TEST=1
PCIE_TBT_T_R2D_C_N<3..0> NO_TEST=1
PEG_GPU_D2R_C_N<7..0> NO_TEST=1
PEG_GPU_D2R_C_P<7..0> NO_TEST=1
PEG_GPU_R2D_N<7..0> NO_TEST=1
PEG_GPU_R2D_P<7..0> NO_TEST=1

PCH/DFR
USB_CAMERA_DFR_N
USB_CAMERA_DFR_P

BAFFIN FRAME BUFFER
FB_A1_CS_L NO_TEST=1
FB_A0_CKE_L NO_TEST=1
FB_A1_CKE_L NO_TEST=1
FB_A0_WE_L NO_TEST=1
FB_A1_WE_L NO_TEST=1
FB_B1_CS_L NO_TEST=1
FB_B0_CKE_L NO_TEST=1
FB_B1_CKE_L NO_TEST=1
FB_B0_WE_L NO_TEST=1
FB_B1_WE_L NO_TEST=1
FB_B0_CLK_P NO_TEST=1
FB_B0_CLK_N NO_TEST=1
FB_B1_CLK_P NO_TEST=1
FB_B0_RAS_L NO_TEST=1
FB_B1_RAS_L NO_TEST=1
FB_B0_CAS_L NO_TEST=1
FB_B1_CAS_L NO_TEST=1
FB_B0_CS_L NO_TEST=1
FB_A0_DQ<31..0> NO_TEST=1
FB_A1_DQ<31..0> NO_TEST=1
FB_A0_A<8..0> NO_TEST=1
FB_A1_A<8..0> NO_TEST=1
FB_A0_WCLK_N<1..0> NO_TEST=1
FB_A0_WCLK_P<1..0> NO_TEST=1
FB_A1_WCLK_N<1..0> NO_TEST=1
FB_A1_WCLK_P<1..0> NO_TEST=1
FB_A0_EDC<3..0> NO_TEST=1
FB_A1_EDC<3..0> NO_TEST=1
FB_A0_DBI_L<3..0> NO_TEST=1
FB_A1_DBI_L<3..0> NO_TEST=1
FB_A0_ABI_L NO_TEST=1
FB_A1_ABI_L NO_TEST=1
FB_A0_CLK_N NO_TEST=1
FB_A0_CLK_P NO_TEST=1
FB_A1_CLK_N NO_TEST=1
FB_A1_CLK_P NO_TEST=1
FB_A0_RAS_L NO_TEST=1
FB_A1_RAS_L NO_TEST=1
FB_A0_CAS_L NO_TEST=1
FB_A1_CAS_L NO_TEST=1
FB_A0_CS_L NO_TEST=1
FB_B1_CLK_N NO_TEST=1
FB_B0_DQ<31..0> NO_TEST=1
FB_B1_DQ<31..0> NO_TEST=1
FB_B0_A<8..0> NO_TEST=1
FB_B1_A<8..0> NO_TEST=1
FB_B0_WCLK_N<1..0> NO_TEST=1
FB_B0_WCLK_P<1..0> NO_TEST=1
FB_B1_WCLK_N<1..0> NO_TEST=1
FB_B1_WCLK_P<1..0> NO_TEST=1
FB_B0_EDC<3..0> NO_TEST=1
FB_B1_EDC<3..0> NO_TEST=1
FB_B0_DBI_L<3..0> NO_TEST=1
FB_B1_DBI_L<3..0> NO_TEST=1
FB_B0_ABI_L NO_TEST=1
FB_B1_ABI_L NO_TEST=1

CPU/PCH CLK
CPU_CLK24M_NSSC_CLK_N NO_TEST=1
CPU_CLK24M_NSSC_CLK_P NO_TEST=1
CPU_CLK100M_PCIBCLK_N NO_TEST=1
CPU_CLK100M_PCIBCLK_P NO_TEST=1
CPU_CLK100M_BCLK_N NO_TEST=1
CPU_CLK100M_BCLK_P NO_TEST=1
PCIE_CLK100M_SSD_N NO_TEST=1
PCIE_CLK100M_SSD_P NO_TEST=1
PCIE_CLK100M_SSD_LB_N NO_TEST=1
PCIE_CLK100M_SSD_LB_P NO_TEST=1
NC_ITPXDP_CLK100MN NO_TEST=1
NC_ITPXDP_CLK100MP NO_TEST=1
PCIE_CLK100M_AP_P NO_TEST=1
PCIE_CLK100M_AP_N NO_TEST=1

MUX
DP_INT_IG_ML_N<3..0> NO_TEST=1
DP_INT_IG_ML_P<3..0> NO_TEST=1
DP_INT_IG_AUX_N NO_TEST=1
DP_INT_IG_AUX_P NO_TEST=1
DP_INT_EG_ML_N<3..0> NO_TEST=1
DP_INT_EG_ML_P<3..0> NO_TEST=1
DP_INT_EG_AUX_N NO_TEST=1
DP_INT_EG_AUX_P NO_TEST=1

DP MUX CRYSTAL
DPMUX_UC_XTAL NO_TEST=1
DPMUX_UC_EXTAL NO_TEST=1
DPMUX_UC_EXTAL_R NO_TEST=1
DPMUX_UC_XTAL_R NO_TEST=1

XTAL
GPU_XTAL_PU_OR_CAP NO_TEST=1
NC_PCH_CLK24M_XTALOUT NO_TEST=1
SYSCLK_CLK24M_X2 NO_TEST=1
SYSCLK_CLK24M_X2_R NO_TEST=1
SYSCLK_CLK24M_X1 NO_TEST=1
TBT_X_XTAL25M_OUT NO_TEST=1
TBT_X_XTAL25M_OUT_R NO_TEST=1
TBT_X_XTAL25M_IN NO_TEST=1
TBT_T_XTAL25M_OUT NO_TEST=1
TBT_T_XTAL25M_OUT_R NO_TEST=1
SOC_XTAL_24M_O NO_TEST=1
TBT_T_XTAL25M_IN NO_TEST=1
SOC_XTAL_24M_I NO_TEST=1
SOC_XTAL_24M_O_R NO_TEST=1
SSD_CLKIN NO_TEST=1
SSD_CLKOUT NO_TEST=1
SSD_CLKOUT_R NO_TEST=1
GPU_XTAL_OUT NO_TEST=1
GPU_XTAL_OR_CLK_IN NO_TEST=1
GPU_XTAL_XTAL_OR_CLK NO_TEST=1

MEM
MEM_A_CAA<9..0> NO_TEST=1
MEM_A_CLK_P<1..0> NO_TEST=1
MEM_A_CLK_N<1..0> NO_TEST=1
MEM_A_DQ<63..0> NO_TEST=1
MEM_B_DQ<63..0> NO_TEST=1
MEM_A_DQS_P<7..0> NO_TEST=1
MEM_A_DQS_N<7..0> NO_TEST=1
MEM_B_DQS_P<7..0> NO_TEST=1
MEM_B_DQS_N<7..0> NO_TEST=1
MEM_A_CAB<9..0> NO_TEST=1
MEM_B_CAA<9..0> NO_TEST=1
MEM_B_CAB<9..0> NO_TEST=1
MEM_B_CLK_P<1..0> NO_TEST=1
MEM_B_CLK_N<1..0> NO_TEST=1

WIFI
50_0_ANT NO_TEST=1
50_1_ANT NO_TEST=1
50_2_ANT NO_TEST=1
50_0_COM NO_TEST=1
50_1_COM NO_TEST=1
50_2_COM NO_TEST=1
50_A_0_DIPLEXER NO_TEST=1
50_A_0_MATCH NO_TEST=1
50_G_0_DIPLEXER NO_TEST=1
50_G_0_MATCH NO_TEST=1
50_A_1_DIPLEXER NO_TEST=1
50_A_1_MATCH NO_TEST=1
50_G_1_DIPLEXER NO_TEST=1
50_G_1_MATCH NO_TEST=1
50_A_2_DIPLEXER NO_TEST=1
50_A_2_MATCH NO_TEST=1
50_G_2_DIPLEXER NO_TEST=1
50_G_2_MATCH NO_TEST=1

USB-C T
USBC_TA_D2R_P<2..1> NO_TEST=1
USBC_TA_D2R_N<2..1> NO_TEST=1
USBC_TA_R2D_C_P<2..1> NO_TEST=1
USBC_TA_R2D_C_N<2..1> NO_TEST=1
DP_TA_AUXCH_P NO_TEST=1
DP_TA_AUXCH_N NO_TEST=1
DP_TA_AUXCH_C_P NO_TEST=1
DP_TA_AUXCH_C_N NO_TEST=1
USBC_TB_D2R_P<2..1> NO_TEST=1
USBC_TB_D2R_N<2..1> NO_TEST=1
USBC_TB_R2D_C_P<2..1> NO_TEST=1
USBC_TB_R2D_C_N<2..1> NO_TEST=1
DP_TB_AUXCH_P NO_TEST=1
DP_TB_AUXCH_N NO_TEST=1
DP_TB_AUXCH_C_P NO_TEST=1
DP_TB_AUXCH_C_N NO_TEST=1
SSD_NAND_FH_DQ4 NO_TEST=1
SSD_NAND_FH_DQ0 NO_TEST=1
SSD_NAND_FH_DQS_P NO_TEST=1
SSD_NAND_FH_DQ5 NO_TEST=1
SSD_NAND_FH_DQ6 NO_TEST=1
SSD_NAND_FH_DQ7 NO_TEST=1
SSD_NAND_FH_DQS_N NO_TEST=1

NAND
SSD_NAND_FA_DQ0 NO_TEST=1
SSD_NAND_FA_DQ1 NO_TEST=1
SSD_NAND_FA_DQ2 NO_TEST=1
SSD_NAND_FA_DQ3 NO_TEST=1
SSD_NAND_FA_DQ4 NO_TEST=1
SSD_NAND_FA_DQ5 NO_TEST=1
SSD_NAND_FA_DQ6 NO_TEST=1
SSD_NAND_FA_DQ7 NO_TEST=1
SSD_NAND_FA_DQS_N NO_TEST=1
SSD_NAND_FA_DQS_P NO_TEST=1
SSD_NAND_FB_DQ0 NO_TEST=1
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SSD_NAND_FB_DQ7 NO_TEST=1
SSD_NAND_FB_DQS_N NO_TEST=1
SSD_NAND_FB_DQS_P NO_TEST=1
SSD_NAND_FC_DQ0 NO_TEST=1
SSD_NAND_FC_DQ1 NO_TEST=1
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SSD_NAND_FC_DQ3 NO_TEST=1
SSD_NAND_FC_DQ4 NO_TEST=1
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SSD_NAND_FC_DQ6 NO_TEST=1
SSD_NAND_FC_DQ7 NO_TEST=1
SSD_NAND_FC_DQS_N NO_TEST=1
SSD_NAND_FC_DQS_P NO_TEST=1
SSD_NAND_FD_DQ0 NO_TEST=1
SSD_NAND_FD_DQ1 NO_TEST=1
SSD_NAND_FD_DQ2 NO_TEST=1
SSD_NAND_FD_DQ3 NO_TEST=1
SSD_NAND_FD_DQ4 NO_TEST=1
SSD_NAND_FD_DQ5 NO_TEST=1
SSD_NAND_FD_DQ6 NO_TEST=1
SSD_NAND_FD_DQ7 NO_TEST=1
SSD_NAND_FD_DQS_N
SSD_NAND_FD_DQS_P
SSD_NAND_FE_DQ0 NO_TEST=1
SSD_NAND_FE_DQ1 NO_TEST=1
SSD_NAND_FE_DQ2 NO_TEST=1
SSD_NAND_FE_DQ3 NO_TEST=1
SSD_NAND_FE_DQ4 NO_TEST=1
SSD_NAND_FE_DQ5 NO_TEST=1
SSD_NAND_FE_DQ6 NO_TEST=1
SSD_NAND_FE_DQ7 NO_TEST=1
SSD_NAND_FE_DQS_N NO_TEST=1
SSD_NAND_FE_DQS_P NO_TEST=1
SSD_NAND_FF_DQ0 NO_TEST=1
SSD_NAND_FF_DQ1 NO_TEST=1
SSD_NAND_FF_DQ2 NO_TEST=1
SSD_NAND_FF_DQ3 NO_TEST=1
SSD_NAND_FF_DQ4 NO_TEST=1
SSD_NAND_FF_DQ5 NO_TEST=1
SSD_NAND_FF_DQ6 NO_TEST=1
SSD_NAND_FF_DQ7 NO_TEST=1
SSD_NAND_FF_DQS_N NO_TEST=1
SSD_NAND_FF_DQS_P NO_TEST=1
SSD_NAND_FG_DQ0 NO_TEST=1
SSD_NAND_FG_DQ2 NO_TEST=1
SSD_NAND_FG_DQ3 NO_TEST=1
SSD_NAND_FG_DQ4 NO_TEST=1
SSD_NAND_FG_DQ5 NO_TEST=1
SSD_NAND_FG_DQ6 NO_TEST=1
SSD_NAND_FG_DQ7 NO_TEST=1
SSD_NAND_FG_DQS_N NO_TEST=1
SSD_NAND_FG_DQS_P NO_TEST=1
SSD_NAND_FH_DQ1 NO_TEST=1
SSD_NAND_FH_DQ2 NO_TEST=1
SSD_NAND_FH_DQ3 NO_TEST=1

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OTHER TEST POINTS / NC

NC with No Testpoint Property

Pages	Net	Pin	Property	Net
13	NC_CLINK_CLK	1	TRUE	NC_CLINK_CLK
13	NC_CLINK_DATA	1	TRUE	NC_CLINK_DATA
13	NC_CLINK_RESET_L	1	TRUE	NC_CLINK_RESET_L
113 18 12	NC_ITPXDP_CLK100MN	1	TRUE	NC_ITPXDP_CLK100MN
113 18 12	NC_ITPXDP_CLK100MP	1	TRUE	NC_ITPXDP_CLK100MP
13	NC_HDA_SDIN1	1	TRUE	NC_HDA_SDIN1

Ref	Pin	Net	Pages
I1	1	NC_SPI_SMC_MOSI	48
I2	1	NC_SPI_SMC_MISO	48
I4	1	NC_SPI_SMC_CS_L	48
I3	1	NC_SPI_SMC_CLK	48
I7		NC_PCH_CLK32K_RTCX2	20

FAN Test Points

Ref	Property	Net	Pages
I25	TRUE	FAN_LT_PWM	43 56 114
I26	TRUE	FAN_LT_TACH	43 56 114
I27	TRUE	PP5V_S0	38 110 115
I28	TRUE	FAN_RT_PWM	43 56 114
I29	TRUE	FAN_RT_TACH	43 56 114
I30	TRUE	PP5V_S0	38 110 115

XDP Test-Points

Ref	Property	Net	Pages
I52	TRUE	XDP_CPU_TCK	6 18
I53	TRUE	XDP_PCH_TCK	13 18
I57	TRUE	XDP_CPU_TDI	6 18
I56	TRUE	XDP_CPU_TDO	6 18
I55	TRUE	XDP_CPU_TRST_L	6 13 18
I54	TRUE	XDP_CPU_TMS	6 18
I58	TRUE	XDP_PCH_TMS	13 18
I62	TRUE	XDP_PCH_TDI	13 18
I61	TRUE	XDP_PCH_TDO	13 18
I60	TRUE	XDP_CPU_PREQ_L	6 13 18
I59	TRUE	XDP_CPU_PRDY_L	6 13 18
I63	TRUE	PM_RSMRST_L	12 18 46 73
I64	TRUE	PM_PCH_PWROK	12 70
I65	TRUE	PM_SYSRST_L	12 18 46
I66	TRUE	CPU_CFG<3>	6 18

CPU (Refer to CPU pages)

CPU_DC_B2_C1 -----------------> TP0501
CPU_DC_B38_C38 -----------------> TP0502
CPU_DC_BR2_BR1 -----------------> TP0503
CPU_DC_C1_B2 -----------------> TP0504
CPU_DC_C38_B38 -----------------> TP0505
CPU_DC_BR1_BR2 -----------------> TP0610
CPU_DC_BR38_BT36 -----------------> TP0900
CPU_DC_BT36_BR38 -----------------> TP0901

PCH (Refer to PCH pages)

XDP_PCH_OBSDATA_A0 -----------------> TP1883
XDP_PCH_OBSDATA_A1 -----------------> TP1884
XDP_PCH_OBSDATA_A2 -----------------> TP1870
XDP_PCH_OBSDATA_A3 -----------------> TP1871
XDP_PCH_OBSDATA_B0 -----------------> TP1872
XDP_PCH_OBSDATA_B1 -----------------> TP1885
XDP_PCH_OBSDATA_B2 -----------------> TP1886
XDP_PCH_OBSDATA_B3 -----------------> TP1887
XDP_PCH_OBSDATA_D0 -----------------> TP1877
XDP_PCH_OBSDATA_D1 -----------------> TP1878
XDP_PCH_OBSDATA_D2 -----------------> TP1879
XDP_PCH_OBSDATA_D3 -----------------> TP1880
XDP_PCH_OBSFN_C0 -----------------> TP1882
XDP_BPM_L<0> -----------------> TP1800
XDP_BPM_L<1> -----------------> TP1801
XDP_BPM_L<2> -----------------> TP1802
XDP_BPM_L<3> -----------------> TP1803
NC_USB_EXTA_OC_L -----------------> TP1873
NC_USB_EXTB_OC_L -----------------> TP1874
NC_USB_EXTC_OC_L -----------------> TP1875
NC_USB_EXTD_OC_L -----------------> TP1876

TPs on BOTTOM to check USB-C Instalation

Ref	Net	Property	Pages
I75	USBC_XA_SBU1	FUNC_TEST=TRUE	30 32
I76	USBC_XB_SBU1	FUNC_TEST=TRUE	31 32
I77	USBC_XA_SBU2	FUNC_TEST=TRUE	30 32
I78	USBC_XB_SBU2	FUNC_TEST=TRUE	31 32
I79	USBC_TA_SBU1	FUNC_TEST=TRUE	104 106
I80	USBC_TB_SBU1	FUNC_TEST=TRUE	105 106
I81	USBC_TA_SBU2	FUNC_TEST=TRUE	104 106
I82	USBC_TB_SBU2	FUNC_TEST=TRUE	105 106
I83	PP20V_USBC_XA_VBUS_CONN	FUNC_TEST=TRUE	32 114 116
I86	PP20V_USBC_XB_VBUS_CONN	FUNC_TEST=TRUE	32 114 116
I85	PP20V_USBC_TA_VBUS_CONN	FUNC_TEST=TRUE	106 114 116
I84	PP20V_USBC_TB_VBUS_CONN	FUNC_TEST=TRUE	106 114 116

TPs to check LifeBoat Instalation

Ref	Net	Property	Pages
I88	PP3V3_S5_POLARIS	FUNC_TEST=TRUE	84 87
I87	PP3V3_S5	FUNC_TEST=TRUE	110
I90	SSD_PWR_EN	FUNC_TEST=TRUE	14 87 114
I89	SSD_PWR_LB_EN	FUNC_TEST=TRUE	84 87
I91	SSD_BOOT_L	FUNC_TEST=TRUE	15 87 114
I92	SSD_BOOT_LB_L	FUNC_TEST=TRUE	77 87 114
I93	SSD_RESET_L	FUNC_TEST=TRUE	14 20 87 114
I95	SSD_RESET_LB_L	FUNC_TEST=TRUE	77 87
I94	SSD_CLKREQ_L	FUNC_TEST=TRUE	20 87
I96	SSD_CLKREQ_LB_L	FUNC_TEST=TRUE	77 87

SYNC_MASTER=X363_BBABADI SYNC_DATE=01/26/2016

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NC & No Test

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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
131S00041	6	CAP,CER,12PF,5%,100V,C0G,0402	CC784,CC785,CC786,CC787,CC788	CRITICAL	

PAGE TITLE

Desense Caps

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BOM_COST_GROUP=WIRELESS

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SYNC_MASTER=X363_ZIFENGSHEN SYNC_DATE=06/02/2016

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Project Specific Constraints

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BOM Variants

44 ULA Configs

BOM NUMBER	BOM NAME	BOM OPTIONS
639-02212	MLB,2.6G,MC-16,ULA-MC,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_MICRON_2133,4GB_MC_BAFFIN,BAFFIN_ULA,SSD_CONFIG:1TB
639-02120	MLB,2.6G,MC-16,ULA-MC,SSD-256,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_MICRON_2133,4GB_MC_BAFFIN,BAFFIN_ULA,SSD_CONFIG:256GB
639-02591	MLB,2.6G,MC-16,ULA-MC,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_MICRON_2133,4GB_MC_BAFFIN,BAFFIN_ULA,SSD_CONFIG:2TB
639-02214	MLB,2.6G,MC-16,ULA-MC,SSD-512,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_MICRON_2133,4GB_MC_BAFFIN,BAFFIN_ULA,SSD_CONFIG:512GB
639-02587	MLB,2.6G,SM-16,ULA-MC,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_SAMSUNG_2133,4GB_MC_BAFFIN,BAFFIN_ULA,SSD_CONFIG:1TB
639-02116	MLB,2.6G,SM-16,ULA-MC,SSD-256,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_SAMSUNG_2133,4GB_MC_BAFFIN,BAFFIN_ULA,SSD_CONFIG:256GB
639-02589	MLB,2.6G,SM-16,ULA-MC,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_SAMSUNG_2133,4GB_MC_BAFFIN,BAFFIN_ULA,SSD_CONFIG:2TB
639-02586	MLB,2.6G,SM-16,ULA-MC,SSD-512,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_SAMSUNG_2133,4GB_MC_BAFFIN,BAFFIN_ULA,SSD_CONFIG:512GB
639-01615	MLB,2.7G,MC-16,ULA-MC,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7,RAM_16G_MICRON_2133,4GB_MC_BAFFIN,BAFFIN_ULA,SSD_CONFIG:1TB
639-02592	MLB,2.7G,MC-16,ULA-MC,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7,RAM_16G_MICRON_2133,4GB_MC_BAFFIN,BAFFIN_ULA,SSD_CONFIG:2TB
639-01614	MLB,2.7G,MC-16,ULA-MC,SSD-512,X36G3	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7,RAM_16G_MICRON_2133,4GB_MC_BAFFIN,BAFFIN_ULA,SSD_CONFIG:512GB
639-01603	MLB,2.7G,SM-16,ULA-MC,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7,RAM_16G_SAMSUNG_2133,4GB_MC_BAFFIN,BAFFIN_ULA,SSD_CONFIG:1TB
639-02594	MLB,2.7G,SM-16,ULA-MC,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7,RAM_16G_SAMSUNG_2133,4GB_MC_BAFFIN,BAFFIN_ULA,SSD_CONFIG:2TB
639-01602	MLB,2.7G,SM-16,ULA-MC,SSD-512,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7,RAM_16G_SAMSUNG_2133,4GB_MC_BAFFIN,BAFFIN_ULA,SSD_CONFIG:512GB
639-01639	MLB,2.9G,MC-16,ULA-MC,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_MICRON_2133,4GB_MC_BAFFIN,BAFFIN_ULA,SSD_CONFIG:1TB
639-01637	MLB,2.9G,MC-16,ULA-MC,SSD-256,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_MICRON_2133,4GB_MC_BAFFIN,BAFFIN_ULA,SSD_CONFIG:256GB
639-02596	MLB,2.9G,MC-16,ULA-MC,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_MICRON_2133,4GB_MC_BAFFIN,BAFFIN_ULA,SSD_CONFIG:2TB
639-01638	MLB,2.9G,MC-16,ULA-MC,SSD-512,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_MICRON_2133,4GB_MC_BAFFIN,BAFFIN_ULA,SSD_CONFIG:512GB
639-01627	MLB,2.9G,SM-16,ULA-MC,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_SAMSUNG_2133,4GB_MC_BAFFIN,BAFFIN_ULA,SSD_CONFIG:1TB
639-01625	MLB,2.9G,SM-16,ULA-MC,SSD-256,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_SAMSUNG_2133,4GB_MC_BAFFIN,BAFFIN_ULA,SSD_CONFIG:256GB
639-02598	MLB,2.9G,SM-16,ULA-MC,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_SAMSUNG_2133,4GB_MC_BAFFIN,BAFFIN_ULA,SSD_CONFIG:2TB
639-01626	MLB,2.9G,SM-16,ULA-MC,SSD-512,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_SAMSUNG_2133,4GB_MC_BAFFIN,BAFFIN_ULA,SSD_CONFIG:512GB
639-02154	MLB,2.6G,MC-16,ULA-SM,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_MICRON_2133,4GB_SM_BAFFIN,BAFFIN_ULA,SSD_CONFIG:1TB
639-02119	MLB,2.6G,MC-16,ULA-SM,SSD-256,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_MICRON_2133,4GB_SM_BAFFIN,BAFFIN_ULA,SSD_CONFIG:256GB
639-02590	MLB,2.6G,MC-16,ULA-SM,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_MICRON_2133,4GB_SM_BAFFIN,BAFFIN_ULA,SSD_CONFIG:2TB
639-02213	MLB,2.6G,MC-16,ULA-SM,SSD-512,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_MICRON_2133,4GB_SM_BAFFIN,BAFFIN_ULA,SSD_CONFIG:512GB
639-02211	MLB,2.6G,SM-16,ULA-SM,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_SAMSUNG_2133,4GB_SM_BAFFIN,BAFFIN_ULA,SSD_CONFIG:1TB
639-02115	MLB,2.6G,SM-16,ULA-SM,SSD-256,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_SAMSUNG_2133,4GB_SM_BAFFIN,BAFFIN_ULA,SSD_CONFIG:256GB
639-02588	MLB,2.6G,SM-16,ULA-SM,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_SAMSUNG_2133,4GB_SM_BAFFIN,BAFFIN_ULA,SSD_CONFIG:2TB
639-02585	MLB,2.6G,SM-16,ULA-SM,SSD-512,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_SAMSUNG_2133,4GB_SM_BAFFIN,BAFFIN_ULA,SSD_CONFIG:512GB
639-01612	MLB,2.7G,MC-16,ULA-SM,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7,RAM_16G_MICRON_2133,4GB_SM_BAFFIN,BAFFIN_ULA,SSD_CONFIG:1TB
639-02593	MLB,2.7G,MC-16,ULA-SM,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7,RAM_16G_MICRON_2133,4GB_SM_BAFFIN,BAFFIN_ULA,SSD_CONFIG:2TB
639-01611	MLB,2.7G,MC-16,ULA-SM,SSD-512,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7,RAM_16G_MICRON_2133,4GB_SM_BAFFIN,BAFFIN_ULA,SSD_CONFIG:512GB
639-01600	MLB,2.7G,SM-16,ULA-SM,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7,RAM_16G_SAMSUNG_2133,4GB_SM_BAFFIN,BAFFIN_ULA,SSD_CONFIG:1TB
639-02595	MLB,2.7G,SM-16,ULA-SM,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7,RAM_16G_SAMSUNG_2133,4GB_SM_BAFFIN,BAFFIN_ULA,SSD_CONFIG:2TB
639-01599	MLB,2.7G,SM-16,ULA-SM,SSD-512,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7,RAM_16G_SAMSUNG_2133,4GB_SM_BAFFIN,BAFFIN_ULA,SSD_CONFIG:512GB
639-01636	MLB,2.9G,MC-16,ULA-SM,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_MICRON_2133,4GB_SM_BAFFIN,BAFFIN_ULA,SSD_CONFIG:1TB
639-01634	MLB,2.9G,MC-16,ULA-SM,SSD-256,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_MICRON_2133,4GB_SM_BAFFIN,BAFFIN_ULA,SSD_CONFIG:256GB
639-02597	MLB,2.9G,MC-16,ULA-SM,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_MICRON_2133,4GB_SM_BAFFIN,BAFFIN_ULA,SSD_CONFIG:2TB
639-01635	MLB,2.9G,MC-16,ULA-SM,SSD-512,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_MICRON_2133,4GB_SM_BAFFIN,BAFFIN_ULA,SSD_CONFIG:512GB
639-01624	MLB,2.9G,SM-16,ULA-SM,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_SAMSUNG_2133,4GB_SM_BAFFIN,BAFFIN_ULA,SSD_CONFIG:1TB
639-01622	MLB,2.9G,SM-16,ULA-SM,SSD-256,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_SAMSUNG_2133,4GB_SM_BAFFIN,BAFFIN_ULA,SSD_CONFIG:256GB
639-02599	MLB,2.9G,SM-16,ULA-SM,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_SAMSUNG_2133,4GB_SM_BAFFIN,BAFFIN_ULA,SSD_CONFIG:2TB
639-01623	MLB,2.9G,SM-16,ULA-SM,SSD-512,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_SAMSUNG_2133,4GB_SM_BAFFIN,BAFFIN_ULA,SSD_CONFIG:512GB

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BOM NUMBER	BOM NAME	BOM OPTIONS
639-02655	MLB,2.7G,MC-16,PROA-HY,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7,RAM_16G_MICRON_2133,2GB_HY_BAFFIN,BAFFIN_PROA,SSD_CONFIG:1TB
639-02656	MLB,2.7G,MC-16,PROA-HY,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7,RAM_16G_MICRON_2133,2GB_HY_BAFFIN,BAFFIN_PROA,SSD_CONFIG:2TB
639-02654	MLB,2.7G,MC-16,PROA-HY,SSD-512,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7,RAM_16G_MICRON_2133,2GB_HY_BAFFIN,BAFFIN_PROA,SSD_CONFIG:512GB
639-02664	MLB,2.7G,SM-16,PROA-HY,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7,RAM_16G_SAMSUNG_2133,2GB_HY_BAFFIN,BAFFIN_PROA,SSD_CONFIG:1TB
639-02665	MLB,2.7G,SM-16,PROA-HY,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7,RAM_16G_SAMSUNG_2133,2GB_HY_BAFFIN,BAFFIN_PROA,SSD_CONFIG:2TB
639-02663	MLB,2.7G,SM-16,PROA-HY,SSD-512,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7,RAM_16G_SAMSUNG_2133,2GB_HY_BAFFIN,BAFFIN_PROA,SSD_CONFIG:512GB
639-02673	MLB,2.9G,MC-16,PROA-HY,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_MICRON_2133,2GB_HY_BAFFIN,BAFFIN_PROA,SSD_CONFIG:1TB
639-02674	MLB,2.9G,MC-16,PROA-HY,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_MICRON_2133,2GB_HY_BAFFIN,BAFFIN_PROA,SSD_CONFIG:2TB
639-02672	MLB,2.9G,MC-16,PROA-HY,SSD-512,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_MICRON_2133,2GB_HY_BAFFIN,BAFFIN_PROA,SSD_CONFIG:512GB
639-02682	MLB,2.9G,SM-16,PROA-HY,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_SAMSUNG_2133,2GB_HY_BAFFIN,BAFFIN_PROA,SSD_CONFIG:1TB
639-02683	MLB,2.9G,SM-16,PROA-HY,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_SAMSUNG_2133,2GB_HY_BAFFIN,BAFFIN_PROA,SSD_CONFIG:2TB
639-02681	MLB,2.9G,SM-16,PROA-HY,SSD-512,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_SAMSUNG_2133,2GB_HY_BAFFIN,BAFFIN_PROA,SSD_CONFIG:512GB
639-02652	MLB,2.7G,MC-16,PROA-MC,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7,RAM_16G_MICRON_2133,2GB_MC_BAFFIN,BAFFIN_PROA,SSD_CONFIG:1TB
639-02653	MLB,2.7G,MC-16,PROA-MC,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7,RAM_16G_MICRON_2133,2GB_MC_BAFFIN,BAFFIN_PROA,SSD_CONFIG:2TB
639-02651	MLB,2.7G,MC-16,PROA-MC,SSD-512,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7,RAM_16G_MICRON_2133,2GB_MC_BAFFIN,BAFFIN_PROA,SSD_CONFIG:512GB
639-02661	MLB,2.7G,SM-16,PROA-MC,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7,RAM_16G_SAMSUNG_2133,2GB_MC_BAFFIN,BAFFIN_PROA,SSD_CONFIG:1TB
639-02662	MLB,2.7G,SM-16,PROA-MC,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7,RAM_16G_SAMSUNG_2133,2GB_MC_BAFFIN,BAFFIN_PROA,SSD_CONFIG:2TB
639-02660	MLB,2.7G,SM-16,PROA-MC,SSD-512,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7,RAM_16G_SAMSUNG_2133,2GB_MC_BAFFIN,BAFFIN_PROA,SSD_CONFIG:512GB
639-02670	MLB,2.9G,MC-16,PROA-MC,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_MICRON_2133,2GB_MC_BAFFIN,BAFFIN_PROA,SSD_CONFIG:1TB
639-02671	MLB,2.9G,MC-16,PROA-MC,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_MICRON_2133,2GB_MC_BAFFIN,BAFFIN_PROA,SSD_CONFIG:2TB
639-02669	MLB,2.9G,MC-16,PROA-MC,SSD-512,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_MICRON_2133,2GB_MC_BAFFIN,BAFFIN_PROA,SSD_CONFIG:512GB
639-02679	MLB,2.9G,SM-16,PROA-MC,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_SAMSUNG_2133,2GB_MC_BAFFIN,BAFFIN_PROA,SSD_CONFIG:1TB
639-02680	MLB,2.9G,SM-16,PROA-MC,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_SAMSUNG_2133,2GB_MC_BAFFIN,BAFFIN_PROA,SSD_CONFIG:2TB
639-02678	MLB,2.9G,SM-16,PROA-MC,SSD-512,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_SAMSUNG_2133,2GB_MC_BAFFIN,BAFFIN_PROA,SSD_CONFIG:512GB
639-02649	MLB,2.7G,MC-16,PROA-SM,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7,RAM_16G_MICRON_2133,2GB_SM_BAFFIN,BAFFIN_PROA,SSD_CONFIG:1TB
639-02650	MLB,2.7G,MC-16,PROA-SM,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7,RAM_16G_MICRON_2133,2GB_SM_BAFFIN,BAFFIN_PROA,SSD_CONFIG:2TB
639-02648	MLB,2.7G,MC-16,PROA-SM,SSD-512,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7,RAM_16G_MICRON_2133,2GB_SM_BAFFIN,BAFFIN_PROA,SSD_CONFIG:512GB
639-02658	MLB,2.7G,SM-16,PROA-SM,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7,RAM_16G_SAMSUNG_2133,2GB_SM_BAFFIN,BAFFIN_PROA,SSD_CONFIG:1TB
639-02659	MLB,2.7G,SM-16,PROA-SM,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7,RAM_16G_SAMSUNG_2133,2GB_SM_BAFFIN,BAFFIN_PROA,SSD_CONFIG:2TB
639-02657	MLB,2.7G,SM-16,PROA-SM,SSD-512,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7,RAM_16G_SAMSUNG_2133,2GB_SM_BAFFIN,BAFFIN_PROA,SSD_CONFIG:512GB
639-02667	MLB,2.9G,MC-16,PROA-SM,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_MICRON_2133,2GB_SM_BAFFIN,BAFFIN_PROA,SSD_CONFIG:1TB
639-02668	MLB,2.9G,MC-16,PROA-SM,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_MICRON_2133,2GB_SM_BAFFIN,BAFFIN_PROA,SSD_CONFIG:2TB
639-02666	MLB,2.9G,MC-16,PROA-SM,SSD-512,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_MICRON_2133,2GB_SM_BAFFIN,BAFFIN_PROA,SSD_CONFIG:512GB
639-02676	MLB,2.9G,SM-16,PROA-SM,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_SAMSUNG_2133,2GB_SM_BAFFIN,BAFFIN_PROA,SSD_CONFIG:1TB
639-02677	MLB,2.9G,SM-16,PROA-SM,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_SAMSUNG_2133,2GB_SM_BAFFIN,BAFFIN_PROA,SSD_CONFIG:2TB
639-02675	MLB,2.9G,SM-16,PROA-SM,SSD-512,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_SAMSUNG_2133,2GB_SM_BAFFIN,BAFFIN_PROA,SSD_CONFIG:512GB

BOM Variants, Power/Socket Configs

BOM NUMBER	BOM NAME	BOM OPTIONS
639-01966	PCBA,MLB,NONE,SM-16,FB4-SM,S256,X363	BASE_BOM,DEVEL_BOM,RAM_16G_SAMSUNG_2133,4GB_SM_BAFFIN,SSD_CONFIG:256GB
639-01967	PCBA,MLB,SKT,VDDC,SM-16,FB4-SM,S256,X363	BASE_BOM,DEVEL_BOM,STARDUST:VDDC,RAM_16G_SAMSUNG_2133,4GB_SM_BAFFIN,SSD_CONFIG:256GB
639-01968	PCBA,MLB,SKT,MVDD,SM-16,FB4-SM,S256,X363	BASE_BOM,DEVEL_BOM,STARDUST:VDDCI_MVDD,RAM_16G_SAMSUNG_2133,4GB_SM_BAFFIN,SSD_CONFIG:256GB
639-01969	PCBA,MLB,SKT,CPU,SM-16,FB4-SM,S256,X363	BASE_BOM,DEVEL_BOM,CPU_SKL:SOCKET,RAM_16G_SAMSUNG_2133,4GB_SM_BAFFIN,SSD_CONFIG:256GB

SYNC_MASTER=J80_MLB SYNC_DATE=07/23/2015

PAGE TITLE: 639 BOM Configuration

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BOM Variants

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BOM NUMBER	BOM NAME	BOM OPTIONS
639-02621	MLB,2.6G,MC-16,LEA-HY,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_MICRON_2133,2GB_HY_BAFFIN,BAFFIN_LEA,SSD_CONFIG:1TB
639-02618	MLB,2.6G,MC-16,LEA-HY,SSD-256,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_MICRON_2133,2GB_HY_BAFFIN,BAFFIN_LEA,SSD_CONFIG:256GB
639-02622	MLB,2.6G,MC-16,LEA-HY,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_MICRON_2133,2GB_HY_BAFFIN,BAFFIN_LEA,SSD_CONFIG:2TB
639-02620	MLB,2.6G,MC-16,LEA-HY,SSD-512,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_MICRON_2133,2GB_HY_BAFFIN,BAFFIN_LEA,SSD_CONFIG:512GB
639-02608	MLB,2.6G,SM-16,LEA-HY,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_SAMSUNG_2133,2GB_HY_BAFFIN,BAFFIN_LEA,SSD_CONFIG:1TB
639-02602	MLB,2.6G,SM-16,LEA-HY,SSD-256,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_SAMSUNG_2133,2GB_HY_BAFFIN,BAFFIN_LEA,SSD_CONFIG:256GB
639-02611	MLB,2.6G,SM-16,LEA-HY,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_SAMSUNG_2133,2GB_HY_BAFFIN,BAFFIN_LEA,SSD_CONFIG:2TB
639-02605	MLB,2.6G,SM-16,LEA-HY,SSD-512,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_SAMSUNG_2133,2GB_HY_BAFFIN,BAFFIN_LEA,SSD_CONFIG:512GB
639-02645	MLB,2.9G,MC-16,LEA-HY,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_MICRON_2133,2GB_HY_BAFFIN,BAFFIN_LEA,SSD_CONFIG:1TB
639-02643	MLB,2.9G,MC-16,LEA-HY,SSD-256,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_MICRON_2133,2GB_HY_BAFFIN,BAFFIN_LEA,SSD_CONFIG:256GB
639-02646	MLB,2.9G,MC-16,LEA-HY,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_MICRON_2133,2GB_HY_BAFFIN,BAFFIN_LEA,SSD_CONFIG:2TB
639-02644	MLB,2.9G,MC-16,LEA-HY,SSD-512,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_MICRON_2133,2GB_HY_BAFFIN,BAFFIN_LEA,SSD_CONFIG:512GB
639-02633	MLB,2.9G,SM-16,LEA-HY,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_SAMSUNG_2133,2GB_HY_BAFFIN,BAFFIN_LEA,SSD_CONFIG:1TB
639-02631	MLB,2.9G,SM-16,LEA-HY,SSD-256,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_SAMSUNG_2133,2GB_HY_BAFFIN,BAFFIN_LEA,SSD_CONFIG:256GB
639-02634	MLB,2.9G,SM-16,LEA-HY,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_SAMSUNG_2133,2GB_HY_BAFFIN,BAFFIN_LEA,SSD_CONFIG:2TB
639-02632	MLB,2.9G,SM-16,LEA-HY,SSD-512,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_SAMSUNG_2133,2GB_HY_BAFFIN,BAFFIN_LEA,SSD_CONFIG:512GB
639-02647	MLB,2.6G,MC-16,LEA-MC,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_MICRON_2133,2GB_MC_BAFFIN,BAFFIN_LEA,SSD_CONFIG:1TB
639-02616	MLB,2.6G,MC-16,LEA-MC,SSD-256,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_MICRON_2133,2GB_MC_BAFFIN,BAFFIN_LEA,SSD_CONFIG:256GB
639-02619	MLB,2.6G,MC-16,LEA-MC,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_MICRON_2133,2GB_MC_BAFFIN,BAFFIN_LEA,SSD_CONFIG:2TB
639-02617	MLB,2.6G,MC-16,LEA-MC,SSD-512,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_MICRON_2133,2GB_MC_BAFFIN,BAFFIN_LEA,SSD_CONFIG:512GB
639-02607	MLB,2.6G,SM-16,LEA-MC,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_SAMSUNG_2133,2GB_MC_BAFFIN,BAFFIN_LEA,SSD_CONFIG:1TB
639-02601	MLB,2.6G,SM-16,LEA-MC,SSD-256,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_SAMSUNG_2133,2GB_MC_BAFFIN,BAFFIN_LEA,SSD_CONFIG:256GB
639-02610	MLB,2.6G,SM-16,LEA-MC,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_SAMSUNG_2133,2GB_MC_BAFFIN,BAFFIN_LEA,SSD_CONFIG:2TB
639-02604	MLB,2.6G,SM-16,LEA-MC,SSD-512,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_SAMSUNG_2133,2GB_MC_BAFFIN,BAFFIN_LEA,SSD_CONFIG:512GB
639-02641	MLB,2.9G,MC-16,LEA-MC,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_MICRON_2133,2GB_MC_BAFFIN,BAFFIN_LEA,SSD_CONFIG:1TB
639-02639	MLB,2.9G,MC-16,LEA-MC,SSD-256,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_MICRON_2133,2GB_MC_BAFFIN,BAFFIN_LEA,SSD_CONFIG:256GB
639-02642	MLB,2.9G,MC-16,LEA-MC,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_MICRON_2133,2GB_MC_BAFFIN,BAFFIN_LEA,SSD_CONFIG:2TB
639-02640	MLB,2.9G,MC-16,LEA-MC,SSD-512,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_MICRON_2133,2GB_MC_BAFFIN,BAFFIN_LEA,SSD_CONFIG:512GB
639-02629	MLB,2.9G,SM-16,LEA-MC,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_SAMSUNG_2133,2GB_MC_BAFFIN,BAFFIN_LEA,SSD_CONFIG:1TB
639-02627	MLB,2.9G,SM-16,LEA-MC,SSD-256,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_SAMSUNG_2133,2GB_MC_BAFFIN,BAFFIN_LEA,SSD_CONFIG:256GB
639-02630	MLB,2.9G,SM-16,LEA-MC,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_SAMSUNG_2133,2GB_MC_BAFFIN,BAFFIN_LEA,SSD_CONFIG:2TB
639-02628	MLB,2.9G,SM-16,LEA-MC,SSD-512,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_SAMSUNG_2133,2GB_MC_BAFFIN,BAFFIN_LEA,SSD_CONFIG:512GB
639-02614	MLB,2.6G,MC-16,LEA-SM,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_MICRON_2133,2GB_SM_BAFFIN,BAFFIN_LEA,SSD_CONFIG:1TB
639-02612	MLB,2.6G,MC-16,LEA-SM,SSD-256,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_MICRON_2133,2GB_SM_BAFFIN,BAFFIN_LEA,SSD_CONFIG:256GB
639-02615	MLB,2.6G,MC-16,LEA-SM,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_MICRON_2133,2GB_SM_BAFFIN,BAFFIN_LEA,SSD_CONFIG:2TB
639-02613	MLB,2.6G,MC-16,LEA-SM,SSD-512,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_MICRON_2133,2GB_SM_BAFFIN,BAFFIN_LEA,SSD_CONFIG:512GB
639-02606	MLB,2.6G,SM-16,LEA-SM,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_SAMSUNG_2133,2GB_SM_BAFFIN,BAFFIN_LEA,SSD_CONFIG:1TB
639-02600	MLB,2.6G,SM-16,LEA-SM,SSD-256,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_SAMSUNG_2133,2GB_SM_BAFFIN,BAFFIN_LEA,SSD_CONFIG:256GB
639-02609	MLB,2.6G,SM-16,LEA-SM,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_SAMSUNG_2133,2GB_SM_BAFFIN,BAFFIN_LEA,SSD_CONFIG:2TB
639-02603	MLB,2.6G,SM-16,LEA-SM,SSD-512,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6,RAM_16G_SAMSUNG_2133,2GB_SM_BAFFIN,BAFFIN_LEA,SSD_CONFIG:512GB
639-02637	MLB,2.9G,MC-16,LEA-SM,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_MICRON_2133,2GB_SM_BAFFIN,BAFFIN_LEA,SSD_CONFIG:1TB
639-02635	MLB,2.9G,MC-16,LEA-SM,SSD-256,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_MICRON_2133,2GB_SM_BAFFIN,BAFFIN_LEA,SSD_CONFIG:256GB
639-02638	MLB,2.9G,MC-16,LEA-SM,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_MICRON_2133,2GB_SM_BAFFIN,BAFFIN_LEA,SSD_CONFIG:2TB
639-02636	MLB,2.9G,MC-16,LEA-SM,SSD-512,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_MICRON_2133,2GB_SM_BAFFIN,BAFFIN_LEA,SSD_CONFIG:512GB
639-02625	MLB,2.9G,SM-16,LEA-SM,SSD-1TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_SAMSUNG_2133,2GB_SM_BAFFIN,BAFFIN_LEA,SSD_CONFIG:1TB
639-02623	MLB,2.9G,SM-16,LEA-SM,SSD-256,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_SAMSUNG_2133,2GB_SM_BAFFIN,BAFFIN_LEA,SSD_CONFIG:256GB
639-02626	MLB,2.9G,SM-16,LEA-SM,SSD-2TB,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_SAMSUNG_2133,2GB_SM_BAFFIN,BAFFIN_LEA,SSD_CONFIG:2TB
639-02624	MLB,2.9G,SM-16,LEA-SM,SSD-512,X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9,RAM_16G_SAMSUNG_2133,2GB_SM_BAFFIN,BAFFIN_LEA,SSD_CONFIG:512GB

Alternate Parts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	
740S0144	740S0118		ALL		
740S00003	740S0135		ALL		
376S1089	376S1128		ALL		
376S1080	376S0820		ALL		
376S00086	376S0761		ALL		
376S00074	376S0855		ALL		
376S00014	376S0761		ALL		
372S0186	372S0185		ALL		
371S0713	371S0558		ALL		
311S00060	311S0273		ALL		
311S00004	311S0370		ALL		
155S0914	155S0897		ALL		
155S0694	155S0387		ALL		
155S0660	155S0513		ALL		
155S00154	155S0398		ALL		
155S00007	155S0667		ALL		
138S0863	138S0853		ALL		
138S0775	138S0860		ALL		<rdar://problem/23155097> for IG <rdar://problem/23564467> for EG
138S0703	138S0648		ALL		<rdar://problem/23118804> for IG <rdar://problem/> for EG
132S00064	132S0409		ALL		
128S0325	128S0397		ALL		
128S00029	128S00007		ALL		
128S00026	128S00011		ALL		
128S00070	128S00007		ALL		<rdar://problem/25329164>
128S00009	128S00007		ALL		
107S0249	107S0251		ALL		
107S00071	107S00053		ALL		
107S00070	107S0085		ALL		
107S00033	107S00034		ALL		BLC
107S00015	107S00011		ALL		
376S1106	376S0678		ALL	Fairchild alt to Vishay	BLC
138S0738	138S1101		ALL	Samsung alt to Murata	
138S0846	138S0811		ALL	Samsung alt to Murata	
376S1053	376S0604		ALL	Diodes alt to Fairchild	
152S00359	152S00253		ALL	Chillisin alt to Cyntec	
740S00027	740S0159		ALL	Bourns alt to Little Fuse	
371S0704	371S00077		ALL	NXP alt to Diodes	
138S00032	138S0831		ALL		
138S00049	138S0831		ALL		

220uF 2.0V D-case, 128S00044 DQ

998-04070	998-04071		OFF	Hynix alt to SS	T208
128S00010	128S00011		ALL		
128S00031	128S00011		ALL		
138S00084	138S00060		ALL		<rdar://problem/23118634> for IG <rdar://problem/23527737> for EG
155S00155	155S0441		OFF		(Stale; delete later after page sync)
155S00190	155S0897		ALL		
353S00107	353S3239		ALL		
353S00525	353S4471		ALL		
376S1193	376S00037		OFF		
740S00028	740S0118		ALL		
152S00369	152S00268		ALL	Cyntec w/ NEC	<rdar://problem/23405116> for IG <rdar://problem/23542193> for EG
128S0296	128S0487		ALL	NEC w/ pana	<rdar://problem/23422212> for IG <rdar://problem/23527282> for EG
128S00012	128S0487		ALL	NEC w/ Rohm	
155S00188	155S0275		ALL	Murata w/ Taiyo	<rdar://problem/23446936> for IG <rdar://problem/23542939> for EG
155S00018	155S0664		ALL	Murata w/ Taiyo	<rdar://problem/23447549> for IG <rdar://problem/23543068> for EG
152S00388	152S00182		ALL		<rdar://problem/23542023>
107S0240	107S0255		ALL		<rdar://problem/23520324>
128S00062	128S00067		ALL	NEC for Panasonic	150uF 6.3V B12
138S0660	138S0684		ALL		<rdar://problem/24316249>
155S00204	155S0731		ALL		<rdar://problem/24310837>

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	
311S0271	311S00008		ALL	NXP w/ Diodes	(Stale; delete later after page sync) <rdar://problem/23117904> for IG <rdar://problem/24316367> for EG
197S00046	197S00036		ALL	Epson w/ TXC	<rdar://problem/23117749> for IG <rdar://problem/23543146> for EG
197S00047	197S00036		ALL	Kyocera w/ TXC	
197S00048	197S00036		ALL	Murata w/ TXC	
197S00053	197S00050		ALL	Kyocera w/ TXC	<rdar://problem/23117849> for IG <rdar://problem/23543858> for EG
197S00054	197S00050		ALL	NDK w/ TXC	
197S00055	197S00050		ALL	Murata w/ TXC	
311S0596	311S0593		ALL	NXP w/ Diodes	<rdar://problem/23118093> for IG <rdar://problem/23544024> for EG
107S0276	107S00020		ALL	Cyntec w/ TFT	<rdar://problem/23117951> for IG <rdar://problem/23542980> for EG
107S00021	107S0284		ALL	TFT w/ Yageo	<rdar://problem/23129685> for IG <rdar://problem/23519943> for EG
132S00012	132S0401		ALL	Taiyo w/ Murata&TDK	<rdar://problem/23118402> for IG <rdar://problem/23527525> for EG
152S00343	152S1682		ALL	NXP w/ Diodes	<rdar://problem/23541694> for IG <rdar://problem/23541871> for EG
107S00087	107S00029		ALL	TFT w/ Yageo	<rdar://problem/23121290> for IG <rdar://problem/23520039> for EG
128S00057	128S00018		ALL	NEC w/ Vishay	<rdar://problem/23363718> for IG <rdar://problem/23526361> for EG
128S00058	128S00018		ALL	NEC w/ Rohm	
128S0364	128S0264		ALL	Kemet w/ Panasonic	<rdar://problem/23366105> for IG <rdar://problem/23527232> for EG
138S0641	138S0700		ALL	Murata w/ SS&Taiyo	<rdar://problem/23118759> for IG <rdar://problem/23528188> for EG
138S0739	138S0706		ALL	NEC w/ Vishay	<rdar://problem/23118915> for IG <rdar://problem/23528364 for EG
138S0945	138S0706		ALL	NEC w/ Rohm	
152S00358	152S00208		ALL	Murata w/ Chillisin	<rdar://problem/23119390> for IG <rdar://problem/23542069> for EG
152S00389	152S00241		OFF	Cyntec w/ Vishay	<rdar://problem/23342644> for IG <rdar://problem/> for EG
152S00390	152S00265		OFF	Cyntec w/ Vishay	<rdar://problem/23342719> for IG <rdar://problem/> for EG
152S00400	152S1872		ALL	Murata w/ Cyntec	
152S1872	152S00361		ALL	Murata w/ Cyntec	<rdar://problem/23132240> for IG <rdar://problem/23542412> for EG
155S00034	155S0706		ALL	Taiyo w/ Murata	<rdar://problem/23118020> for IG <rdar://problem/23543018> for EG

371S00082	371S00046		ALL	On-Semi w/ Diodes	<rdar://problem/23341512> for IG <rdar://problem/23545301> for EG
376S00146	376S1061		ALL	NXP w/ Diodes	<rdar://problem/23341674> for IG <rdar://problem/23545469> for EG
353S00711	353S2073		ALL	On Semi w/ TI	<rdar://problem/23605939> for IG <rdar://problem/23605957> for EG

740S00019	740S00007		ALL	Bourns w/ Polytronics	<rdar://problem/23137505> for IG <rdar://problem/23545479> for EG
155S00189	155S0342		ALL	Murata w/ Taiyo	<rdar://problem/23117986> for IG <rdar://problem/23543015> for EG
132S0438	132S0428		ALL	Murata w/ Taiyo&TDK	<rdar://problem/23118548> for IG <rdar://problem/23527609> for EG
138S0714	138S0713		ALL	Murata w/ Samsung	<rdar://problem/23118842> for IG <rdar://problem/23528300> for EG
138S0715	138S0732		ALL	Murata w/ Samsung	<rdar://problem/23118873> for IG <rdar://problem/23528349> for EG
107S00086	107S00056		ALL	TFT w/ Cyntec	<rdar://problem/23121312> for IG <rdar://problem/23520074> for EG
138S0875	138S0678		ALL	Taiyo w/ Mur&SS	<rdar://problem/23129763> for IG <rdar://problem/23528106> for EG
138S0786	138S0705		ALL	Murata w/ Samsung	<rdar://problem/23129934> for IG <rdar://problem/23528216> for EG
155S0382	155S0659		ALL	Murata w/ TDK	<rdar://problem/23132624> for IG <rdar://problem/23543045> for EG
152S2052	152S1954		ALL	Taiyo w/ Cyntec	<rdar://problem/23135833> for IG <rdar://problem/23542478> for EG
152S2015	152S1958		ALL	Taiyo w/ Cyntec	<rdar://problem/23137154> for IG <rdar://problem/23542579> for EG
128S00055	128S00002		ALL	Kemet w/ Panasonic	<rdar://problem/23148659> for IG <rdar://problem/23526236> for EG
138S0748	138S0751		ALL	Murata w/ SS	<rdar://problem/23155052> for IG <rdar://problem/23528417> for EG
138S00102	138S0773		ALL	Murata w/ Taiyo	<rdar://problem/23155089> for IG <rdar://problem/23529046> for EG
138S0789	138S0941		ALL	Murata w/ SS	<rdar://problem/23155132> for IG <rdar://problem/23529142> for EG
107S00101	107S00005		ALL	Cyntec w/ Yageo	<rdar://problem/23193801> for IG <rdar://problem/23519872> for EG
107S00102	107S00017		ALL	Cyntec w/ Yageo	<rdar://problem/23194019> for IG <rdar://problem/23519914> for EG
107S00100	107S00057		ALL	Cyntec w/ TFT	<rdar://problem/23224358> for IG <rdar://problem/23520119> for EG
107S00103	107S00058		ALL	Cyntec w/ Yageo	<rdar://problem/23228523> for IG <rdar://problem/23520163> for EG
107S00104	107S00061		ALL	Cyntec w/ Yageo	<rdar://problem/23229217> for IG <rdar://problem/23520193> for EG
107S00105	107S00062		ALL	Cyntec w/ Yageo	<rdar://problem/23230371> for IG <rdar://problem/23520241> for EG
152S00403	152S00322		ALL	Murata w/ Chillisin	<rdar://problem/23341061> for IG <rdar://problem/23542282> for EG
353S00852	353S4262		ALL	TI w/ OnSemi	<rdar://problem/23341158> for IG <rdar://problem/23542282> for EG

138S00104	138S0978		ALL		<rdar://problem/23583487>
128S00069	128S00067		ALL	Rohm for Panasonic	150uF 6.3V B12 <rdar://problem/23527054>
138S0759	138S0762		ALL		<rdar://problem/23582324>
377S0077	377S0183		ALL		<rdar://problem/23640745>
152S00363	152S00048		ALL		<rdar://problem/23541533>
138S00111	138S00036		ALL		<rdar://problem/23546425>
138S00097	138S0750		ALL		<rdar://problem/23581953>
155S00203	155S0894		ALL		<rdar://problem/24285018>
116S00006	116S0175		ALL		<rdar://problem/23524119>
311S00104	311S00091		ALL		<rdar://problem/23640575>

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	
311S00013	311S0508		ALL	Part will be stuffed in production	<rdar://problem/24316400>
311S00118	311S0489		ALL		<rdar://problem/27623384>
311S00072	311S0657		ALL		<rdar://problem/23595055>
311S00090	311S00028		ALL		<rdar://problem/23602138>
353S00750	353S00877		ALL		<rdar://problem/23545068>
353S00878	353S00599		ALL		<rdar://problem/23545040>
311S00105	311S0233		ALL		<rdar://problem/24316381>
353S00880	353S3452		ALL		<rdar://problem/23545099>
311S00007	311S0426		ALL		<rdar://problem/23597412>
128S0445	128S0392		ALL		<rdar://problem/23566873>
152S00415	152S00140		ALL		<rdar://problem/23766197>
128S0436	128S0392		ALL		<rdar://problem/23566873>
343S00135	343S00136		ALL	T208	
343S00137	343S00136		ALL	T208	
343S00138	343S00136		ALL	T208	
138S00105	138S00037		ALL		<rdar://problem/23640133>
152S00434	152S1829		ALL		<rdar://problem/23825631>
353S3527	353S3528		ALL		<rdar://problem/24492801>
353S3526	353S3528		ALL		<rdar://problem/24492801>
353S00135	353S00034		ALL		<rdar://problem/24534121>
353S2220	353S00034		ALL		<rdar://problem/24534121>
353S00769	353S4398		ALL		<rdar://problem/24534714>
353S00879	353S00754		ALL		<rdar://problem/23545080>
104S00012	155S0398		ALL		<rdar://problem/25180014>
353S4342	353S00854		ALL		<rdar://problem/23621422>
152S00543	152S00484		ALL		<rdar://problem/24780681>
371S00089	371S00085		ALL		<rdar://problem/24806090>
107S00111	107S00110		ALL		<rdar://problem/24968171>
335S00213	335S0888		ALL		<rdar://problem/24985900>
311S0437	311S00112		ALL		<rdar://problem/24747570>
377S0178	377S00031		ALL		<rdar://problem/25118595>
371S00091	371S00083		ALL		<rdar://problem/24827333>
197S00069	197S00068		ALL		<rdar://problem/24473517>
138S0698	138S00113		ALL		<rdar://problem/23528153>
152S00461	152S00112		ALL		<rdar://problem/24417973>
371S00074	371S0602		ALL		<rdar://problem/25919460>
197S00082	197S00081		ALL		
353S00991	353S00920		ALL		
131S00134	131S00041		ALL		<rdar://problem/24316117>